Course Notes

Lie Theory

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Contents

Chapter 1

Lecture 1: Where Lie algebras come from

1.1 Introduction

Lie algebras were originally introduced by Sophus Lie in the 1870s to study the concept of continuous transformation groups. Lie's motivation was to develop a theory for differential equations analogous to Galois theory for algebraic equations. He discovered that the infinitesimal transformations of a continuous group form a linear space equipped with a bracket operation, now known as a Lie algebra. This linearization allows complex global geometric problems to be translated into algebraic problems, making them more tractable.

Let F be a field (mostly $\operatorname{char} F = 0$, even $F = \mathbb{C}$).

Definition 1.1.1. *A linear algebra A is a vector space over F, with a binary bilinear operation:*

- *binary operation: $A \times A \to A$, $(a, b) \mapsto ab$.*
- *bilinear: this operation is linear in each variable. In other word, $\forall \alpha \in F$ and $a, a', a'', b, b', b'' \in A$, we have:*

$$(\alpha a)b = a(\alpha b) = \alpha(ab)$$
$$(a' + a'')b = a'b + a''b$$
$$a(b' + b'') = ab' + ab''$$

The algebra A is associative if $\forall a, b, c \in A$, $(ab)c = a(bc)$.

Example 1.1.2. *$M_n(F)$, the algebra of $n \times n$ matrices over F.*

Example 1.1.3. *Let V be a vector space, $\operatorname{End}_F(V) = \{$linear transformations $\varphi : V \to V\}$ is an associative algebra with the composition operation$(\phi\psi)(v) = \phi(\psi(v))$. If $\dim V < \infty$, when we fix a basis, then we fet a matrix representation of φ in our fix basis.*

The most important thing everyone should know is Wedderburn's theorem.

Theorem 1.1.4 (Wedderburn-Artin). *Let A be a semisimple Artinian ring (with 1). Then A is a direct sum of matrix rings over division rings.*
Specifically, if ${}_AA \simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$ where $S_1, \ldots, S_r$ are non-isomorphic simple modules occurring with multiplicities $n_1, \ldots, n_r$, then

$$A \simeq M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$$

where $D_i = \mathrm{End}_A(S_i)^{op}$. This is a ring isomorphism, and each $M_{n_i}(D_i)$ is a 2-sided ideal.

Proof. First, since A is a semisimple ring, its left regular module ${}_AA$ can be decomposed into a direct sum of simple left A-modules:

$${}_AA \simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$$

where $S_1, \ldots, S_r$ are pairwise non-isomorphic simple modules.

By Schur's lemma, the endomorphism ring of a simple module is a division ring, so $D_i = \mathrm{End}_A(S_i)^{op}$ is a division ring.

Next, we compute the endomorphism ring of the left regular module $\mathrm{End}_A({}_AA)$. On one hand, for any ring A with 1, we have the ring isomorphism $\mathrm{End}_A({}_AA) \simeq A^{op}$. On the other hand, using our direct sum decomposition:

$$\mathrm{End}_A({}_AA) \simeq \mathrm{End}_A(S_1^{n_1}) \oplus \cdots \oplus \mathrm{End}_A(S_r^{n_r})$$

This decomposition holds because $\mathrm{Hom}_A(S_i^{n_i}, S_j^{n_j}) = 0$ whenever $i \neq j$.

For each component in the direct sum, we can rewrite the endomorphism ring using matrices over D_i:

$$\mathrm{End}_A(S_i^{n_i}) \simeq M_{n_i}(D_i)^{op} \simeq M_{n_i}(D_i^{op})$$

Substituting this back into our equation for A^{op}, we obtain:

$$A^{op} \simeq M_{n_1}(D_1^{op}) \oplus \cdots \oplus M_{n_r}(D_r^{op})$$

To recover the ring A, we take the opposite ring on both sides. Recall the property that $M_n(R)^{op} \simeq M_n(R^{op})$. Applying this to each block yields:

$$A \simeq (A^{op})^{op} \simeq \left(\bigoplus_{i=1}^{r} M_{n_i}(D_i^{op}) \right)^{op} \simeq \bigoplus_{i=1}^{r} M_{n_i}(D_i)$$

Thus, the isomorphism is established. $\square$

Remark 1.1.5. *If the base field F is algebraically closed, every finite-dimensional division* *algebra over F is equal to F. Hence the theorem simplifies to*

$$A \cong \prod_{i=1}^{r} M_{n_i}(F).$$

Example 1.1.6. *Let G be a finite group and let $\mathbb{C}[G]$ be its group algebra. By Maschke's theorem $\mathbb{C}[G]$ is semisimple. Therefore*

$$\mathbb{C}[G] \cong \prod_{i=1}^{r} M_{n_i}(\mathbb{C}),$$

where n_i are the dimensions of the irreducible complex representations of G.

Definition 1.1.7. *Suppose F is algebraically closed. An algebra A is simple if*

1. $\nexists\ I \lhd A$ *and* $I \neq 0$.
2. $A \cdot A \neq (0)$.

Corollary 1.1.8. *Any simple finite dimensional associative algebra is isomorphic to $M_n(F)$ for some n.*

Definition 1.1.9 (Jacobson Radical)**.** *Let A be an associative algebra with identity over a field F. The* Jacobson radical *of A, denoted by* $\mathrm{Rad}(A)$ *or $J(A)$, is defined as the intersection of all maximal left ideals of A:*

$$\mathrm{Rad}(A) = \bigcap_{M \text{ maximal left ideal of } A} M.$$

Proposition 1.1.10. *Let A be an associative algebra with identity. Then*

$$\mathrm{Rad}(A) = \bigcap_{S \text{ simple left } A\text{-module}} \mathrm{Ann}_A(S),$$

where

$$\mathrm{Ann}_A(S) = \{a \in A \mid aS = 0\}$$

is the annihilator of S in A.

Proposition 1.1.11 (Equivalent Characterization)**.** *Let A be an associative algebra with identity. Then an element $a \in A$ lies in* $\mathrm{Rad}(A)$ *if and only if for every $x \in A$ the element*

$$1 - ax$$

is invertible in A.

Proposition 1.1.12. *The Jacobson radical* $\mathrm{Rad}(A)$ *is a two-sided ideal of A.*

Definition 1.1.13 (Semisimple Algebra)**.** *An associative algebra A is called* semisimple *if*

$$\mathrm{Rad}(A) = 0.$$

1.2 Where do Lie algebras come from?

Definition 1.2.1. *A Lie algebra is a vector space L over a field F, with a binary operation $[a, b] \in L$, such that*

1. $[a, b]$ *is bilinear.*
2. $[a, b] = -[b, a]$*.*
3. $[a, [b, c]] + [c, [a, b]] + [b, [c, a]] = 0$*, this is called Jacobi identity.*

1.2.1 They come from associative algebras

Let A be an associative algebra. $\forall a, b \in A$, define $[a, b] = ab - ba$. Check that $(A, [\ ,\ ])$ is a Lie algebra:

Proof. Bilinearity and skew-symmetry are clear. For Jacobi identity:

$$\begin{aligned}
&[a, [b, c]] + [b, [c, a]] + [c, [a, b]] \\
&\qquad = a(bc - cb) - (bc - cb)a + b(ca - ac) - (ca - ac)b + c(ab - ba) - (ab - ba)c \\
&\qquad = abc - acb - bca + cba + bca - bac - cab + acb + cab - cba - abc + bac \\
&\qquad = 0.
\end{aligned}$$

□

We denote this new algebra $(A, [\ ,\ ])$ by $A^{(-)}$.
Suppose that $L \subseteq A$ and $[L, L] \subseteq L$. Then $L \leq A^{(-)}$。

Example 1.2.2. $M_n(F)^{(-)} = \mathfrak{gl}(n)$*. We call it the general linear Lie algebra.*
$[L, L] = \{\sum_i [a_i, b_i]\} \trianglelefteq L$*. Clearly* $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq \mathfrak{gl}(n)$ *and for* $n \geq 2$*,* $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq 0$*.*

Example 1.2.3. *Focus on* $[\mathfrak{gl}(n), \mathfrak{gl}(n)]$*, all matrices are of trace 0. This is called the special linear Lie algebra, denoted by* $\mathfrak{sl}(n)$*.*

Example 1.2.4. *If the base field changed, some properties of the Lie algebra may change. For example, if* $\mathrm{char}\, F = 2$*, then* $[a, b] = [b, a]$*. The Lie bracket is no longer skew-symmetric but actually symmetric.*
Consider $[\mathfrak{gl}(n), \mathfrak{gl}(n)] = \mathfrak{sl}(n)$*. If* $\mathrm{char}\, F = 0$ *or* $\mathrm{char}\, F \nmid n$*, then* $\mathfrak{sl}(n)$ *is a simple Lie algebra. If* $\mathrm{char}\, F \mid n$*, then* $\mathfrak{sl}(n)$ *is not simple.*

Definition 1.2.5. *Z is the center of Lie algebra L if $Z = \{z \in L | [z, L] = 0\}$.*

1.2.2 They come from involutions of central simple algebras

We can also find Lie algebras from the involution of a central simple algebra. Let A be central simple algebra and $* : A \to A$ be a linear transformation. $*$ is called a an involution if $(a^*)^* = a$ and $(ab)^* = b^*a^*$, $\forall a, b \in A$.

Proposition 1.2.6. *Let A be a central simple algebra over a field F equipped with an involution $* : A \to A$. Then the set*

$$L = \{x \in A \mid *(x) = -x\}$$

forms a Lie algebra under the commutator bracket

$$[x, y] = xy - yx.$$

Proof. First observe that any associative algebra A becomes a Lie algebra with the commutator bracket

$$[x, y] = xy - yx.$$

Let

$$L = \{x \in A \mid *(x) = -x\}.$$

We show that L is closed under the Lie bracket.

Take $x, y \in L$. Then

$$*(x) = -x, \qquad *(y) = -y.$$

Since $*$ is an involution, it satisfies

$$*(xy) = *(y) * (x).$$

Hence

$$*([x, y]) = *(xy - yx) = *(y) * (x) - *(x) * (y).$$

Substituting $*(x) = -x$ and $*(y) = -y$, we obtain

$$*([x, y]) = (-y)(-x) - (-x)(-y) = yx - xy = -[x, y].$$

Thus $[x, y] \in L$. Therefore L is closed under the commutator bracket.

Since the commutator bracket in any associative algebra satisfies bilinearity, antisymmetry, and the Jacobi identity, it follows that L is a Lie algebra. □

This construction explains why many classical Lie algebras arise from central simple algebras with involution. In particular, the classical Lie algebras of Cartan–Killing type can be obtained in this way.

Example 1.2.7 (Orthogonal involution). *Let*

$$A = M_n(F)$$

and define the involution

$$*(X) = X^T.$$

Then

$$L = \{X \in M_n(F) \mid X^T = -X\}.$$

This is the Lie algebra of skew-symmetric matrices, which is the orthogonal Lie algebra

$$\mathfrak{so}_n(F).$$

Example 1.2.8 (Symplectic involution). *Let*

$$A = M_{2n}(F)$$

and let

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Define

$$*(X) = J^{-1} X^T J.$$

Then

$$L = \{X \in M_{2n}(F) \mid *(X) = -X\}$$

is the symplectic Lie algebra

$$\mathfrak{sp}_{2n}(F).$$

Example 1.2.9 (Unitary involution). *Let K/F be a quadratic field extension with Galois involution $a \mapsto \overline{a}$. Consider*

$$A = M_n(K)$$

with involution

$$*(X) = \overline{X}^T.$$

Then

$$L = \{X \in M_n(K) \mid *(X) = -X\}$$

is the unitary Lie algebra

$$\mathfrak{su}_n.$$

Therefore classical Lie algebras arise from pairs $(A, *)$ where A is a central simple algebra and $*$ is an involution. This provides an algebraic framework for the Cartan–Killing classification of classical Lie algebras.

Example 1.2.10. *The Quaternion algebra is an associative algebra with the involution $a \mapsto a^*$. It is 4 dimensional central simple algebra with standard basis $1, i, j, k$, where the relations are $i^2 = j^2 = k^2 = -1, ij = k, jk = i, ki = j$ as the following graph:*

$$\begin{tikzcd} i \arrow[r] & j \arrow[d] \\ & k \arrow[ul] \end{tikzcd}$$

We use $\mathbb{H}$ to denote the Quaternion algebra when we choose $\mathbb{R}$ as the base field. i.e. $\mathbb{H} = \{a_0 1 + a_1 i + a_2 j + a_3 k | a_0, a_1, a_2, a_3 \in \mathbb{R}\}$. $\forall x \in \mathbb{H}$ where $x = a_0 1 + a_1 i + a_2 j + a_3 k$, $*$ is defined as $x^* = a_0 1 - a_1 i - a_2 j - a_3 k$ and $x^{-1} = \frac{1}{a_0^2 + a_1^2 + a_2^2 + a_3^2}(a_0 1 - a_1 i - a_2 j - a_3 k)$. Hence $\mathbb{H}$ is a division algebra.
If the base field is algebraically closed, $F = \overline{F}$. We can denote the Quaternion algebra by 2×2 matrices

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \xi \end{pmatrix}$$

Where $\alpha, \beta, \gamma, \xi \in F$. In this case it is no longer a division algebra ($a_0^2 + a_1^2 + a_2^2 + a_3^2$ may equal to 0), but just $M_2(F)$. And in this case the involution $*$ is defined as:

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \xi \end{pmatrix} \mapsto \begin{pmatrix} \xi & -\beta \\ -\gamma & \alpha \end{pmatrix}$$

Which is the Symplectic involution.

Example 1.2.11. *From what we discussed. For a $2n \times 2n$ matrix $A = \begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix}$ in which A_1, A_2, A_3, A_4 are $n \times n$ matrices, the unitary involution is defined as:*

$$A^* = \begin{pmatrix} A_4^t & -A_3^t \\ -A_2^t & A_1^t \end{pmatrix}$$

And the unitary involution is defined as:

$$A^* = \overline{A}^t$$

When $n = 1$ we have two different type, and we will get two different Lie algebras. But they are isomorphic.

Proposition 1.2.12. *Let F be a field of characteristic $\neq 2$, and let*

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in M_2(F).$$

Define an involution $*$ on $M_2(F)$ by

$$X^* = J^{-1}X^tJ.$$

Then the Lie algebra of $*$-skew elements,

$$K(M_2(F), *) = \{X \in M_2(F) \mid X^* = -X\},$$

is exactly $\mathfrak{sl}_2(F)$.

Proof. Let

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(F).$$

Since

$$J^{-1} = -J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

we compute

$$X^* = J^{-1}X^tJ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Now the condition $X^* = -X$ becomes

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix}.$$

Hence $d = -a$, while b and c are arbitrary. Therefore

$$K(M_2(F), *) = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \middle| \, a, b, c \in F \right\}.$$

But this is precisely the set of all 2×2 matrices of trace 0, namely

$$\mathfrak{sl}_2(F) = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \middle| \, a, b, c \in F \right\}.$$

Thus

$$K(M_2(F), *) = \mathfrak{sl}_2(F).$$

Since $K(M_2(F), *)$ is closed under the commutator bracket, it follows that the Lie algebra obtained from the symplectic involution on $M_2(F)$ is $\mathfrak{sl}_2(F)$. □

Example 1.2.13. *Let K/F be a quadratic field extension, and let $a \mapsto \overline{a}$ denote the nontrivial F-automorphism of K. Consider the algebra $M_2(K)$ and define an involution $*$ by*

$$X^* = \overline{X}^t.$$

Then $*$ is a unitary involution on $M_2(K)$.
The Lie algebra of $*$-skew elements is

$$K(M_2(K),*) = \{X \in M_2(K) \mid X^* = -X\}.$$

If

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then

$$X^* = \overline{X}^t = \begin{pmatrix} \overline{a} & \overline{c} \\ \overline{b} & \overline{d} \end{pmatrix}.$$

Thus the condition $X^* = -X$ is equivalent to

$$\overline{a} = -a, \qquad \overline{d} = -d, \qquad \overline{c} = -b.$$

Hence every element of $K(M_2(K),*)$ has the form

$$X = \begin{pmatrix} a & b \\ -\overline{b} & d \end{pmatrix}, \qquad \text{with } \overline{a} = -a,\ \overline{d} = -d.$$

This is the unitary Lie algebra associated with the standard hermitian form.
Its derived Lie algebra consists of those elements with trace 0, namely

$$\mathfrak{su}_2(K/F) = \left\{ \begin{pmatrix} a & b \\ -\overline{b} & -a \end{pmatrix} \,\middle|\, a, b \in K,\ \overline{a} = -a \right\}.$$

After extending scalars to a splitting field (for instance to an algebraic closure of F), this Lie algebra becomes isomorphic to $\mathfrak{sl}_2$. Therefore the Lie algebra arising from a unitary involution in the 2×2 case is of type A_1.

Remark 1.2.14. *Why does this Lie algebra become $\mathfrak{sl}_2$? The reason is that unitary Lie algebras are forms of special linear Lie algebras. In the present 2×2 case, the corresponding type is A_1. Over a field where the quadratic extension splits, the involution becomes equivalent to the ordinary transpose situation, and the resulting special unitary Lie algebra becomes the split simple Lie algebra of type A_1, namely*

$$\mathfrak{sl}_2.$$

Thus, although over the ground field one naturally writes the Lie algebra as $\mathfrak{su}_2$, its split form is precisely $\mathfrak{sl}_2$.

Example 1.2.15 (The unitary involution on $M_2(\mathbb{C})$ and the Lie algebra $\mathfrak{su}_2$)**.** *Let $F = \mathbb{R}$ and $K = \mathbb{C}$, and let $^-$ denote complex conjugation. Consider the algebra $M_2(\mathbb{C})$ equipped with the involution*

$$X^* = \overline{X}^t.$$

This is the standard unitary involution on $M_2(\mathbb{C})$ relative to the extension $\mathbb{C}/\mathbb{R}$.
We first determine the Lie algebra of $*$-skew elements:

$$K(M_2(\mathbb{C}),*) = \{X \in M_2(\mathbb{C}) \mid X^* = -X\}.$$

Let

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{C}).$$

Then

$$X^* = \overline{X}^t = \begin{pmatrix} \overline{a} & \overline{c} \\ \overline{b} & \overline{d} \end{pmatrix}.$$

Thus the condition $X^* = -X$ is equivalent to

$$\overline{a} = -a, \qquad \overline{d} = -d, \qquad \overline{c} = -b.$$

Hence a and d are purely imaginary, and $c = -\overline{b}$. Therefore

$$K(M_2(\mathbb{C}),*) = \left\{ \begin{pmatrix} a & b \\ -\overline{b} & d \end{pmatrix} \middle| \, a, d \in i\mathbb{R},\ b \in \mathbb{C} \right\}.$$

This is the unitary Lie algebra $\mathfrak{u}_2$.
Its derived Lie algebra is the trace-zero part, namely

$$\mathfrak{su}_2 = \left\{ \begin{pmatrix} a & b \\ -\overline{b} & -a \end{pmatrix} \middle| \, a \in i\mathbb{R},\ b \in \mathbb{C} \right\}.$$

Indeed, $\operatorname{tr}([X,Y]) = 0$ for all $X, Y \in M_2(\mathbb{C})$, so every commutator in $\mathfrak{u}_2$ has trace 0. Conversely, in this 2×2 case, the trace-zero subspace is generated by commutators, and hence it is exactly the derived Lie algebra.
To see that $\mathfrak{su}_2$ is 3-dimensional over $\mathbb{R}$, write

$$a = it, \qquad b = u + iv, \qquad t, u, v \in \mathbb{R}.$$

Then every element of $\mathfrak{su}_2$ has the form

$$\begin{pmatrix} it & u + iv \\ -u + iv & -it \end{pmatrix}, \qquad t, u, v \in \mathbb{R}.$$

Therefore

$$\dim_{\mathbb{R}} \mathfrak{su}_2 = 3.$$

A convenient real basis is

$$H = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

One checks directly that

$$[H,X]=2Y, \qquad [H,Y]=-2X, \qquad [X,Y]=2H,$$

so $\mathfrak{su}_2$ is a 3-dimensional real Lie algebra.
Now we explain why this Lie algebra becomes $\mathfrak{sl}_2(\mathbb{C})$ after extending scalars from $\mathbb{R}$ to $\mathbb{C}$. Consider the complexification

$$\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C}.$$

Since H, X, Y form a real basis of $\mathfrak{su}_2$, they also form a complex basis of $\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C}$. Define

$$h=-iH=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \; e=\frac{1}{2}(X-iY)=\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}, \; f=\frac{1}{2}(-X-iY)=\begin{pmatrix}0 & 0\\ -1 & 0\end{pmatrix}.$$

These elements lie in $\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C}$ and satisfy

$$[h,e]=2e, \qquad [h,f]=-2f, \qquad [e,f]=h.$$

Thus they form a standard $\mathfrak{sl}_2$-triple, and therefore

$$\mathfrak{su}_2 \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{sl}_2(\mathbb{C}).$$

In other words, $\mathfrak{su}_2$ is a real form of $\mathfrak{sl}_2(\mathbb{C})$. Hence the Lie algebra arising from the unitary involution on $M_2(\mathbb{C})$ is of type A_1.

Remark 1.2.16. *This explains why, in the 2×2 case, the Lie algebra arising from a symplectic involution is isomorphic to the one of type A_1. Equivalently, in this lowest-rank case one has the accidental isomorphism*

$$C_1 \cong A_1.$$

1.2.3 They come from the derivation of any algebra

Let A be any algebra.

Definition 1.2.17. *A derivation of A is a linear map $\mathrm{d}: A \to A$ such that $\mathrm{d}(ab) = \mathrm{d}(a)b + a\mathrm{d}(b)$ for all $a, b \in A$.*

Example 1.2.18. *Let $A = F[t_1, \ldots, t_n]$ be the polynomial algebra. We have the formal derivative ∂ where $\partial(t_i^n) = nt_i^{n-1}$ and $\partial(fg) = f\partial(g) + \partial(f)g$.*
Let $\mathrm{d} = \sum_{i=1}^n f_i(t_1, \ldots, t_n)\frac{\partial}{\partial t_i}$, this is a derivation of A.

$$\mathrm{d}(f) = \sum_{i=1}^n f_i(t_1, \ldots, t_n)\frac{\partial f}{\partial t_i}$$

If $\mathrm{d}_1, \mathrm{d}_2$ are derivations of A, then $[\mathrm{d}_1, \mathrm{d}_2]$ is again a derivation of A. We can check *it by the following calculation:*

$$\begin{aligned}
[\mathrm{d}_1, \mathrm{d}_2](ab) &= \mathrm{d}_1(\mathrm{d}_2(ab)) - \mathrm{d}_2(\mathrm{d}_1(ab)) \\
&= \mathrm{d}_1(\mathrm{d}_2(a)b + a\mathrm{d}_2(b)) - \mathrm{d}_2(\mathrm{d}_1(a)b + a\mathrm{d}_1(b)) \\
&= \mathrm{d}_1(\mathrm{d}_2(a))b + \mathrm{d}_1(a\mathrm{d}_2(b)) + \mathrm{d}_2(\mathrm{d}_1(a))b + \mathrm{d}_2(a\mathrm{d}_1(b)) \\
&= \mathrm{d}_1(\mathrm{d}_2(a))b + \mathrm{d}_2(\mathrm{d}_1(a))b + \mathrm{d}_1(a\mathrm{d}_2(b)) + \mathrm{d}_2(a\mathrm{d}_1(b)) \\
&= \mathrm{d}_1(\mathrm{d}_2(a))b + \mathrm{d}_2(\mathrm{d}_1(a))b + a[\mathrm{d}_1, \mathrm{d}_2](b) \\
&= [\mathrm{d}_1, \mathrm{d}_2](a)b + a[\mathrm{d}_1, \mathrm{d}_2](b) \\
&= [\mathrm{d}_1, \mathrm{d}_2](ab)
\end{aligned}$$

Therefore $[\mathrm{d}_1, \mathrm{d}_2]$ *is a derivation of* A*.*

The above example means that:

$$\mathrm{Der}(A) = \{\ \mathrm{d} : A \to A \mid \mathrm{d} \text{ is a derivation}\} < \mathrm{End}_F(A)^{(-)}$$

In particular, when $A = F[t_1, \ldots, t_n]$, we have:

$$\mathrm{Der}(A) = \{\ \sum_{i=1}^{n} f_i(t_1, \ldots, t_n)\frac{\partial}{\partial t_i} \mid f_i(t_1, \ldots, t_n) \in A\} < \mathrm{End}_F(A)^{(-)}$$

We denote this algebra by $W(n)$. It is called the n-th Witt algebra. It is a simple, infinite dimensional Lie algebra (when $\operatorname{char} F = 0$).

Thus every element of $W(n)$ has the form

$$D = \sum_{i=1}^{n} f_i\, \partial_i, \qquad f_i \in A.$$

Definition 1.2.19. *The divergence map is defined by*

$$\mathrm{div} : W(n) \to A, \qquad \mathrm{div}\left(\sum_{i=1}^{n} f_i\partial_i\right) = \sum_{i=1}^{n} \partial_i(f_i).$$

Its kernel is the special *Lie algebra*

$$S(n) := \ker(\mathrm{div}) = \left\{ \sum_{i=1}^{n} f_i\partial_i \in W(n) \;\middle|\; \sum_{i=1}^{n} \partial_i(f_i) = 0 \right\}.$$

Equivalently, $S(n)$ *is the Lie algebra of polynomial vector fields preserving the standard volume form*

$$\omega = dx_1 \wedge \cdots \wedge dx_n.$$

A basic family of elements of $S(n)$ is given by

$$D_{ij}(f) := \partial_j(f)\partial_i - \partial_i(f)\partial_j, \qquad 1 \le i < j \le n, \quad f \in A,$$

since

$$\operatorname{div}(D_{ij}(f)) = \partial_i\partial_j(f) - \partial_j\partial_i(f) = 0.$$

One can check

$$S(n) =< D_{ij}(f) \mid 1 \leq i < j \leq n, f \in A >_{Lie}$$

These elements generate $S(n)$ as a Lie algebra.

Definition 1.2.20. *The grading of $W(n)$ is defined by Give A its standard grading by declaring $\deg x_i = 1$. Then $W(n)$ becomes a $\mathbb{Z}$-graded Lie algebra with*

$$\deg(x^a\partial_i) = |a| - 1, \qquad |a| := a_1 + \cdots + a_n,$$

so that

$$W(n) = \bigoplus_{k \geq -1} W(n)_k.$$

and

$$W(n)_0 = \operatorname{span}_F\{x_j\partial_i \mid 1 \leq i, j \leq n\} \cong \mathfrak{gl}_n(F).$$

Since the divergence map is homogeneous of degree 0, the subalgebra $S(n)$ is graded:

$$S(n) = \bigoplus_{k \geq -1} S(n)_k, \qquad S(n)_k = S(n) \cap W(n)_k.$$

Moreover,

$$S(n)_{-1} = W(n)_{-1},$$

and

$$S(n)_0 = \left\{ \sum_{i,j} a_{ij}x_j\partial_i \;\middle|\; \sum_i a_{ii} = 0 \right\} \cong \mathfrak{sl}_n(F),$$

because

$$\operatorname{div}\left(\sum_{i,j} a_{ij}x_j\partial_i\right) = \sum_i a_{ii}.$$

In characteristic 0, the special algebra is perfect:

$$[S(n), S(n)] = S(n).$$

Hence

$$S(n)^{(1)} = S(n).$$

For $n \geq 2$, the Lie algebra $S(n)$ is simple. Therefore, in characteristic 0 one has

$$W(n) \supset S(n) = S(n)^{(1)},$$

where both algebras are infinite-dimensional, $W(n)$ is the full Lie algebra of polynomial vector fields, and $S(n)$ is its simple divergence-zero subalgebra.

Remark 1.2.21. *In characteristic 0, the chain $W(n) \supset S(n) \supset S(n)^{(1)}$ is thus less infor-*

mative than in positive characteristic, since the last inclusion is an equality.

Let A be an associative algebra. Fix an element $a \in A$, we define : $\mathrm{ad}(a) : A \to A : b \mapsto [a, b]$. This is a derivation of A.

Definition 1.2.22. *Let L be a Lie algebra. The adjoint map is defined as:*

$$\mathrm{ad} : L \to \mathrm{End}_F(L)^{(-)}, a \mapsto \mathrm{ad}(a)$$

where $\mathrm{ad}(a) : L \to L : b \mapsto [a, b]$ is an inner derivation of L. We also call $\mathrm{ad}(a)$ an adjoint operator.

Note that $\mathrm{ad}(a) \in \mathrm{Der}(L)$ check it by the following calculation:

$$\begin{aligned}
\mathrm{ad}(a)([b, c]) &= \mathrm{ad}(a)(bc - cb) \\
&= [a, bc] - [a, cb] \\
&= [a, b]c + b[a, c] - [a, c]b - c[a, b] \\
&= [\mathrm{ad}(a)(b), c] + [b, \mathrm{ad}(a)(c)]
\end{aligned}$$

"Inner" here comes from that $\mathrm{ad}(L) \triangleleft \mathrm{Der}(L)$ since $\forall \mathrm{d} \in \mathrm{Der}(L)$, we have $[\mathrm{d}, \mathrm{ad}(a)] = \mathrm{ad}(\mathrm{d}(a))$. Check it by the following calculation:

$$\begin{aligned}
[d, \mathrm{ad}(a)](x) &= d\big([a, x]\big) - [a, d(x)] \\
&= [d(a), x] + [a, d(x)] - [a, d(x)] \\
&= [d(a), x] \\
&= \mathrm{ad}(d(a))(x).
\end{aligned}$$

Remark 1.2.23 (Exponentials of nilpotent inner derivations)**.** *Let L be a Lie algebra over a field F, and let*

$$d \in \mathrm{ad}(L) \subseteq \mathrm{Der}(L).$$

Thus there exists $a \in L$ such that

$$d = \mathrm{ad}(a), \qquad d(x) = [a, x] \quad \textit{for all } x \in L.$$

Assume that d is nilpotent, so that $d^N = 0$ for some integer $N \geq 1$. Then one defines the exponential of d by

$$\exp(d) := \sum_{n=0}^{\infty} \frac{d^n}{n!}.$$

Since d is nilpotent, this is actually a finite sum:

$$\exp(d) = I + d + \frac{d^2}{2!} + \cdots + \frac{d^{N-1}}{(N-1)!}.$$

Hence $\exp(d)$ is a well-defined linear endomorphism of L.

Moreover, $\exp(d)$ *is invertible, with inverse*

$$\exp(d)^{-1} = \exp(-d).$$

Since d is a derivation, $\exp(d)$ *is in fact a Lie algebra automorphism:*

$$\exp(d)([x,y]) = [\exp(d)(x), \exp(d)(y)] \qquad \text{for all } x, y \in L.$$

Therefore

$$\exp(d) \in \mathrm{Aut}(L).$$

In particular, if $d = \mathrm{ad}(a)$, then

$$\exp(\mathrm{ad}(a))(x) = x + [a,x] + \frac{1}{2!}[a,[a,x]] + \frac{1}{3!}[a,[a,[a,x]]] + \cdots.$$

When $\mathrm{ad}(a)$ *is nilpotent, this sum is finite. The automorphism*

$$\exp(\mathrm{ad}(a))$$

is called an inner automorphism arising from the nilpotent inner derivation $\mathrm{ad}(a)$*.*

A fundamental example is the matrix Lie algebra $L = \mathfrak{gl}_n(F)$. For $A \in \mathfrak{gl}_n(F)$, one has

$$\mathrm{ad}(A)(X) = [A,X] = AX - XA,$$

and

$$\exp(\mathrm{ad}(A))(X) = e^A X e^{-A}.$$

Thus, in the matrix case, the exponential of an inner derivation is exactly conjugation by the matrix exponential.

Proposition 1.2.24. *Let L be a Lie algebra over a field F, and let $d \in \mathrm{Der}(L)$ be nilpotent. Then*

$$\exp(d) := \sum_{n=0}^{\infty} \frac{d^n}{n!}$$

is a Lie algebra automorphism of L.

Proof. Since d is nilpotent, there exists $N \geq 1$ such that $d^N = 0$. Hence

$$\exp(d) = I + d + \frac{d^2}{2!} + \cdots + \frac{d^{N-1}}{(N-1)!}$$

is a finite sum, so it is a well-defined linear endomorphism of L.

Moreover,

$$\exp(d)\exp(-d) = \exp(-d)\exp(d) = I,$$

because d commutes with itself and the usual exponential identity holds for polynomials in d. Thus $\exp(d)$ is invertible, with inverse $\exp(-d)$.

It remains to show that $\exp(d)$ preserves the Lie bracket. Define

$$\phi_t := \exp(td) = \sum_{n=0}^{\infty} \frac{t^n d^n}{n!},$$

where the sum is again finite since d is nilpotent. For $x, y \in L$, set

$$F(t) := \phi_t([x, y]), \qquad G(t) := [\phi_t(x), \phi_t(y)].$$

We show that $F(t) = G(t)$.

First,

$$F'(t) = d(\phi_t([x, y])) = \phi_t(d([x, y])).$$

Since d is a derivation,

$$d([x, y]) = [d(x), y] + [x, d(y)].$$

Therefore

$$F'(t) = \phi_t([d(x), y] + [x, d(y)]) = [\phi_t(d(x)), \phi_t(y)] + [\phi_t(x), \phi_t(d(y))].$$

On the other hand,

$$G'(t) = [\phi_t'(x), \phi_t(y)] + [\phi_t(x), \phi_t'(y)] = [\phi_t(d(x)), \phi_t(y)] + [\phi_t(x), \phi_t(d(y))].$$

Hence

$$F'(t) = G'(t).$$

Also,

$$F(0) = [x, y] = G(0).$$

It follows that $F(t) = G(t)$ for all t, and in particular at $t = 1$ we get

$$\exp(d)([x, y]) = [\exp(d)(x), \exp(d)(y)].$$

Thus $\exp(d)$ is a Lie algebra automorphism of L. □

1.2.4 They come from brackets

Definition 1.2.25 (Poisson bracket)**.** *Let A be a commutative associative algebra over a field $\Bbbk$. A* Poisson bracket *on A is a bilinear map*

$$\{\cdot, \cdot\} : A \times A \to A$$

such that for all $f, g, h \in A$ the following conditions hold:

1. ***Skew-symmetry:***
$$\{f, g\} = -\{g, f\}.$$

2. ***Jacobi identity:***

$$\{f,\{g,h\}\}+\{g,\{h,f\}\}+\{h,\{f,g\}\}=0.$$

3. ***Leibniz rule:***

$$\{f,gh\}=\{f,g\}h+g\{f,h\}.$$

In other words, $(A,\{\cdot,\cdot\})$ is a Lie algebra, and for each fixed $f\in A$, the map

$$\mathrm{ad}_f : A\to A, \qquad \mathrm{ad}_f(g)=\{f,g\},$$

is a derivation of the algebra A. An algebra A equipped with such a bracket is called a Poisson algebra*.*

Remark 1.2.26. *The Leibniz rule in the second variable follows automatically from skew-symmetry:*

$$\{fg,h\}=f\{g,h\}+g\{f,h\}.$$

Thus a Poisson bracket is precisely a Lie bracket compatible with the commutative multiplication on A by the derivation property.

Example 1.2.27 (Canonical Poisson bracket on $\mathbb{R}^{2n}$)**.** *Let*

$$A=C^\infty(\mathbb{R}^{2n}), \qquad (q_1,\dots,q_n,p_1,\dots,p_n)$$

be the standard coordinates. Define

$$\{f,g\}=\sum_{i=1}^{n}\left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i}-\frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i}\right).$$

Then $\{\cdot,\cdot\}$ is a Poisson bracket on A. In particular,

$$\{q_i,q_j\}=0, \qquad \{p_i,p_j\}=0, \qquad \{q_i,p_j\}=\delta_{ij}.$$

Definition 1.2.28. *Let F be a field of characteristic 0, and let $\mathcal{A}$ be an algebra of differentiable functions in one variable x. For $f,g\in\mathcal{A}$, define*

$$[f,g]:=fg'-f'g,$$

where $f'=\dfrac{df}{dx}$ and $g'=\dfrac{dg}{dx}$.

Definition 1.2.29. *To each $f\in\mathcal{A}$, associate the vector field*

$$X_f:=f(x)\frac{d}{dx}.$$

Proposition 1.2.30. *For $f, g \in \mathcal{A}$, one has*

$$[X_f, X_g] = X_{[f,g]}.$$

Equivalently,

$$[X_f, X_g] = \left(fg' - gf'\right) \frac{d}{dx}.$$

Proof. For any test function u, we have

$$X_f(u) = fu', \qquad X_g(u) = gu'.$$

Thus

$$X_f(X_g(u)) = f(gu')' = fg'u' + fgu'',$$

and similarly

$$X_g(X_f(u)) = g(fu')' = gf'u' + gfu''.$$

Subtracting, we obtain

$$[X_f, X_g](u) = X_f(X_g(u)) - X_g(X_f(u)) = (fg' - gf')u'.$$

Therefore

$$[X_f, X_g] = \left(fg' - gf'\right) \frac{d}{dx} = X_{[f,g]}.$$

□

Proposition 1.2.31. *The operation*

$$[f, g] = fg' - f'g$$

is bilinear and skew-symmetric.

Proof. Bilinearity follows immediately from the linearity of differentiation. Also,

$$[g, f] = gf' - g'f = -(fg' - f'g) = -[f, g].$$

Hence the bracket is skew-symmetric. □

Theorem 1.2.32. *The bracket*

$$[f, g] = fg' - f'g$$

defines a Lie algebra structure on $\mathcal{A}$.

Proof. By the previous proposition, it remains only to verify the Jacobi identity. This follows from the fact that the map

$$f \longmapsto X_f = f \frac{d}{dx}$$

identifies this bracket with the usual Lie bracket of vector fields on the line, and the latter always satisfies the Jacobi identity. Hence

$$[f,[g,h]]+[g,[h,f]]+[h,[f,g]]=0$$

for all $f,g,h\in\mathcal{A}$. $\square$

1.2.5 They come from Lie groups

Definition 1.2.33. *Let G be a Lie group with identity element e. The* Lie algebra *of G, denoted by* $\mathrm{Lie}(G)$ *or $\mathfrak{g}$, is the tangent space*

$$\mathfrak{g}:=T_eG$$

equipped with the bracket induced by left-invariant vector fields.

Definition 1.2.34. *For each $X\in T_eG$, the corresponding* left-invariant vector field *$\widetilde{X}$ on G is defined by*

$$\widetilde{X}_g:=(dL_g)_e(X),\qquad g\in G,$$

where $L_g:G\to G$ is left translation, $L_g(h)=gh$.

Proposition 1.2.35. *The assignment $X\mapsto\widetilde{X}$ gives a bijection between T_eG and the space of left-invariant vector fields on G.*

Definition 1.2.36. *For $X,Y\in\mathfrak{g}=T_eG$, define*

$$[X,Y]:=[\widetilde{X},\widetilde{Y}](e),$$

where on the right-hand side $[\widetilde{X},\widetilde{Y}]$ is the usual Lie bracket of vector fields on G.

Proposition 1.2.37. *The bracket defined above makes T_eG into a Lie algebra. In particular, it is bilinear, alternating, and satisfies the Jacobi identity.*

Example 1.2.38. *If $G=GL_n(F)$, where $F=\mathbb{R}$ or $\mathbb{C}$, then*

$$\mathrm{Lie}(G)=T_IGL_n(F)\cong M_n(F),$$

and the Lie bracket is

$$[X,Y]=XY-YX.$$

Hence

$$\mathrm{Lie}(GL_n(F))=\mathfrak{gl}_n(F).$$

Example 1.2.39. *If $G = SL_n(F)$, then*

$$\mathrm{Lie}(SL_n(F)) = \mathfrak{sl}_n(F) = \{X \in M_n(F) \mid \mathrm{tr}(X) = 0\}.$$

Indeed, differentiating the defining equation $\det(g) = 1$ at the identity gives the trace-zero condition.

Example 1.2.40. *If $G = O_n(\mathbb{R})$, then*

$$\mathrm{Lie}(O_n(\mathbb{R})) = \mathfrak{so}_n(\mathbb{R}) = \{X \in M_n(\mathbb{R}) \mid X^T = -X\}.$$

This follows by differentiating the relation $g^T g = I$ at the identity.

1.3 Basic concepts of Lie algebra

Let L be a Lie algebra over a field F.

Definition 1.3.1. *The* derived series *of L is defined inductively by*

$$L^{(0)} = L, \qquad L^{(1)} = [L, L], \qquad L^{(i)} = [L^{(i-1)}, L^{(i-1)}] \quad (i \geq 1).$$

The Lie algebra L is called solvable *if*

$$L^{(n)} = 0$$

for some integer $n \geq 0$.

Definition 1.3.2. *The* descending central series *of L is defined by*

$$L^1 = L, \qquad L^2 = [L, L], \qquad L^i = [L, L^{i-1}] \quad (i \geq 2).$$

The Lie algebra L is called nilpotent *if*

$$L^n = 0$$

for some integer $n \geq 1$.

Proposition 1.3.3. *If L is nilpotent, then L is solvable.*

Proof. It is enough to show that

$$L^{(i)} \subseteq L^{2^i} \qquad (i \geq 0).$$

We argue by induction on i. The case $i = 0$ is clear. If

$$L^{(i)} \subseteq L^{2^i},$$

then

$$L^{(i+1)} = [L^{(i)}, L^{(i)}] \subseteq [L^{2^i}, L^{2^i}] \subseteq L^{2^{i+1}}.$$

Hence the claim follows for all i. If $L^n = 0$ for some n, then for i sufficiently large one has $2^i \geq n$, so

$$L^{(i)} \subseteq L^{2^i} = 0.$$

Therefore L is solvable. $\square$

Definition 1.3.4. *A* solvable ideal *of L is an ideal which is solvable as a Lie algebra in its own right. Since the sum of two solvable ideals is again solvable, there exists a unique maximal solvable ideal of L, called the* radical *of L, denoted by*

$$\mathrm{Rad}(L).$$

The Lie algebra L is called semisimple *if*

$$\mathrm{Rad}(L) = 0.$$

Remark 1.3.5. *Thus the three notions fit into the chain*

$$\textit{nilpotent} \implies \textit{solvable},$$

while semisimplicity means that L has no nonzero solvable ideals. In particular, a semisimple Lie algebra cannot contain any nonzero abelian ideal.

Example 1.3.6. *Any abelian Lie algebra is nilpotent, hence solvable. On the other hand, the Lie algebra of upper triangular matrices is solvable in general, but need not be nilpotent.*

Example 1.3.7. *Let*

$$L = \left\{ \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \middle| \, a, b \in F \right\} \subseteq \mathfrak{gl}_2(F),$$

Then L is solvable but not nilpotent.
Indeed, if

$$x = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}, \qquad y = \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix},$$

then

$$[x, y] = \begin{pmatrix} 0 & ad - bc \\ 0 & 0 \end{pmatrix}.$$

Hence

$$[L, L] = \left\{ \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \middle| \, t \in F \right\}, \qquad [[L, L], [L, L]] = 0.$$

Thus the derived series terminates:

$$L^{(1)} = [L, L], \qquad L^{(2)} = [L^{(1)}, L^{(1)}] = 0,$$

so L is solvable.

On the other hand, L is not nilpotent, since its descending central series does not reach 0. In fact,

$$L^2 = [L, L] = \left\{ \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \middle| \, t \in F \right\},$$

and for any

$$x = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \in L, \qquad y = \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} \in L^2,$$

we have

$$[x, y] = \begin{pmatrix} 0 & at \\ 0 & 0 \end{pmatrix} \in L^2.$$

Therefore

$$L^3 = [L, L^2] = L^2,$$

and inductively

$$L^n = L^2 \neq 0 \qquad (n \geq 2).$$

So L is not nilpotent.

Chapter 2

Lecture 2: Engel Theorem

2.1 Engel Theorem the first type

Definition 2.1.1. *Let L be a Lie algebra over a field F. An* enveloping algebra *of L is an associative F-algebra A together with a Lie algebra homomorphism*

$$\iota : L \to A^-,$$

where A^- denotes the Lie algebra obtained from A by equipping the underlying vector space of A with the bracket

$$[x, y] = xy - yx,$$

such that the associative algebra A is generated by $\iota(L)$ (or just writing $A =< L >_{assoc}$ or simply $A =< L >$).
If, in addition, the pair (A, ι) satisfies the following universal property: for every associative F-algebra B and every Lie algebra homomorphism

$$\varphi : L \to B^-,$$

there exists a unique associative algebra homomorphism

$$\widetilde{\varphi} : A \to B$$

such that

$$\varphi = \widetilde{\varphi} \circ \iota,$$

then A is called the universal enveloping algebra *of L.*

Theorem 2.1.2. *Let A be a finite dimensional enveloping algebra of L, i.e. $L \subseteq A^{(-)}$, $A =< L >$.If $\exists n \geq 1$ such that $\forall a \in L$, $a^n = 0$. Then A is nilpotent.*

2.2 Engel Theorem the second type

In fact we can prove a stronger version.

Definition 2.2.1. *A subset $S \subseteq L$ is called a* Lie subset *if*

$$[a, b] \in S \qquad \text{for all } a, b \in S.$$

Note that a Lie subset need not be a subspace, and hence need not be a Lie subalgebra.

Theorem 2.2.2. *Let A be a finite-dimensional associative algebra over F, and let*

$$L \subseteq A^{(-)}$$

be a Lie subalgebra such that A is generated by L as an associative algebra. Suppose that

$$L = \mathrm{span}(S)$$

for some Lie subset $S \subseteq L$, and that every element of S is nilpotent in A. Then A is a nilpotent associative algebra.

Remark 2.2.3. *In the finite-dimensional situation, the existence of a uniform exponent n is automatic once every element of S is nilpotent in A. Indeed, let*

$$m = \dim_F A,$$

and let $a \in A$ be nilpotent. Consider the left multiplication operator

$$L_a : A \to A, \qquad x \mapsto ax.$$

Since a is nilpotent, L_a is a nilpotent linear transformation on the m-dimensional vector space A. Hence

$$L_a^m = 0.$$

But

$$L_a^m = L_{a^m},$$

so $a^m = 0$. Therefore every nilpotent element of A has nilpotency index at most $\dim_F A$. In particular, if every element of S is nilpotent in A, then one may take

$$n = \dim_F A$$

and obtain

$$a^n = 0 \qquad \text{for all } a \in S.$$

Thus, in the theorem above, the hypothesis that every element of S is nilpotent in A already implies the existence of a common exponent n.

Theorem 2.2.4 (Zorn's Lemma)**.** *Let $(P, \leq)$ be a partially ordered set. Assume that every chain in P has an upper bound in P. Then P contains a maximal element.*

Proof. We prove the result assuming the Well-Ordering Theorem.

Suppose, for contradiction, that P has no maximal element. Then for every $x \in P$ there exists some $y \in P$ such that

$$x < y.$$

By the Well-Ordering Theorem, we may choose a well-ordering $\preceq$ on P. For each $x \in P$, let $f(x)$ be the $\preceq$-least element of the set

$$\{y \in P : x < y\}.$$

Then

$$x < f(x) \qquad \text{for all } x \in P.$$

We now define, by transfinite recursion, a family (x_α) indexed by ordinals. Suppose that for all $\beta < \alpha$ the elements x_β have been defined and form a chain. By hypothesis, the chain

$$C_\alpha := \{x_\beta : \beta < \alpha\}$$

has an upper bound $u_\alpha \in P$. Define

$$x_\alpha := f(u_\alpha).$$

Since u_α is an upper bound of C_α, for every $\beta < \alpha$ we have

$$x_\beta \leq u_\alpha < x_\alpha.$$

Hence (x_α) is a strictly increasing family:

$$\beta < \alpha \implies x_\beta < x_\alpha.$$

Therefore the map

$$\alpha \longmapsto x_\alpha$$

is injective from the class of all ordinals into the set P. This is impossible, since there is no injection from the class of all ordinals into a set.

This contradiction shows that P must contain a maximal element. □

Lemma 2.2.5. *Let*

$$\mathcal{P} = \{T \subseteq S \mid T \textit{ is a Lie subset and } a^n = 0 \textit{ for all } a \in T\},$$

partially ordered by inclusion. Then $\mathcal{P}$ has a maximal element.

Proof. We apply Zorn's lemma. Let $\{T_i\}_{i \in I}$ be a chain in $\mathcal{P}$, and set

$$T = \bigcup_{i \in I} T_i.$$

We claim that $T \in \mathcal{P}$.

First, T is a Lie subset. Indeed, if $a, b \in T$, then $a \in T_i$ and $b \in T_j$ for some $i, j \in I$. Since $\{T_i\}_{i \in I}$ is a chain, one of T_i, T_j is contained in the other; say $T_i \subseteq T_j$. Then $a, b \in T_j$, and since T_j is a Lie subset,

$$[a, b] \in T_j \subseteq T.$$

Second, if $a \in T$, then $a \in T_i$ for some $i \in I$, so $a^n = 0$ because $T_i \in \mathcal{P}$.

Thus $T \in \mathcal{P}$, and clearly T is an upper bound of the chain. Therefore every chain in $\mathcal{P}$ has an upper bound. By Zorn's lemma, $\mathcal{P}$ has a maximal element. $\square$

Proof of the theorem. Consider the collection

$$\mathcal{P} = \{T \subseteq S \mid T \text{ is a Lie subset of } L \text{ and } \langle T \rangle \text{ is a nilpotent associative subalgebra of } A\},$$

partially ordered by inclusion. Here $\langle T \rangle$ denotes the associative subalgebra of A generated by T.

We first show that every chain in $\mathcal{P}$ has an upper bound. Let $\{T_i\}_{i \in I}$ be a chain in $\mathcal{P}$, and set

$$T = \bigcup_{i \in I} T_i.$$

Since the T_i form a chain, T is again a Lie subset of L. Let $N = \dim_F A$. Since each $\langle T_i \rangle$ is a nilpotent subalgebra of the finite-dimensional algebra A, we have

$$\langle T_i \rangle^N = 0 \qquad \text{for all } i \in I.$$

Now take any $x_1, \ldots, x_N \in \langle T \rangle$. Each x_j belongs to $\langle T_{i_j} \rangle$ for some $i_j \in I$. Since the family $\{T_i\}$ is totally ordered, there exists $k \in I$ such that

$$T_{i_1} \cup \cdots \cup T_{i_N} \subseteq T_k.$$

Hence

$$x_1, \ldots, x_N \in \langle T_k \rangle,$$

and therefore

$$x_1 \cdots x_N = 0.$$

Thus $\langle T \rangle^N = 0$, so $T \in \mathcal{P}$. By Zorn's lemma, $\mathcal{P}$ has a maximal element, say S_1.

Set

$$B = \langle S_1 \rangle.$$

Then B is nilpotent, so there exists $t \geq 1$ such that

$$B^t = 0.$$

We claim that if $S_1 \neq S$, then there exists $s \in S \setminus S_1$ such that

$$[S_1, s] \subseteq S_1.$$

Assume the contrary. Choose $s_1 \in S \setminus S_1$. By assumption, there exists $x_1 \in S_1$ such that

$$s_2 = [x_1, s_1] \notin S_1.$$

Since S is a Lie subset, $s_2 \in S$. Repeating this argument, we construct inductively elements

$$x_1, x_2, \dots \in S_1, \qquad s_1, s_2, \dots \in S \setminus S_1$$

such that

$$s_{r+1} = [x_r, s_r] \qquad (r \geq 1).$$

Thus

$$s_{r+1} = \mathrm{ad}(x_r) \cdots \mathrm{ad}(x_1)(s_1).$$

We now show that this process cannot continue for $2t$ steps. Indeed, for any $y_1, \dots, y_r \in B$, the iterated commutator

$$\mathrm{ad}(y_r) \cdots \mathrm{ad}(y_1)(s_1)$$

is a sum of terms of the form

$$\pm u\, s_1\, v,$$

where $u, v \in B$ and the total number of factors from B occurring in u and v is r. Therefore, if $r \geq 2t$, then in each such term either $u \in B^t$ or $v \in B^t$, so the term is 0. Hence

$$\mathrm{ad}(y_r) \cdots \mathrm{ad}(y_1)(s_1) = 0 \qquad \text{for all } r \geq 2t.$$

Applying this with $y_i = x_i \in S_1 \subseteq B$, we obtain

$$s_{2t+1} = 0,$$

which is impossible since $0 \in S_1$. This contradiction proves the claim.

Choose $s \in S \setminus S_1$ such that

$$[S_1, s] \subseteq S_1.$$

Let

$$S_2 = S_1 \cup \{s\}.$$

Then S_2 is a Lie subset of L: indeed, brackets of elements of S_1 remain in S_1, $[S_1, s] \subseteq S_1$, $[s, S_1] \subseteq S_1$ by antisymmetry, and $[s, s] = 0 \in S_1$.

We next show that $\langle S_2 \rangle$ is nilpotent. Since $[S_1, s] \subseteq S_1$, we have

$$[B, s] \subseteq B.$$

Hence for every $b \in B$,

$$bs = sb + [b, s], \qquad [b, s] \in B.$$

By repeated use of this identity, every product in the algebra generated by B and s can be written as a linear combination of terms of the form

$$s^r b, \qquad b \in B,$$

with $r \geq 0$. Since $s^m = 0$, we may assume $0 \leq r \leq m-1$.

Now consider a product of length mt in $\langle S_2 \rangle$. After rewriting it as above, each term has the form $s^r b$ with $0 \leq r \leq m-1$. Since the original length is mt, such a term must contain at least

$$mt - r \geq mt - (m-1) \geq t$$

factors coming from B. Therefore $b \in B^t = 0$, and every such term is zero. It follows that

$$\langle S_2 \rangle^{mt} = 0.$$

Thus $\langle S_2 \rangle$ is nilpotent.

But $S_1 \subsetneq S_2$ and $S_2 \in \mathcal{P}$, contradicting the maximality of S_1. Hence $S_1 = S$. Therefore

$$A = \langle L \rangle = \langle S \rangle = \langle S_1 \rangle$$

is nilpotent. □

2.3 Engel Theorem the third type

Definition 2.3.1. *Let L be a Lie algebra over a field F. An L-module is a vector space V over F together with a bilinear map*

$$L \times V \to V, \qquad (x, v) \mapsto x \cdot v,$$

such that

$$[x, y] \cdot v = x \cdot (y \cdot v) - y \cdot (x \cdot v)$$

for all $x, y \in L$ and $v \in V$. In other words, V is an L-module if and only if there exists a Lie algebra homomorphism

$$\phi : L \to \mathfrak{gl}(V).$$

such that

$$\phi(x)(v) = x \cdot v$$

for all $x \in L$ and $v \in V$.

Proposition 2.3.2. *Let L be a Lie algebra over a field F, and let M be a vector space over F. Suppose a bilinear operation*

$$L \times M \to M, \qquad (a, m) \mapsto a \cdot m$$

is given. Then the following conditions are equivalent:

1. *The vector space $L \oplus M$ becomes a Lie algebra under the bracket defined by*

$$[a+m,\ b+n] = [a,b] + a \cdot n - b \cdot m \qquad (a,b \in L,\ m,n \in M),$$

   equivalently,

$$[a,m] = a \cdot m, \qquad [m,a] = -\, a \cdot m$$

   for all $a \in L$, $m \in M$, and

$$[m,n] = 0$$

   for all $m,n \in M$.

2. *For all $a,b \in L$ and $m \in M$,*

$$[a,b] \cdot m = a \cdot (b \cdot m) - b \cdot (a \cdot m).$$

3. *The map*

$$\rho : L \to \mathfrak{gl}_F(M), \qquad \rho(a)(m) = a \cdot m,$$

   is a Lie algebra homomorphism.

When these equivalent conditions hold, we say that M is an L-module.

Proof. We prove (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1).

(1) $\Rightarrow$ (2). Assume that $L \oplus M$ is a Lie algebra with the stated bracket. Since every Lie algebra satisfies the Jacobi identity, for any $a, b \in L$ and $m \in M$ we have

$$[a,[b,m]] + [b,[m,a]] + [m,[a,b]] = 0.$$

Now

$$[b,m] = b \cdot m, \qquad [m,a] = -\, a \cdot m, \qquad [a,b] \in L.$$

Because M is abelian, this gives

$$[a, b \cdot m] + [b, -\, a \cdot m] + [m,[a,b]] = 0.$$

Using again the definition of the bracket,

$$a \cdot (b \cdot m) - b \cdot (a \cdot m) - [a,b] \cdot m = 0.$$

Hence

$$[a,b] \cdot m = a \cdot (b \cdot m) - b \cdot (a \cdot m).$$

(2) $\Rightarrow$ (3). Define

$$\rho : L \to \mathfrak{gl}_F(M), \qquad \rho(a)(m) = a \cdot m.$$

Since the action is bilinear, ρ is linear. For any $a, b \in L$ and $m \in M$, condition (2) gives

$$\rho([a,b])(m) = [a,b] \cdot m = a \cdot (b \cdot m) - b \cdot (a \cdot m) = \rho(a)\rho(b)(m) - \rho(b)\rho(a)(m).$$

Therefore

$$\rho([a,b]) = [\rho(a), \rho(b)],$$

so ρ is a Lie algebra homomorphism.

(3) $\Rightarrow$ (1). Assume that $\rho : L \to \mathfrak{gl}_F(M)$ is a Lie algebra homomorphism, and define a bracket on $L \oplus M$ by

$$[a+m,\ b+n] = [a,b] + \rho(a)(n) - \rho(b)(m).$$

This bracket is bilinear and skew-symmetric. It remains to verify the Jacobi identity.

Take $a, b, c \in L$ and $m, n, p \in M$. Write

$$x = a + m, \qquad y = b + n, \qquad z = c + p.$$

A direct calculation shows that the L-component of

$$[x,[y,z]] + [y,[z,x]] + [z,[x,y]]$$

is

$$[a,[b,c]] + [b,[c,a]] + [c,[a,b]],$$

which is zero because L is a Lie algebra.

For the M-component, expanding the definition of the bracket yields

$$\rho(a)\rho(b)(p) - \rho(a)\rho(c)(n) - \rho([b,c])(m) + \rho(b)\rho(c)(m) - \rho(b)\rho(a)(p) - \rho([c,a])(n)$$
$$+\rho(c)\rho(a)(n) - \rho(c)\rho(b)(m) - \rho([a,b])(p).$$

Grouping the terms by m, n, p, we obtain

$$\big(\rho(b)\rho(c) - \rho(c)\rho(b) - \rho([b,c])\big)(m)$$
$$+\big(\rho(c)\rho(a) - \rho(a)\rho(c) - \rho([c,a])\big)(n)$$
$$+\big(\rho(a)\rho(b) - \rho(b)\rho(a) - \rho([a,b])\big)(p).$$

Since ρ is a Lie algebra homomorphism, each bracketed expression is zero. Hence the M-component also vanishes, so the Jacobi identity holds.

Therefore $L \oplus M$ is a Lie algebra with the stated bracket.

This proves that the three conditions are equivalent. $\square$

Theorem 2.3.3. *Let L be a finite dimensional Lie algebra over a field F. If $\forall a \in L$, $\mathrm{ad}(a)$ is nilpotent, then L is nilpotent.*

Proof. Note that since L is finite-dimensional, say $\dim L = n < \infty$, and for every $a \in L$ we have

$$(\operatorname{ad} a)^n = 0.$$

Hence we may apply the previous theorem: if for all $a \in L$,

$$(\operatorname{ad} a)^n = 0,$$

then

$$(\operatorname{ad} L)^n = 0,$$

and therefore

$$L^{n+1} = 0.$$

So L is nilpotent. $\square$

Remark 2.3.4. *If L is infinite-dimensional, one should distinguish two cases:*

$$\begin{cases} \operatorname{ad} a \textit{ nilpotent with a uniform bound } n \implies L \textit{ nilpotent}, \\ \textit{no uniform bound on the nilpotency indices } \not\Rightarrow L \textit{ nilpotent.} \end{cases}$$

Remark 2.3.5. *There are three related versions, with*

$$\textit{type } (2) \Longrightarrow \textit{type } (1) \Longrightarrow \textit{type } (3).$$

Type (2): Let $S \subseteq L$ be a Lie subset. And S generates L as a Lie algebra. Assume there exists n such that for every $a \in S$

$$a^n = 0.$$

Then $A =< L >$ is a nilpotent algebra. (When A is a finite-dimensional associative algebra).

In particular, if we choose S to be L, then we get type (1).

Type (1): Let A be a finite dimensional enveloping algebra of L, i.e. $L \subseteq A^{(-)}$, $A =< L >$.If $\exists n \geq 1$ such that $\forall a \in L$, $a^n = 0$. Then A is nilpotent.

And consider A as $\operatorname{ad}(L)$, we get type (3).

Type (3): Let L be a finite dimensional Lie algebra over a field F. If $\forall a \in L$, $\operatorname{ad}(a)$ is nilpotent, then L is nilpotent.

We also have the following truth:

Corollary 2.3.6. *Let M be an L-module with $\dim M < \infty$, that is, let*

$$\varphi : L \to \mathfrak{gl}_F(M)$$

be a representation. Assume that for every $a \in L$, the operator $\varphi(a)$ is nilpotent. Equivalently, for every $a \in L$ there exists n such that

$$\underbrace{a\big(a(\cdots(a}_{n \textit{ times}}\ M)\cdots)\big) = 0.$$

Then there exists N such that

$$\underbrace{L\cdots L}_{N\ times} M = 0.$$

In other words, if every $\varphi(a)$ is nilpotent, then the Lie algebra $\varphi(L)$ is nilpotent. Hence there exists a basis of M in which every element of $\varphi(L)$ is represented by a strictly upper triangular matrix:

$$\varphi(L) \subseteq \left\{ \begin{pmatrix} 0 & * & \cdots & * \\ 0 & 0 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix} \right\}.$$

Remark 2.3.7. *This corollary is the module-theoretic form of Engel's theorem. Indeed, the representation*

$$\varphi : L \to \mathfrak{gl}_F(M)$$

has image $\varphi(L)$, which is a Lie subalgebra of $\mathfrak{gl}_F(M)$. By assumption, every element of $\varphi(L)$ is a nilpotent endomorphism of the finite-dimensional vector space M. Hence, by Engel's theorem, the Lie algebra $\varphi(L)$ is nilpotent.

The conclusion

$$\underbrace{L\cdots L}_{N\ times} M = 0$$

means that there exists a uniform integer N such that for all $a_1, \dots, a_N \in L$ and all $m \in M$,

$$a_1\big(a_2(\cdots(a_N m)\cdots)\big) = 0.$$

Thus the action is nilpotent not only for each fixed $a \in L$, but uniformly for all sufficiently long iterated actions.

A further consequence of Engel's theorem is that there exists a basis of M with respect to which every operator $\varphi(a)$ is represented by a strictly upper triangular matrix. Equivalently, M admits a chain of subspaces

$$0 = M_0 \subset M_1 \subset \cdots \subset M_r = M, \qquad \dim M_i = i,$$

such that

$$L \cdot M_i \subseteq M_{i-1} \qquad (1 \le i \le r).$$

Explanation: By Engel's theorem, $\varphi(L)$ is nilpotent. Hence

$$\underbrace{L\cdots L}_{N\ times} M = 0$$

Take the minimal N such that $L^N M = 0$ and $L^{N-1}M \neq 0$. Then we have a non-zero vector v_1 in $L^{N-1}M$ which is killed by $\varphi(L)$. Denote $M_1 = \operatorname{span}(v_1)$. Then $L \cdot M_1 \subseteq M_0$. Now consider M/M_1. By Engel's theorem and by induction hypothesis, $\varphi(L)$ is nilpotent on M/M_1. Hence there exists a basis of M/M_1 with respect to which every element of $\varphi(L)$

is represented by a strictly upper triangular matrix. Equivalently, M/M_1 admits a chain of subspaces

$$0 = M_1/M_1 \subset M_2/M_1 \subset \cdots \subset M_r/M_1 = M/M_1, \qquad \dim M_i/M_1 = i,$$

such that

$$L \cdot (M_i/M_1) \subseteq M_{i-1}/M_1 \qquad (1 \leq i \leq r).$$

Now we can take a basis of M_1 and M/M_1 to get a basis of M with respect to which every element of $\varphi(L)$ is represented by a strictly upper triangular matrix.

In particular, this corollary reduces to the usual form of Engel's theorem when one takes $M = L$ and $\varphi = \mathrm{ad}$.

Remark 2.3.8. *According to this corollary and type (2), we can rewrite it as:*

Let M be an L-module with $\dim M < \infty$, that is, let

$$\varphi : L \to \mathfrak{gl}_F(M)$$

be a representation. Assume that for every $a \in S$, where S is a Lie subset of L and $< S >= L$, the operator $\varphi(a)$ is nilpotent. Equivalently, for every $a \in S$ there exists n such that

$$\underbrace{a\big(a(\cdots(a}_{n\ times}\ M)\cdots)\big) = 0.$$

Then there exists N such that

$$\underbrace{L \cdots L}_{N\ times} M = 0.$$

For this result, we do not need L or the enveloping algebra $A =< L >$ to be finite-dimensional now, we only need such n is a universal bound.

Chapter 3

Lecture 3: Lie's theorem and Generalized Subspaces

3.1 Lie's theorem

Lemma 3.1.1. *Let F be a field with* $\operatorname{char} F = 0$ *or* $\operatorname{char} F > n$. *Let* $a, b \in M_n(F)$ *satisfy*

$$[a, [a, b]] = 0.$$

Then $[a, b]$ *is nilpotent. In fact,*

$$[a, b]^n = 0.$$

Proof. Set

$$\delta := \operatorname{ad}(a), \qquad \delta(x) = [a, x].$$

Since δ is an inner derivation of the associative algebra $M_n(F)$, we have

$$\delta(xy) = \delta(x)y + x\delta(y) \qquad \text{for all } x, y \in M_n(F).$$

Our assumption is precisely

$$\delta^2(b) = 0.$$

Now let us prove the following two facts:

Claim 1. For every $m \geq 1$ and every $r > m$,

$$\operatorname{End} \delta^r(b^m) = 0.$$

Indeed, each application of δ to the product b^m differentiates one of the m factors. Since $\delta^2(b) = 0$, no factor can be differentiated twice. Hence after more than m applications, every term vanishes.

Claim 2. For every $m \geq 1$,

$$\delta^m(b^m) = m!\, \delta(b)^m = m!\, [a, b]^m.$$

We prove this by induction on m. The case $m = 1$ is clear. Assume the formula holds for $m-1$. Using the derivation rule,

$$\delta(b^m) = \delta(b)b^{m-1} + b\,\delta(b^{m-1}).$$

Applying δ^{m-1} and using Leibniz repeatedly, together with $\delta^2(b) = 0$, one finds that the only surviving terms are those in which each of the m copies of b is hit exactly once. There are exactly $m!$ such terms, and each contributes

$$\delta(b)^m.$$

Hence

$$\delta^m(b^m) = m!\,\delta(b)^m.$$

Now apply Cayley–Hamilton to b:

$$b^n + \beta_{n-1}b^{n-1} + \cdots + \beta_1 b + \beta_0 I = 0.$$

Apply δ^n to both sides. By Claim 1, for every $i < n$ we have

$$\delta^n(b^i) = 0,$$

and also $\delta^n(I) = 0$. Therefore

$$\delta^n(b^n) = 0.$$

By Claim 2 with $m = n$, this becomes

$$n!\,[a,b]^n = 0.$$

Since $\operatorname{char} F = 0$ or $\operatorname{char} F > n$, we have $n! \neq 0$ in F. Hence

$$[a,b]^n = 0.$$

Therefore $[a,b]$ is nilpotent. $\square$

Theorem 3.1.2 (Lie's theorem)**.** *Assume that* $\operatorname{char} F = 0$ *or* $\operatorname{char} F > \dim_F M$*. F is algebraically closed. Let L be a solvable Lie algebra and let M be an L-module with*

$$\dim_F M = n < \infty.$$

Then there exists a basis of M such that, relative to this basis, the image of L (from here when talk about L acting on its module M, we simply use L instead of $\varphi(L)$)

consists of upper triangular matrices; that is,

$$L \subseteq \left\{ \begin{pmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & * \end{pmatrix} \right\}.$$

Proof. We argue first that M has a common eigenvector for the action of L.

Assume first that M is irreducible as an L-module. Since we are mainly concerned with the image of

$$L \to \mathfrak{gl}_F(M),$$

we may replace L by its image and assume that $L \subseteq \mathfrak{gl}_F(M)$.

Because L is solvable, its derived series

$$L \supseteq L^{[1]} \supseteq L^{[2]} \supseteq \cdots$$

terminates at 0. Hence there exists an integer $s \geq 0$ such that

$$L^{[s]} \neq 0, \qquad L^{[s+1]} = [L^{[s]}, L^{[s]}] = 0.$$

If $s = 0$, then L is abelian, and since F is algebraically closed, there exists an eigenvector $v \in M$ for some $x \in L$. As L is abelian, every element of L preserves the eigenspace of x, and by irreducibility this gives a common eigenvector for all $x \in L$.

Now assume $s \geq 1$. Choose $a \in L^{[s]}$ and $b \in L$. Then

$$[a, [a, b]] \in [L^{[s]}, L^{[s]}] = L^{[s+1]} = 0,$$

so

$$(\operatorname{ad} a)^2(b) = 0.$$

Consider the set

$$S = \{[a, b] \mid a \in L^{[s]},\ b \in L\}.$$

This is a Lie subset of L, and every element of S is nilpotent in $\mathfrak{gl}_F(M)$. By Engel's theorem,

$$\operatorname{span}_F(S) = [L^{[s]}, L]$$

is a nilpotent Lie algebra of linear transformations on M.

Now let I be any ideal of L. Then IM is an L-submodule of M, because for $x \in L$, $y \in I$, and $m \in M$,

$$x \cdot (y \cdot m) = y \cdot (x \cdot m) + [x, y] \cdot m \in IM,$$

since $[x, y] \in I$. As M is irreducible, it follows that

$$IM = 0 \quad \text{or} \quad IM = M.$$

Applying this to the ideal $I = L^{[s]}$, we cannot have $L^{[s]}M = M$, because $L^{[s]}$ acts nilpotently on M. Hence

$$L^{[s]}M = 0,$$

that is, $L^{[s]}$ acts trivially on M.

Thus every element of $L^{[s]}$ commutes with the action of L on M. In particular, for any $c \in L^{[s-1]}$ and $d \in L^{[s-1]}$,

$$[c, d] \in L^{[s]}$$

acts as 0 on M. Therefore $L^{[s]} = 0$ in the image of L in $\mathfrak{gl}_F(M)$, contrary to the choice of s. This contradiction shows that $s = 0$, so L is abelian (on M), and hence M contains a common eigenvector for all elements of L.

Now let $0 \neq v_1 \in M$ be such a common eigenvector. Then Fv_1 is a one-dimensional L-submodule of M. Consider the quotient M/Fv_1, which is again a finite-dimensional L-module. By induction on $\dim_F M$, there exists a basis of M/Fv_1 with respect to which the induced action of L is upper triangular. Lifting this basis to M and adjoining v_1, we obtain a basis of M relative to which every element of L is represented by an upper triangular matrix.

Therefore there exists a basis of M such that the image of L consists of upper triangular matrices. □

Proposition 3.1.3. *Let L be a solvable Lie algebra over F with $\operatorname{char} F = 0$ or $\operatorname{char} F > \dim_F M$. F is algebraically closed. Then the followings are equivalent:*

1. *Any irreducible finite dimensional L-module is 1-dimensional.*
2. *If M is a finite dimensional L-module, then there exist a basis of M such that the image of L consists of upper triangular matrices.*
3. *If M is any finite dimensional L-module. Then $[L, L]$ acts nilpotently on M.*

Proof. We prove

$$(2) \Rightarrow (1), \qquad (1) \Rightarrow (2), \qquad (2) \Rightarrow (3), \qquad (3) \Rightarrow (1).$$

(2) $\Rightarrow$ (1). Let M be an irreducible finite-dimensional L-module. By (2), there exists a basis of M relative to which every operator in the image of L is upper triangular. Then the subspace spanned by the first basis vector is stable under L, hence is a nonzero submodule of M. Since M is irreducible, this submodule must be all of M. Therefore $\dim_F M = 1$.

(1) $\Rightarrow$ (2). We argue by induction on $\dim_F M$.

If $\dim_F M = 1$, there is nothing to prove. Assume $\dim_F M > 1$, and suppose the statement is known for all modules of smaller dimension. Since M is finite-dimensional, it contains an irreducible submodule N. By (1), we have $\dim_F N = 1$.

Choose $0 \neq v_1 \in N$, so that $N = Fv_1$. Then N is an L-submodule, and hence M/N is again a finite-dimensional L-module. By the induction hypothesis, there exists a basis

$$\overline{v_2}, \ldots, \overline{v_n}$$

of M/N such that the induced action of L on M/N is given by upper triangular matrices. Choose representatives $v_2, \ldots, v_n \in M$ of these cosets. Then

$$v_1, v_2, \ldots, v_n$$

is a basis of M.

Since Fv_1 is L-stable, every element of L sends v_1 into Fv_1. Since the induced action on M/Fv_1 is upper triangular with respect to

$$\overline{v_2}, \ldots, \overline{v_n},$$

it follows that the action of L on M is upper triangular with respect to

$$v_1, v_2, \ldots, v_n.$$

Thus (2) holds.

(2) $\Rightarrow$ (3). Let M be a finite-dimensional L-module, and choose a basis of M such that every element of the image of L is represented by an upper triangular matrix. Then for any $x, y \in L$,

$$\rho([x, y]) = [\rho(x), \rho(y)].$$

Since the commutator of two upper triangular matrices has zero diagonal entries, $\rho([x, y])$ is strictly upper triangular. Hence every element of $\rho([L, L])$ is strictly upper triangular, and therefore nilpotent. Thus $[L, L]$ acts nilpotently on M.

(3) $\Rightarrow$ (1). Let M be an irreducible finite-dimensional L-module. By (3), every element of $[L, L]$ acts nilpotently on M. Hence the Lie algebra $\rho([L, L]) \subseteq \mathfrak{gl}(M)$ consists entirely of nilpotent endomorphisms. By Engel's theorem, there exists a nonzero vector $v \in M$ such that

$$\rho([L, L])v = 0.$$

Thus $[L, L]$ annihilates v.

Let $N = U(L)v$ be the submodule generated by v. Since $[L, L]v = 0$, the action of L on N factors through the abelian Lie algebra $L/[L, L]$. Therefore N is a finite-dimensional module for the abelian Lie algebra $L/[L, L]$.

Because F is algebraically closed, every irreducible finite-dimensional module over an abelian Lie algebra is 1-dimensional. Hence N contains a 1-dimensional submodule. Since M is irreducible and $0 \neq N \subseteq M$, we have $N = M$. Therefore M itself contains a 1-dimensional submodule, and irreducibility forces $\dim_F M = 1$.

This proves (1).

Therefore

$$(1) \iff (2) \iff (3).$$

$\square$

Remark 3.1.4. *The assumption that F is algebraically closed is used in the last step of the proof of* (3) $\Rightarrow$ (1)*, namely in the fact that every irreducible finite-dimensional module over* *the abelian Lie algebra $L/[L,L]$ is 1-dimensional. Equivalently, one needs the existence of eigenvalues and eigenvectors for commuting linear operators.*

Example 3.1.5. *Assume that* $\operatorname{char} F = 2$. *Then* $\mathfrak{gl}(2,F)$ *is solvable.*
Indeed, every element of $\mathfrak{gl}(2,F)$ *has the form*

$$\begin{pmatrix} \alpha & \gamma \\ \beta & \delta \end{pmatrix}, \qquad \alpha, \beta, \gamma, \delta \in F.$$

A direct computation shows that the commutator of two such matrices is always of the form

$$\left[\begin{pmatrix} \alpha & \gamma \\ \beta & \delta \end{pmatrix}, \begin{pmatrix} \alpha' & \gamma' \\ \beta' & \delta' \end{pmatrix} \right] = \begin{pmatrix} d & x \\ y & d \end{pmatrix},$$

for some $d, x, y \in F$. *Hence*

$$\mathfrak{gl}(2,F)^{(1)} = [\mathfrak{gl}(2,F), \mathfrak{gl}(2,F)] \subseteq \left\{ \begin{pmatrix} d & x \\ y & d \end{pmatrix} : \ d, x, y \in F \right\}.$$

Now if

$$A = \begin{pmatrix} \alpha & \gamma \\ \beta & \alpha \end{pmatrix} \qquad (\textit{since } \operatorname{char} F = 2),$$

and similarly

$$B = \begin{pmatrix} \alpha' & \gamma' \\ \beta' & \alpha' \end{pmatrix},$$

then their commutator is of the form

$$[A, B] = \begin{pmatrix} d & 0 \\ 0 & d \end{pmatrix}$$

for some $d \in F$. *Therefore*

$$\mathfrak{gl}(2,F)^{(2)} \subseteq \left\{ \begin{pmatrix} d & 0 \\ 0 & d \end{pmatrix} : \ d \in F \right\},$$

and so

$$\mathfrak{gl}(2,F)^{(3)} = 0.$$

Thus $\mathfrak{gl}(2,F)$ *is solvable.*

3.2 Generalized eigenvectors and generalized eigenspaces

Definition 3.2.1. *Let V be a finite-dimensional vector space over a field F where F is algebraically closed, and let*

$$T \in \operatorname{End}_F(V).$$

Suppose that the characteristic polynomial of T splits as

$$f(x) = (x - \lambda_1)^{a_1} \cdots (x - \lambda_k)^{a_k},$$

where $\lambda_1, \ldots, \lambda_k$ are distinct. For each i, define

$$V_{\lambda_i} := \ker(T - \lambda_i I)^{a_i}.$$

This is called the generalized eigenspace of T corresponding to λ_i.

Theorem 3.2.2. *With the notation above,*

$$V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_k}.$$

Proof. For each i, let

$$P_i(x) = \frac{f(x)}{(x - \lambda_i)^{a_i}} = \prod_{j \neq i} (x - \lambda_j)^{a_j}.$$

Since the polynomials $P_1(x), \ldots, P_k(x)$ have greatest common divisor 1, there exist polynomials

$$Q_1(x), \ldots, Q_k(x) \in F[x]$$

such that

$$Q_1(x)P_1(x) + \cdots + Q_k(x)P_k(x) = 1.$$

Replacing x by T, we obtain

$$Q_1(T)P_1(T) + \cdots + Q_k(T)P_k(T) = I_V.$$

Hence every $v \in V$ can be written as

$$v = \sum_{i=1}^{k} Q_i(T)P_i(T)v.$$

Now

$$(T - \lambda_i I)^{a_i} Q_i(T)P_i(T) = Q_i(T)f(T) = 0,$$

because $f(T) = 0$ by the Cayley–Hamilton theorem. Therefore

$$Q_i(T)P_i(T)v \in \ker(T - \lambda_i I)^{a_i} = V_{\lambda_i},$$

and so

$$V = V_{\lambda_1} + \cdots + V_{\lambda_k}.$$

To show that the sum is direct, let

$$v_1 + \cdots + v_k = 0, \qquad v_i \in V_{\lambda_i}.$$

Fix i. Since $P_i(\lambda_i) \neq 0$, the polynomial $P_i(x)$ is relatively prime to $(x - \lambda_i)^{a_i}$. Hence there exist polynomials $A_i(x), B_i(x) \in F[x]$ such that

$$A_i(x)P_i(x) + B_i(x)(x - \lambda_i)^{a_i} = 1.$$

Replacing x by T and applying to $v_i \in V_{\lambda_i}$, we get

$$A_i(T)P_i(T)v_i = v_i,$$

because $(T - \lambda_i I)^{a_i} v_i = 0$. Thus $P_i(T)$ is invertible on V_{λ_i}.

On the other hand, if $j \neq i$, then

$$P_i(T)v_j = 0,$$

because $P_i(x)$ contains the factor $(x - \lambda_j)^{a_j}$ and

$$(T - \lambda_j I)^{a_j} v_j = 0.$$

Applying $P_i(T)$ to the relation

$$v_1 + \cdots + v_k = 0,$$

we obtain

$$P_i(T)v_i = 0.$$

Since $P_i(T)$ is invertible on V_{λ_i}, it follows that $v_i = 0$. As i was arbitrary, all v_i are zero. Therefore the sum is direct. $\square$

Chapter 4

Lecture 4: Jordan-Chevalley decomposition

4.1 Review

Let L be a finite-dimensional solvable Lie algebra over an algebraically closed field F of characteristic 0. Let V be a finite-dimensional L-module.

- $\forall$ representation $\rho : L \to \mathfrak{gl}(V)$, there exists a basis of V such that

$$\rho(L) \subseteq \left\{ \begin{pmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & * \end{pmatrix} \right\} (\dim V < \infty)$$

- All irreducible finite-dimensional L-modules are one-dimensional.

Lemma 4.1.1. *Let $I \triangleleft L$ be an ideal of L. If both I and L/I are solvable, then L is solvable. (In this lemma we do not need L to be finite-dimensional, also F can be any field.)*

Proof. Since L/I is solvable, there exists an integer $m \geq 0$ such that

$$(L/I)^{(m)} = 0.$$

Using the compatibility of the derived series with quotients, we have

$$(L/I)^{(m)} = \big(L^{(m)} + I\big)/I,$$

hence

$$L^{(m)} \subseteq I.$$

Since I is solvable, there exists an integer $n \geq 0$ such that

$$I^{(n)} = 0.$$

We obtain

$$L^{(m+n)} \subseteq I^{(n)} = 0.$$

Therefore $L^{(m+n)} = 0$, so L is solvable. □

Example 4.1.2. *No similar result for nilpotent Lie algebras. Indeed, Let L be the 2-dimensional Lie algebra over a field F with basis $\{x, y\}$ and bracket*

$$[x, y] = y.$$

Let

$$I := Fy.$$

Then $I \triangleleft L$, since

$$[x, y] = y \in I, \qquad [y, y] = 0.$$

Now I is 1-dimensional abelian, hence nilpotent. Also

$$L/I$$

is 1-dimensional abelian, hence nilpotent.
However, L itself is not nilpotent. Indeed, its lower central series is

$$\gamma_1(L) = L, \qquad \gamma_2(L) = [L, L] = Fy,$$

and for every $n \geq 2$,

$$\gamma_{n+1}(L) = [L, \gamma_n(L)] = [L, Fy] = Fy.$$

Thus the lower central series never reaches 0, so L is not nilpotent.
Therefore, even if I and L/I are both nilpotent, L need not be nilpotent.

Lemma 4.1.3. *Let $I, J \triangleleft L$ both I, J are solvable. Then $I + J$ is solvable.*

Proof. Since $J \triangleleft I + J$, the quotient

$$(I + J)/J$$

is naturally isomorphic to

$$I/(I \cap J).$$

Because I is solvable, every quotient of I is solvable; hence

$$(I + J)/J$$

is solvable. By assumption, J is solvable. Therefore, applying the previous lemma to the Lie algebra $I + J$ and its ideal J, we conclude that $I + J$ is solvable. □

Let $\operatorname{Rad}(L)$ be the largest solvable ideal of L ($\dim_F L < \infty$). $\operatorname{Rad}(L)$ is called the *radical* of L. $\operatorname{Rad}(L)$ contains all other solvable ideals and $L/\operatorname{Rad}(L)$ does not contain any non-zero solvable ideals.

Definition 4.1.4. *A finite-dimensional Lie algebra L is called* semisimple *if* $\operatorname{Rad}(L) = 0$.

4.2 Jordan-Chevalley decomposition

From now on we need over base field F to be algebraically closed. (At least we say the base field F contains all the roots of the characteristic polynomial of the linear transformation we need.)

Definition 4.2.1. *A finite-dimensional linear transformation $\varphi : V \to V$ is called* semisimple *if*

- *$\exists$ a basis of V such that φ is represented by a diagonal matrix.*
- *Or $(\varphi - \alpha I)^2 v = 0$ if and only if $(\varphi - \alpha I)v = 0$ for all $\alpha \in F, v \in V$.*

For any $V' \subseteq V$, V' is φ-invariant.

Lemma 4.2.2 (Chinese Remainder Theorem for commutative rings)**.** *Let R be a commutative ring with 1, and let $I_1, \dots, I_n$ be ideals of R such that*

$$I_i + I_j = R \qquad \text{for all } i \neq j.$$

Then the natural map

$$\varphi : R \to \prod_{i=1}^{n} R/I_i, \qquad r \mapsto (r + I_1, \dots, r + I_n)$$

is surjective, and

$$\ker(\varphi) = \bigcap_{i=1}^{n} I_i.$$

Hence it induces an isomorphism

$$R \Big/ \bigcap_{i=1}^{n} I_i \;\cong\; \prod_{i=1}^{n} R/I_i.$$

Proof. We first determine the kernel of φ. For $r \in R$,

$$\varphi(r) = 0$$

if and only if

$$r + I_i = 0 + I_i \qquad \text{for all } i = 1, \dots, n,$$

which is equivalent to $r \in I_i$ for all i. Thus

$$\ker(\varphi) = \bigcap_{i=1}^{n} I_i.$$

It remains to prove that φ is surjective. Let

$$(a_1 + I_1, \ldots, a_n + I_n) \in \prod_{i=1}^{n} R/I_i$$

be arbitrary. We shall construct $r \in R$ such that

$$r \equiv a_i \pmod{I_i} \qquad (i = 1, \ldots, n).$$

Fix i. Since the ideals are pairwise comaximal, for every $j \neq i$ we have

$$I_i + I_j = R.$$

We claim that

$$I_i + \prod_{j \neq i} I_j = R.$$

Indeed, for each $j \neq i$, choose $u_j \in I_i$ and $v_j \in I_j$ such that

$$u_j + v_j = 1.$$

Then

$$\prod_{j \neq i} (u_j + v_j) = 1.$$

After expanding this product, every term except $\prod_{j \neq i} v_j$ contains at least one factor $u_j \in I_i$, hence lies in I_i. On the other hand,

$$\prod_{j \neq i} v_j \in \prod_{j \neq i} I_j.$$

Therefore

$$1 = x_i + y_i$$

for some

$$x_i \in I_i, \qquad y_i \in \prod_{j \neq i} I_j.$$

In particular,

$$y_i \equiv 1 \pmod{I_i}, \qquad y_i \equiv 0 \pmod{I_j} \quad (j \neq i).$$

Now set

$$r := \sum_{i=1}^{n} a_i y_i.$$

Fix $k \in \{1, \ldots, n\}$. Modulo I_k, we have

$$r \equiv a_k y_k + \sum_{i \neq k} a_i y_i.$$

Since $y_k \equiv 1 \pmod{I_k}$ and $y_i \equiv 0 \pmod{I_k}$ for $i \neq k$, it follows that

$$r \equiv a_k \pmod{I_k}.$$

Thus

$$\varphi(r) = (a_1 + I_1, \ldots, a_n + I_n).$$

So φ is surjective.

By the First Isomorphism Theorem,

$$R/\ker(\varphi) \cong \operatorname{im}(\varphi) = \prod_{i=1}^{n} R/I_i.$$

Since $\ker(\varphi) = \bigcap_{i=1}^{n} I_i$, we obtain

$$R \Big/ \bigcap_{i=1}^{n} I_i \cong \prod_{i=1}^{n} R/I_i.$$

This completes the proof. □

Corollary 4.2.3. *Under the assumptions of the theorem,*

$$I_1 \cap \cdots \cap I_n = I_1 \cdots I_n.$$

Hence

$$R/(I_1 \cdots I_n) \cong \prod_{i=1}^{n} R/I_i.$$

Proof. For any ideals $J_1, \ldots, J_n$ of a commutative ring, one always has

$$J_1 \cdots J_n \subseteq J_1 \cap \cdots \cap J_n.$$

So it remains to prove the reverse inclusion.

We first prove the case of two ideals. Suppose $I + J = R$. Let $x \in I \cap J$. Since $I + J = R$, there exist $a \in I$ and $b \in J$ such that

$$a + b = 1.$$

Then

$$x = x(a + b) = xa + xb.$$

Now $xa \in IJ$ because $x \in J$ and $a \in I$, and similarly $xb \in IJ$ because $x \in I$ and $b \in J$.

Hence $x \in IJ$. Therefore
$$I \cap J = IJ.$$

Now proceed by induction on n. Let

$$J := I_1 \cdots I_{n-1}.$$

Since the ideals $I_1, \ldots, I_n$ are pairwise comaximal, one has

$$I_n + I_k = R \qquad (k = 1, \ldots, n-1),$$

and this implies
$$I_n + J = R.$$

Hence, by the two-ideal case,
$$J \cap I_n = JI_n.$$

By the induction hypothesis,

$$I_1 \cap \cdots \cap I_{n-1} = I_1 \cdots I_{n-1} = J.$$

Therefore

$$I_1 \cap \cdots \cap I_n = (I_1 \cap \cdots \cap I_{n-1}) \cap I_n = J \cap I_n = JI_n = I_1 \cdots I_n.$$

The desired isomorphism now follows from the theorem. □

Theorem 4.2.4. *An arbitrary linear transformation φ can be decomposed as $\varphi = \varphi_s + \varphi_n$, where φ_s is semisimple and φ_n is nilpotent.*
Furthermore, there exists a polynomial $p(t) \in F[t]$ with zero constant term such that $\varphi_s = p(\varphi)$ and $\varphi_n = \varphi - p(\varphi)$. In other words, φ_s and φ_n are polynomials in φ.

Proof. Decompose V into a direct sum of generalized eigenspaces $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_k}$. For each λ_i, let V_{λ_i} be the generalized eigenspace of φ corresponding to λ_i.

Define φ_s as $\varphi_s|_{V_{\lambda_i}} = \lambda_i I$. Then φ_s is semisimple and $\varphi_n = \varphi - \varphi_s$ is nilpotent. We only need to show that φ_s and φ_n are polynomials in φ.

Suppose the characteristic polynomial of φ is $P(t) = (t-\lambda_1)^{a_1} \cdots (t-\lambda_k)^{a_k}$. By Chinese Remainder Theorem, there exists a polynomial $p(t) \in F[t]$ such that $p(t) \equiv \lambda_i \pmod{(t-\lambda_i)^{a_i}}$ for all i.

If $0 \in \{\lambda_1, \ldots, \lambda_k\}$, then $p(0) = 0$. If $0 \notin \{\lambda_1, \ldots, \lambda_k\}$, then there exists $p(t)$ such that $p(t) \equiv \lambda_i \pmod{(t-\lambda_i)^{a_i}}$ and $p(t) \equiv 0 \pmod{t}$. Hence we can always confirm that $p(t)$ is a polynomial with zero constant term.

Now we have $p(\varphi) = \varphi_s$ and $\varphi - p(\varphi) = \varphi_n$. □

Theorem 4.2.5. *For all linear transformation φ there exists a unique decomposition $\varphi = \varphi_s + \varphi_n$ such that φ_s is semisimple and φ_n is nilpotent. And φ_s and φ_n are polynomials in φ, in particular, φ_s and φ_n commute.*

Proof. We only need to show the uniqueness of the decomposition. Suppose

$$\varphi = \varphi_s' + \varphi_n' = \varphi_s'' + \varphi_n'',$$

And $[\varphi_s', \varphi_n'] = [\varphi_s'', \varphi_n''] = 0$. By former theorem we already have a decomposition $\varphi = \varphi_s + \varphi_n$ such that φ_s is semisimple and φ_n is nilpotent. And φ_s and φ_n are polynomials in φ, in particular, φ_s and φ_n commute.

First, we prove $\varphi_s' = \varphi_s$ and $\varphi_n' = \varphi_n$.

Since $[\varphi_s', \varphi_n'] = 0$, we have $[\varphi_s', \varphi_n' + \varphi_s'] = 0$ which means that $[\varphi_s', \varphi] = 0$. Hence $[\varphi_s', f(\varphi)] = 0$ for all polynomial $f(t) \in F[t]$. In particular, $[\varphi_s', \varphi_s] = 0$. Similarly, $[\varphi_n', \varphi_n] = 0$.

And $\varphi_s' + \varphi_n' = \varphi_s + \varphi_n$ which means that $\varphi_s' - \varphi_s = \varphi_n' - \varphi_n$. Since the difference of two commutative semisimple linear transformations is still semisimple, we have $\varphi_s' - \varphi_s$ is semisimple and for the same reason $\varphi_n' - \varphi_n$ is nilpotent. Hence $\varphi_s' - \varphi_s$ and $\varphi_n' - \varphi_n$ are both zero.

By the same reason, we have $\varphi_s'' = \varphi_s$ and $\varphi_n'' = \varphi_n$. Thus

$$\varphi_s' = \varphi_s = \varphi_s'', \varphi_n' = \varphi_n = \varphi_n''.$$

Therefore the decomposition is unique. □

Definition 4.2.6. *Consider*

$$\varphi_s = \begin{pmatrix} \alpha_1 & 0 & \cdots & 0 \\ 0 & \alpha_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_n \end{pmatrix}.$$

We call a linear transformation ψ is a replica of φ if

$$\psi = \begin{pmatrix} \beta_1 & 0 & \cdots & 0 \\ 0 & \beta_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \beta_n \end{pmatrix}.$$

Where $\alpha_i = \alpha_j$ implies $\beta_i = \beta_j$. Then if $\varphi_s = p(\varphi)$, notice that $\psi = f(\varphi_s) = f(p(\varphi))$ for some polynomial $f(t) \in F[t]$.

Proposition 4.2.7. *Let φ be a linear transformation, $\varphi \in \mathrm{End}_F(V) = \mathfrak{gl}(V)$, consider $\mathrm{ad}(\varphi) : \mathfrak{gl}(V) \to \mathfrak{gl}(V)$, $x \mapsto [\varphi, x]$.*
If φ is semisimple then $\mathrm{ad}(\varphi)$ is a semisimple linear transformation.

Proof. Since φ is semisimple, there exists a basis of V in which φ is diagonal. Thus we may write

$$V = \bigoplus_{i=1}^{r} V_{\lambda_i},$$

where $\lambda_1, \ldots, \lambda_r \in F$ are the distinct eigenvalues of φ, and

$$\varphi|_{V_{\lambda_i}} = \lambda_i \operatorname{id}_{V_{\lambda_i}}.$$

Now consider $\mathfrak{gl}(V) = \operatorname{End}_F(V)$. For $1 \leq i, j \leq r$, define

$$E_{ij} := \operatorname{Hom}_F(V_{\lambda_j}, V_{\lambda_i}) \subseteq \operatorname{End}_F(V).$$

Using the decomposition of V, we have

$$\operatorname{End}_F(V) = \bigoplus_{i,j=1}^{r} E_{ij}.$$

We claim that each E_{ij} is an eigenspace of $\operatorname{ad}(\varphi)$. Indeed, let

$$x \in E_{ij} = \operatorname{Hom}_F(V_{\lambda_j}, V_{\lambda_i}).$$

Then for any $v \in V_{\lambda_j}$,

$$(\operatorname{ad}(\varphi)(x))(v) = (\varphi x - x\varphi)(v) = \varphi(x(v)) - x(\varphi(v)).$$

Since $x(v) \in V_{\lambda_i}$, we have

$$\varphi(x(v)) = \lambda_i x(v),$$

and since $v \in V_{\lambda_j}$,

$$x(\varphi(v)) = x(\lambda_j v) = \lambda_j x(v).$$

Therefore

$$(\operatorname{ad}(\varphi)(x))(v) = (\lambda_i - \lambda_j)x(v).$$

Hence

$$\operatorname{ad}(\varphi)(x) = (\lambda_i - \lambda_j)x,$$

so every $x \in E_{ij}$ is an eigenvector of $\operatorname{ad}(\varphi)$ with eigenvalue

$$\lambda_i - \lambda_j.$$

Thus $\operatorname{End}_F(V)$ is a direct sum of eigenspaces of $\operatorname{ad}(\varphi)$:

$$\operatorname{End}_F(V) = \bigoplus_{i,j=1}^{r} E_{ij}.$$

It follows that $\operatorname{ad}(\varphi)$ is diagonalizable, hence semisimple. $\square$

Theorem 4.2.8. *Suppose $\varphi = \varphi_s + \varphi_n$ is a Jordan-Chevalley decomposition of φ. Then $\operatorname{ad}(\varphi) = \operatorname{ad}(\varphi_s) + \operatorname{ad}(\varphi_n)$ is a Jordan-Chevalley decomposition of $\operatorname{ad}(\varphi)$.*

Proof. We already know $\operatorname{ad}(\varphi_s)$ is semisimple. And by easy discussion, we know $\operatorname{ad}(\varphi_n)$ is nilpotent. According to Jordan-Chevalley decomposition theorem, we only need to show that $[\mathrm{ad}(\varphi_s), \mathrm{ad}(\varphi_n)] = 0$. Then $\mathrm{ad}(\varphi) = \mathrm{ad}(\varphi_s) + \mathrm{ad}(\varphi_n)$ is a Jordan-Chevalley decomposition. The commutativity comes from $[\mathrm{ad}(\varphi_s), \mathrm{ad}(\varphi_n)] = \mathrm{ad}([\varphi_s, \varphi_n]) = 0$ $\square$

4.3 Preparation for Cartan's criterion

Consider the base field F. If $\operatorname{char} F = 0$ then $\mathbb{Q} \subseteq F$. Let $\pi : \sum_{i=1}^{n} \mathbb{Q}\alpha_i \to \mathbb{Q}$ be a $\mathbb{Q}$-linear map.

Consider

$$\psi = \begin{pmatrix} \pi(\alpha_1) & 0 & \cdots & 0 \\ 0 & \pi(\alpha_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \pi(\alpha_n) \end{pmatrix}.$$

This ψ is a replica of φ_s. Immediately we know that $\mathrm{ad}(\psi)$ is a replica of $\mathrm{ad}(\varphi_s)$.

Lemma 4.3.1. *Let A and B be subspaces of $\mathfrak{gl}(V)$ $(\dim_F(V) < \infty)$. let*

$$M = \{\varphi \in \mathfrak{gl}(V) \mid [\varphi, B] \subseteq A\}$$

Then M is a Lie subalgebra of $\mathfrak{gl}(V)$. If $\varphi \in M$, forall $m \in M$, $\mathrm{Tr}(\varphi m) = (0)$, then φ is nilpotent.

Proof. We shall prove that $\varphi_s = 0$, hence $\varphi = \varphi_n$ is nilpotent.

Choose a basis of V such that

$$\varphi_s = \begin{pmatrix} \alpha_1 & 0 & \cdots & 0 \\ 0 & \alpha_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_n \end{pmatrix}, \qquad \alpha_i \in F.$$

Since $\operatorname{char} F = 0$, we have $\mathbb{Q} \subseteq F$. Set

$$W := \sum_{i=1}^{n} \mathbb{Q}\alpha_i \subseteq F.$$

Let

$$\pi : W \to \mathbb{Q}$$

be any $\mathbb{Q}$-linear map, and define

$$\psi = \begin{pmatrix} \pi(\alpha_1) & 0 & \cdots & 0 \\ 0 & \pi(\alpha_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \pi(\alpha_n) \end{pmatrix}.$$

Then ψ is a replica of φ_s.

We claim that $\psi \in M$. Indeed, since $\varphi \in M$, we have

$$[\varphi, B] \subseteq A.$$

Passing to the quotient $\mathfrak{gl}(V)/A$, this means that the induced map

$$\operatorname{ad}(\varphi) : B + A/A \to \mathfrak{gl}(V)/A$$

is zero. Hence its semisimple part

$$\operatorname{ad}(\varphi_s)$$

also acts as zero on $B + A/A$. Since $\operatorname{ad}(\psi)$ is a replica of $\operatorname{ad}(\varphi_s)$, it follows that $\operatorname{ad}(\psi)$ also vanishes on $B + A/A$. Therefore

$$[\psi, B] \subseteq A,$$

so $\psi \in M$.

By the hypothesis of the lemma, for every $m \in M$ one has

$$\operatorname{Tr}(\varphi m) = 0.$$

Applying this to $m = \psi$, we get

$$\operatorname{Tr}(\varphi\psi) = 0.$$

Now

$$\operatorname{Tr}(\varphi\psi) = \operatorname{Tr}(\varphi_s\psi) + \operatorname{Tr}(\varphi_n\psi).$$

Since ψ is a polynomial in φ_s, it commutes with φ_s, and hence also with φ_n. Therefore $\varphi_n\psi$ is nilpotent, so

$$\operatorname{Tr}(\varphi_n\psi) = 0.$$

Consequently,

$$0 = \operatorname{Tr}(\varphi_s\psi) = \sum_{i=1}^{n} \alpha_i\, \pi(\alpha_i).$$

This holds for every $\mathbb{Q}$-linear map $\pi : W \to \mathbb{Q}$.

Now choose a $\mathbb{Q}$-basis $\beta_1, \ldots, \beta_r$ of W, and write

$$\alpha_i = \sum_{j=1}^{r} a_{ij}\beta_j, \qquad a_{ij} \in \mathbb{Q}.$$

Let $\pi_k : W \to \mathbb{Q}$ be the coordinate functional defined by

$$\pi_k(\beta_j) = \delta_{jk}.$$

Then

$$0 = \sum_{i=1}^{n} \alpha_i\, \pi_k(\alpha_i) = \sum_{i=1}^{n} \alpha_i a_{ik} = \sum_{i=1}^{n} \left(\sum_{j=1}^{r} a_{ij}\beta_j \right) a_{ik} = \sum_{j=1}^{r} \left(\sum_{i=1}^{n} a_{ij}a_{ik} \right) \beta_j.$$

Since $\beta_1, \ldots, \beta_r$ are linearly independent over $\mathbb{Q}$, we obtain

$$\sum_{i=1}^{n} a_{ij} a_{ik} = 0 \qquad \text{for all } 1 \leq j, k \leq r.$$

In particular, taking $j = k$, we get

$$\sum_{i=1}^{n} a_{ik}^2 = 0 \qquad \text{for each } k.$$

As each $a_{ik} \in \mathbb{Q}$, this implies

$$a_{ik} = 0 \qquad \text{for all } i, k.$$

Hence $\alpha_i = 0$ for every i, so $\varphi_s = 0$.

Therefore

$$\varphi = \varphi_n$$

is nilpotent. This proves the lemma. $\square$

Chapter 5

Lecture 5: Cartan's criterion and Killing form

5.1 Cartan Criterion for Solvability

Proposition 5.1.1. *Let $L \subseteq \mathfrak{gl}(V)$ with $\dim_F V < \infty$. Assume*

$$\mathrm{Tr}(xy) = 0 \quad \textit{for all } x \in L,\ y \in [L, L]$$

(equivalently, $\mathrm{Tr}(L[L, L]) = 0$). Then every element of $[L, L]$ is nilpotent as an endomorphism of V. In particular, $[L, L]$ acts nilpotently on V.

Proof. Apply the main lemma with

$$A := [L, L], \qquad B := L, \qquad M := \{\varphi \in \mathfrak{gl}(V) \mid [\varphi, B] \subseteq A\}.$$

Then M is a Lie subalgebra, and $L \subseteq M$.

Fix $a, b \in L$, and set $c \in M$. To use the lemma, it suffices to prove

$$\mathrm{Tr}([a, b]c) = 0.$$

For any $c \in M$, we compute

$$[a, b]c = [ac, b] - a[c, b].$$

Hence

$$\mathrm{Tr}([a, b]c) = \mathrm{Tr}([ac, b]) - \mathrm{Tr}(a[c, b]).$$

The first term is 0 because the trace of a commutator is 0. For the second term, $[c, b] \in A = [L, L]$ (since $c \in M$, $b \in B = L$), and $a \in L$; by the assumption $\mathrm{Tr}(L[L, L]) = 0$, we get $\mathrm{Tr}(a[c, b]) = 0$.

Therefore $\mathrm{Tr}([a, b]c) = 0$ for all $c \in M$. By the main lemma, each $[a, b]$ is nilpotent.

Now let

$$S := \{[a, b] \mid a, b \in L\} \subseteq [L, L].$$

Since $[L, L] = \langle S \rangle_{\mathrm{Lie}}$, and every element of S is nilpotent on V, Engel's theorem implies that $[L, L]$ is a nilpotent Lie algebra. Consequently, $[L, L]$ is solvable, so L is solvable. $\square$

5.2 Killing form

$\forall a, b \in L$, denote $\kappa(a|b) = \mathrm{Tr}(\mathrm{ad}\, a\, \mathrm{ad}\, b)$. Obviously, κ is a symmetric bilinear form on L.

Definition 5.2.1. *Let V be a vector space over a field F, and let*

$$B : V \times V \to F$$

be a bilinear form. We say that B is nondegenerate *if either of the following equivalent conditions holds:*

1. *If $v \in V$ satisfies*
$$B(v, w) = 0 \qquad \forall\, w \in V,$$
then $v = 0$.

2. *If $w \in V$ satisfies*
$$B(v, w) = 0 \qquad \forall\, v \in V,$$
then $w = 0$.

Equivalently, the induced linear map

$$V \to V^*, \qquad v \mapsto B(v, -)$$

is injective; if $\dim V < \infty$, this is equivalent to this map being an isomorphism.

Proposition 5.2.2. *Killing form is associative.*

Proof. Let L be a finite-dimensional Lie algebra, and let

$$\kappa(x, y) = \mathrm{Tr}(\mathrm{ad}\, x\ \mathrm{ad}\, y) \qquad (x, y \in L)$$

be its Killing form. We claim that for all $x, y, z \in L$,

$$\kappa([x, y], z) = \kappa(x, [y, z]).$$

Indeed, using

$$\mathrm{ad}([x, y]) = [\mathrm{ad}\, x, \mathrm{ad}\, y] = \mathrm{ad}\, x\ \mathrm{ad}\, y - \mathrm{ad}\, y\ \mathrm{ad}\, x,$$

we compute

$$\begin{aligned}
\kappa([x, y], z) &= \mathrm{Tr}\big(\mathrm{ad}([x, y])\ \mathrm{ad}\, z\big) \\
&= \mathrm{Tr}\big((\mathrm{ad}\, x\ \mathrm{ad}\, y - \mathrm{ad}\, y\ \mathrm{ad}\, x)\, \mathrm{ad}\, z\big) \\
&= \mathrm{Tr}(\mathrm{ad}\, x\ \mathrm{ad}\, y\ \mathrm{ad}\, z) - \mathrm{Tr}(\mathrm{ad}\, y\ \mathrm{ad}\, x\ \mathrm{ad}\, z).
\end{aligned}$$

By cyclicity of trace,
$$\mathrm{Tr}(\mathrm{ad}\, y\ \mathrm{ad}\, x\ \mathrm{ad}\, z) = \mathrm{Tr}(\mathrm{ad}\, x\ \mathrm{ad}\, z\ \mathrm{ad}\, y),$$
hence

$$\begin{aligned}\kappa([x,y],z) &= \mathrm{Tr}(\mathrm{ad}\, x\ \mathrm{ad}\, y\ \mathrm{ad}\, z) - \mathrm{Tr}(\mathrm{ad}\, x\ \mathrm{ad}\, z\ \mathrm{ad}\, y)\\ &= \mathrm{Tr}\big(\mathrm{ad}\, x(\mathrm{ad}\, y\ \mathrm{ad}\, z - \mathrm{ad}\, z\ \mathrm{ad}\, y)\big)\\ &= \mathrm{Tr}\big(\mathrm{ad}\, x\, [\mathrm{ad}\, y, \mathrm{ad}\, z]\big)\\ &= \mathrm{Tr}\big(\mathrm{ad}\, x\ \mathrm{ad}([y,z])\big)\\ &= \kappa(x,[y,z]).\end{aligned}$$

Therefore the Killing form is associative. $\square$

Now we prove the theorem.

($\Rightarrow$): Let $I = \{a \in L | (a|L) = (0)\} \triangleleft L$. Since $([a,b]|L) = (a|[b,L]) = (0)$. We consider $\mathrm{ad}(L) \subseteq \mathfrak{gl}(L)$. Then we have $\mathrm{Tr}(\mathrm{ad}(I)\mathrm{ad}(I)) = 0$. Hence $\mathrm{ad}(I) \subseteq \mathfrak{gl}(L)$ is solvable. And $\mathrm{ad} : I \to \mathrm{ad}(I)$ the image is solvable. $\ker \mathrm{ad} = I \cap Z(L)$ is abelian, hence solvable. If L is semisimple, then we have $I = 0$. Which shows that κ is nondegenerate.

($\Leftarrow$): Suppose L is not semisimple. Then $\mathrm{Rad}(L) \neq 0$. Let $I \triangleleft L$ be a non-zero solvable ideal, then L mast have a non-zero abelian ideal J. Then $\kappa(J|L) = 0$. Contradiction.

Proposition 5.2.3. *Let V be a vector space over F, not necessarily finite-dimensional, and let*
$$\kappa : V \times V \to F$$
be a symmetric bilinear form. Let $W \subseteq V$ be a finite-dimensional subspace. Assume that the restriction
$$\kappa|_{W\times W}$$
is nondegenerate. Define
$$W^\perp := \{\, v \in V \mid \kappa(v,w) = 0 \text{ for all } w \in W \,\}.$$
Then
$$V = W \oplus W^\perp.$$

Proof. It is enough to prove that
$$V = W + W^\perp,$$
since $W \cap W^\perp = 0$ follows immediately from the nondegeneracy of $\kappa|_{W\times W}$.

Let $w_1, \ldots, w_n$ be a basis of W. Consider the Gram matrix
$$G = (\kappa(w_i, w_j))_{1\le i,j\le n}.$$
Since $\kappa|_{W\times W}$ is nondegenerate, the matrix G is invertible.

Take any $v \in V$. We seek $u = \sum_{i=1}^{n} c_i w_i \in W$ such that

$$v - u \in W^{\perp}.$$

This is equivalent to requiring

$$\kappa(v - u, w_j) = 0 \qquad (j = 1, \dots, n),$$

that is,

$$\kappa(v, w_j) = \sum_{i=1}^{n} c_i \, \kappa(w_i, w_j) \qquad (j = 1, \dots, n).$$

This is a linear system for the coefficients c_i, whose coefficient matrix is G. Since G is invertible, the system has a unique solution. Hence there exists $u \in W$ such that $v - u \in W^{\perp}$.

Therefore every $v \in V$ can be written as

$$v = u + (v - u) \in W + W^{\perp},$$

so $V = W + W^{\perp}$. As noted above, $W \cap W^{\perp} = 0$, hence

$$V = W \oplus W^{\perp}.$$

□

Proposition 5.2.4. *Let L be a semisimple Lie algebra, and let*

$$I \lhd L, \qquad I \neq 0.$$

Then the restriction of the Killing form κ of L to $I \times I$ is nondegenerate.

Proof. Suppose, to the contrary, that $\kappa|_{I \times I}$ is degenerate. Then there exists $0 \neq a \in I$ such that

$$\kappa(a, I) = 0.$$

We claim that in fact

$$\kappa([a, I], L) = 0.$$

Indeed, let $x \in I$ and $y \in L$. By associativity of the Killing form,

$$\kappa([a, x], y) = \kappa(a, [x, y]).$$

Since $I \lhd L$, we have $[x, y] \in I$, and since $\kappa(a, I) = 0$, it follows that

$$\kappa([a, x], y) = 0 \qquad \text{for all } x \in I, \; y \in L.$$

Thus

$$[a, I] \subseteq L^{\perp},$$

where

$$L^\perp = \{ z \in L \mid \kappa(z, L) = 0 \}.$$

But L is semisimple, so its Killing form is nondegenerate. Hence $L^\perp = 0$, and therefore

$$[a, I] = 0.$$

So $a \in C_L(I)$, the centralizer of I in L.

Now $I \cap C_L(I)$ is a nonzero ideal of L: it is nonzero because $a \in I \cap C_L(I)$, and it is an ideal since both I and $C_L(I)$ are ideals of L. Moreover, it is abelian, because for $x, y \in I \cap C_L(I)$ we have $y \in I$ and x centralizes I, hence

$$[x, y] = 0.$$

Thus L contains a nonzero abelian ideal, contradicting the semisimplicity of L.

Therefore $\kappa|_{I\times I}$ must be nondegenerate. □

Theorem 5.2.5. *Every semisimple Lie algebra is a direct sum of simple ideals.*

Proof. We argue by induction on $\dim L$. If L is simple, there is nothing to prove.

Assume that L is semisimple but not simple. Then there exists a nonzero proper ideal

$$0 \neq I \lhd L, \qquad I \neq L.$$

By the previous proposition, the restriction of the Killing form to I is nondegenerate. Since I is finite-dimensional, the preceding orthogonal decomposition proposition gives

$$L = I \oplus I^\perp,$$

where

$$I^\perp = \{ x \in L \mid \kappa(x, I) = 0 \}.$$

We claim that $I^\perp$ is also an ideal of L. Let $x \in I^\perp$, $y \in L$, and $z \in I$. Then, by associativity of the Killing form,

$$\kappa([y, x], z) = \kappa(y, [x, z]).$$

Since $I \lhd L$, we have $[x, z] \in I$. Because $x \in I^\perp$, it follows that

$$\kappa(x, [z, y]) = 0,$$

and hence

$$\kappa([y, x], z) = 0 \qquad \text{for all } z \in I.$$

Therefore $[y, x] \in I^\perp$, proving that $I^\perp \lhd L$.

Both I and $I^\perp$ are semisimple, being ideals of the semisimple Lie algebra L. Moreover,

$$\dim I < \dim L, \qquad \dim I^\perp < \dim L.$$

By the induction hypothesis, each of I and $I^{\perp}$ is a direct sum of simple ideals. Hence so is

$$L = I \oplus I^{\perp}.$$

This completes the proof. $\square$

5.3 Cartan Criterion for Semisimplicity

Theorem 5.3.1 (Cartan criterion for semisimplicity)**.** *Let L be a finite-dimensional Lie algebra over a field F of characteristic 0. Then L is semisimple if and only if its Killing form*

$$\kappa(x, y) = \mathrm{Tr}(\mathrm{ad}\, x \; \mathrm{ad}\, y) \qquad (x, y \in L)$$

is nondegenerate.

Proof. We prove the two implications separately.

($\Rightarrow$) Assume that L is semisimple. Let

$$L^{\perp} := \{x \in L \mid \kappa(x, y) = 0 \text{ for all } y \in L\}$$

be the radical of κ. We first show that $L^{\perp}$ is an ideal of L. Indeed, if $x \in L^{\perp}$ and $z \in L$, then for every $y \in L$,

$$\kappa([z, x], y) = \kappa(z, [x, y]) = -\kappa(x, [z, y]) = 0,$$

where we used the associativity of the Killing form. Hence

$$[z, x] \in L^{\perp},$$

so $L^{\perp} \triangleleft L$.

Now let $a \in [L^{\perp}, L^{\perp}]$. Then $a \in L^{\perp}$, so

$$\kappa(a, L) = 0.$$

Thus

$$\kappa([L^{\perp}, L^{\perp}], L^{\perp}) = 0.$$

By Cartan's criterion for solvability, $L^{\perp}$ is solvable. Since L is semisimple, it has no nonzero solvable ideals. Therefore

$$L^{\perp} = 0.$$

Hence κ is nondegenerate.

($\Leftarrow$) Assume that κ is nondegenerate. Let $R = \mathrm{Rad}(L)$ be the solvable radical of L. We shall show that $R = 0$.

Since R is solvable, its derived algebra $[R, R]$ is nilpotent, hence solvable. For every

$$x \in [R, R], \qquad y \in L,$$

the operator $\operatorname{ad} x \, \operatorname{ad} y$ has trace 0. Therefore

$$\kappa([R, R], L) = 0.$$

By the nondegeneracy of κ, this implies

$$[R, R] = 0.$$

So R is abelian.

Now let $x \in R$ and $y \in L$. Since R is an abelian ideal, we have

$$[R, R] = 0 \qquad \text{and} \qquad [L, R] \subseteq R.$$

Choose a basis of L adapted to R, say

$$L = R \oplus U.$$

Relative to this decomposition, $\operatorname{ad} x$ has the block form

$$\operatorname{ad} x = \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix},$$

because $\operatorname{ad} x(L) \subseteq R$ and $\operatorname{ad} x(R) = [x, R] = 0$. Hence for every $y \in L$, the product $\operatorname{ad} x \, \operatorname{ad} y$ also has trace 0. Thus

$$\kappa(x, y) = 0 \qquad \text{for all } y \in L.$$

So $x \in L^{\perp}$. Since κ is nondegenerate, $L^{\perp} = 0$, hence $x = 0$. Therefore

$$R = 0.$$

Thus L is semisimple. $\square$

Chapter 6

Lecture 6: Basics of Representation

6.1 Inner Derivation of Lie algebra

For any algebra A, $\mathrm{d} : A \to A$ is a derivation if $\mathrm{d}(ab) = \mathrm{d}(a)b + a\mathrm{d}(b)$ for all $a, b \in A$. Hence $\mathrm{Der}(A)$ is a Lie subalgebra of $\mathrm{End}_F(A)^{(-)} = \mathfrak{gl}(A)$.

Now let L be a Lie algebra (not necessarily semisimple or finite dimensional). Let $a \in L$, $\mathrm{ad}(a) : L \to L : b \mapsto [a, b]$ is a derivation. Such derivation is called inner derivation.

Theorem 6.1.1. *If L is a finite dimensional semisimple Lie algebra, then* $\mathrm{Der}(L) = \mathrm{ad}(L)$.

Remark 6.1.2. *An example of not inner derivation: Let $A = \begin{pmatrix} f(t) & g(t) \\ h(t) & k(t) \end{pmatrix}$. Then $\mathrm{d} : A \to A : \begin{pmatrix} f(t) & g(t) \\ h(t) & k(t) \end{pmatrix} \mapsto \begin{pmatrix} f'(t) & g'(t) \\ h'(t) & k'(t) \end{pmatrix}$ is a derivation. But it is not inner derivation.*

Another example for Lie algebra: Let A be an abelian Lie algebra. Then any linear transformation is a derivation. But it is not inner derivation.

Proof. Let $d \in \mathrm{Der}(L)$. We must show that d is inner.

Consider the vector space

$$\widetilde{L} := L \oplus Fd,$$

and define a Lie bracket on $\widetilde{L}$ by requiring that:

- the bracket on L is the original one;
- for $x \in L$,

$$[d, x] := d(x), \qquad [x, d] := -d(x);$$

- $[d, d] = 0$.

Since d is a derivation of L, the Jacobi identity holds, so $\widetilde{L}$ is a Lie algebra and $L \triangleleft \widetilde{L}$ is an ideal of codimension 1.

Let $\mathrm{Rad}(\widetilde{L})$ be the solvable radical of $\widetilde{L}$. Since L is semisimple, it contains no nonzero solvable ideals. Hence

$$\mathrm{Rad}(\widetilde{L}) \cap L = 0.$$

On the other hand, $\mathrm{Rad}(\widetilde{L}) + L$ is a subspace of $\widetilde{L}$ containing L, and since $\dim(\widetilde{L}/L) = 1$, either $\mathrm{Rad}(\widetilde{L}) \subseteq L$ or

$$\widetilde{L} = L + \mathrm{Rad}(\widetilde{L}).$$

The first possibility would imply $\mathrm{Rad}(\widetilde{L}) = 0$, hence in any case we may write

$$\widetilde{L} = L + \mathrm{Rad}(\widetilde{L}).$$

Therefore there exist $a \in L$ and $r \in \mathrm{Rad}(\widetilde{L})$ such that

$$d = a + r.$$

Equivalently,

$$r = d - a.$$

Now let $b \in L$. Since $L \triangleleft \widetilde{L}$, we have $[r, b] \in L$. Also $r \in \mathrm{Rad}(\widetilde{L})$ and $\mathrm{Rad}(\widetilde{L})$ is an ideal, so $[r, b] \in \mathrm{Rad}(\widetilde{L})$. Thus

$$[r, b] \in L \cap \mathrm{Rad}(\widetilde{L}) = 0,$$

hence $[r, b] = 0$ for all $b \in L$.

But

$$[r, b] = [d - a, b] = [d, b] - [a, b] = d(b) - [a, b].$$

Therefore

$$d(b) = [a, b] = -[b, a] = \mathrm{ad}(-a)(b) \qquad \text{for all } b \in L.$$

So

$$d = \mathrm{ad}(-a) \in \mathrm{ad}(L).$$

Since every derivation of L is inner, we conclude that

$$\mathrm{Der}(L) = \mathrm{ad}(L).$$

□

6.2 Some Basics of Representation of Lie algebra

Definition 6.2.1. *Let L be a Lie algebra over a field F, and let V be a vector space over F. An L-*module structure *on V is a bilinear map*

$$L \times V \to V, \qquad (x, v) \mapsto x \cdot v,$$

such that

$$[x, y] \cdot v = x \cdot (y \cdot v) - y \cdot (x \cdot v)$$

for all $x, y \in L$ and $v \in V$.

Definition 6.2.2. *Let V and W be L-modules. The* tensor product module $V \otimes W$ *is the vector space $V \otimes_F W$ equipped with the action*

$$x \cdot (v \otimes w) = (x \cdot v) \otimes w + v \otimes (x \cdot w), \qquad x \in L,\ v \in V,\ w \in W,$$

extended linearly to all of $V \otimes W$.

Proof. We verify that this defines an L-module structure. Since the formula is bilinear in x, v, and w, it extends uniquely to a linear action on $V \otimes W$. It remains to check the bracket identity.

For $x, y \in L$, $v \in V$, and $w \in W$, we have

$$\begin{aligned} x \cdot \big(y \cdot (v \otimes w)\big) &= x \cdot \big((y \cdot v) \otimes w + v \otimes (y \cdot w)\big) \\ &= (x \cdot (y \cdot v)) \otimes w + (y \cdot v) \otimes (x \cdot w) + (x \cdot v) \otimes (y \cdot w) + v \otimes (x \cdot (y \cdot w)), \end{aligned}$$

and similarly

$$y \cdot \big(x \cdot (v \otimes w)\big) = (y \cdot (x \cdot v)) \otimes w + (x \cdot v) \otimes (y \cdot w) + (y \cdot v) \otimes (x \cdot w) + v \otimes (y \cdot (x \cdot w)).$$

Subtracting, we obtain

$$\begin{aligned} x \cdot \big(y \cdot (v \otimes w)\big) - y \cdot \big(x \cdot (v \otimes w)\big) &= \big([x, y] \cdot v\big) \otimes w + v \otimes \big([x, y] \cdot w\big) \\ &= [x, y] \cdot (v \otimes w). \end{aligned}$$

Thus $V \otimes W$ is an L-module. □

Definition 6.2.3. *Let V and W be L-modules. The vector space*

$$\mathrm{Hom}_F(V, W)$$

becomes an L-module by defining

$$(x \cdot f)(v) = x \cdot f(v) - f(x \cdot v), \qquad x \in L,\ f \in \mathrm{Hom}_F(V, W),\ v \in V.$$

This is called the Hom-module.

Proof. We first note that for fixed $x \in L$ and $f \in \mathrm{Hom}_F(V, W)$, the map $x \cdot f$ is again F-linear in v, so $x \cdot f \in \mathrm{Hom}_F(V, W)$.

Now let $x, y \in L$, $f \in \mathrm{Hom}_F(V, W)$, and $v \in V$. Then

$$\begin{aligned}\big(x \cdot (y \cdot f)\big)(v) &= x \cdot \big((y \cdot f)(v)\big) - (y \cdot f)(x \cdot v) \\ &= x \cdot \big(y \cdot f(v) - f(y \cdot v)\big) - \big(y \cdot f(x \cdot v) - f(y \cdot (x \cdot v))\big) \\ &= x \cdot (y \cdot f(v)) - x \cdot f(y \cdot v) - y \cdot f(x \cdot v) + f(y \cdot (x \cdot v)),\end{aligned}$$

and similarly

$$\big(y \cdot (x \cdot f)\big)(v) = y \cdot (x \cdot f(v)) - y \cdot f(x \cdot v) - x \cdot f(y \cdot v) + f(x \cdot (y \cdot v)).$$

Subtracting gives

$$\begin{aligned}\big(x \cdot (y \cdot f) - y \cdot (x \cdot f)\big)(v) &= [x, y] \cdot f(v) - f([x, y] \cdot v) \\ &= \big([x, y] \cdot f\big)(v).\end{aligned}$$

Hence

$$[x, y] \cdot f = x \cdot (y \cdot f) - y \cdot (x \cdot f),$$

so $\mathrm{Hom}_F(V, W)$ is an L-module. □

Definition 6.2.4. *Let V be an L-module. The dual space*

$$V^* = \mathrm{Hom}_F(V, F)$$

is called the dual module *of V, where F is regarded as the trivial L-module. Its action is given by*

$$(x \cdot \varphi)(v) = -\,\varphi(x \cdot v), \qquad x \in L,\ \varphi \in V^*,\ v \in V.$$

Proof. This is the special case $W = F$ of the Hom-module construction, where the action of L on F is trivial. Indeed, if $\varphi \in \mathrm{Hom}_F(V, F)$, then

$$(x \cdot \varphi)(v) = x \cdot \varphi(v) - \varphi(x \cdot v) = 0 - \varphi(x \cdot v) = -\varphi(x \cdot v).$$

Thus V^* is an L-module. □

Remark 6.2.5. *The subspace of L-invariants in $\mathrm{Hom}_F(V, W)$ is precisely*

$$\mathrm{Hom}_L(V, W) = \{f \in \mathrm{Hom}_F(V, W) \mid x \cdot f = 0 \textit{ for all } x \in L\}.$$

Indeed, the condition $x \cdot f = 0$ means

$$x \cdot f(v) = f(x \cdot v) \qquad \textit{for all } x \in L,\ v \in V,$$

which is exactly the condition that f be an L-module homomorphism.

Proposition 6.2.6. *Assume that V is finite-dimensional. Then there is a natural vector space isomorphism*

$$V^* \otimes W \cong \mathrm{Hom}_F(V, W),$$

given by

$$\varphi \otimes w \longmapsto \big(v \mapsto \varphi(v)w\big).$$

This is in fact an isomorphism of L-modules.

Proof. Let

$$\Phi : V^* \otimes W \to \mathrm{Hom}_F(V, W)$$

be defined by

$$\Phi(\varphi \otimes w)(v) = \varphi(v)w.$$

It is the usual canonical isomorphism of vector spaces when V is finite-dimensional. We check that it is L-equivariant.

For $x \in L$, $\varphi \in V^*$, $w \in W$, and $v \in V$, we have

$$\begin{aligned} \Phi\big(x \cdot (\varphi \otimes w)\big)(v) &= \Phi\big((x \cdot \varphi) \otimes w + \varphi \otimes (x \cdot w)\big)(v) \\ &= (x \cdot \varphi)(v)\, w + \varphi(v)\, (x \cdot w) \\ &= -\varphi(x \cdot v)\, w + \varphi(v)\, (x \cdot w). \end{aligned}$$

On the other hand,

$$\begin{aligned} \big(x \cdot \Phi(\varphi \otimes w)\big)(v) &= x \cdot \big(\Phi(\varphi \otimes w)(v)\big) - \Phi(\varphi \otimes w)(x \cdot v) \\ &= x \cdot (\varphi(v)w) - \varphi(x \cdot v)w \\ &= \varphi(v)(x \cdot w) - \varphi(x \cdot v)w. \end{aligned}$$

Thus

$$\Phi\big(x \cdot (\varphi \otimes w)\big) = x \cdot \Phi(\varphi \otimes w),$$

so Φ is an L-module isomorphism. □

Remark 6.2.7. *Under the identification*

$$\mathrm{End}_F(V) = \mathrm{Hom}_F(V, V),$$

the L-action is

$$(x \cdot T)(v) = x \cdot T(v) - T(x \cdot v), \qquad x \in L,\ T \in \mathrm{End}_F(V),\ v \in V.$$

Thus L acts on $\mathrm{End}_F(V)$ *by commutator.*

Remark 6.2.8. *The natural pairing*

$$V^* \times V \to F, \qquad (\varphi, v) \mapsto \varphi(v),$$

is L-invariant in the sense that

$$(x \cdot \varphi)(v) + \varphi(x \cdot v) = 0$$

for all $x \in L$, $\varphi \in V^*$, and $v \in V$.

6.3 Schur's Lemma

Lemma 6.3.1 (Schur's Lemma)**.** *Let L be a Lie algebra over a field F, and let V, W be irreducible L-modules. Suppose*

$$\varphi : V \to W$$

is an L-module homomorphism. Then either $\varphi = 0$, or φ is an isomorphism.

Proof. Since φ is an L-module homomorphism, its kernel

$$\ker \varphi = \{v \in V \mid \varphi(v) = 0\}$$

is a submodule of V, and its image

$$\operatorname{Im} \varphi = \varphi(V)$$

is a submodule of W.

Because V is irreducible, $\ker \varphi$ is either 0 or V. If $\ker \varphi = V$, then $\varphi = 0$. So if $\varphi \neq 0$, we must have $\ker \varphi = 0$, hence φ is injective.

Likewise, since W is irreducible and $\operatorname{Im} \varphi \neq 0$, we must have

$$\operatorname{Im} \varphi = W.$$

Thus φ is surjective.

Therefore, if $\varphi \neq 0$, then φ is both injective and surjective, hence an isomorphism. □

Corollary 6.3.2 (Schur's Lemma, endomorphism form)**.** *Let V be an irreducible L-module. Then every nonzero element of*

$$\operatorname{End}_L(V)$$

is invertible. In particular, $\operatorname{End}_L(V)$ *is a division algebra over F.*

Proof. Apply Schur's Lemma with $W = V$. Any nonzero L-module endomorphism

$$\varphi : V \to V$$

must be an isomorphism, hence invertible. Therefore $\operatorname{End}_L(V)$ is a division algebra. □

Corollary 6.3.3. *Assume moreover that F is algebraically closed and $\dim_F V < \infty$. If V is an irreducible L-module, then every L-module endomorphism of V is a scalar multiple of the identity:*

$$\mathrm{End}_L(V) = F\,\mathrm{id}_V\,.$$

Proof. Let $\varphi \in \mathrm{End}_L(V)$. Since F is algebraically closed and V is finite-dimensional, φ has an eigenvalue $\lambda \in F$. Then

$$\psi := \varphi - \lambda\,\mathrm{id}_V$$

is an L-module endomorphism of V, and ψ is not injective because λ is an eigenvalue of φ. By Schur's Lemma, a nonzero L-module endomorphism of an irreducible module must be an isomorphism, hence injective. Therefore $\psi = 0$. So

$$\varphi = \lambda\,\mathrm{id}_V\,.$$

This proves that every L-module endomorphism of V is scalar. □

Remark 6.3.4. *The last corollary is the form most often used in representation theory. It says that, over an algebraically closed field, an irreducible finite-dimensional module has only scalar L-equivariant endomorphisms.*

6.4 Casimir element

Definition 6.4.1. *Let L be a finite-dimensional Lie algebra over F, and let $B(\cdot,\cdot)$ be a nondegenerate invariant symmetric bilinear form on L. Choose a basis $x_1, \dots, x_n$ of L, and let $x^1, \dots, x^n$ be the B-dual basis, so that*

$$B(x_i, x^j) = \delta_{ij}.$$

The element

$$\Omega_B = \sum_{i=1}^{n} x_i x^i \in U(L)$$

is called the Casimir element associated with B.

Remark 6.4.2. *The definition is independent of the choice of basis. Indeed, Ω is the element corresponding to the identity map of L under the natural identification*

$$L \otimes L \cong \mathrm{End}_F(L), \qquad x \otimes y \mapsto \big(z \mapsto (y,z)x\big).$$

Proposition 6.4.3. *Let L be a finite-dimensional Lie algebra equipped with a nondegenerate invariant symmetric bilinear form B, and let*

$$\Omega_B = \sum_{i=1}^{n} x_i x^i$$

be the associated Casimir element. If $\rho : L \to \mathfrak{gl}(V)$ is a representation of L, define

$$C_V := \sum_{i=1}^{n} \rho(x_i)\rho(x^i) \in \mathrm{End}_F(V).$$

Then

$$\rho(a)C_V = C_V\rho(a) \qquad \text{for all } a \in L.$$

Proof. For any $a \in L$, we compute

$$[\rho(a), C_V] = \sum_{i=1}^{n} \big([\rho(a), \rho(x_i)]\rho(x^i) + \rho(x_i)[\rho(a), \rho(x^i)]\big).$$

Since ρ is a representation, this becomes

$$[\rho(a), C_V] = \sum_{i=1}^{n} \big(\rho([a, x_i])\rho(x^i) + \rho(x_i)\rho([a, x^i])\big).$$

Write

$$[a, x_i] = \sum_{j=1}^{n} c_{ji}x_j.$$

We claim that

$$[a, x^i] = -\sum_{j=1}^{n} c_{ij}x^j.$$

Indeed, for every k,

$$B([a, x^i], x_k) = -B(x^i, [a, x_k])$$

by invariance and symmetry of B. Since

$$[a, x_k] = \sum_{j=1}^{n} c_{jk}x_j,$$

we get

$$B([a, x^i], x_k) = -\sum_{j=1}^{n} c_{jk}B(x^i, x_j) = -c_{ik}.$$

If

$$[a, x^i] = \sum_{j=1}^{n} d_{ji}x^j,$$

then

$$d_{ki} = B([a, x^i], x_k) = -c_{ik},$$

because $B(x^j, x_k) = \delta_{jk}$. This proves the claim.

Therefore

$$\sum_{i=1}^{n} \rho([a, x_i])\rho(x^i) = \sum_{i,j=1}^{n} c_{ji}\rho(x_j)\rho(x^i),$$

while

$$\sum_{i=1}^{n} \rho(x_i)\rho([a, x^i]) = -\sum_{i,j=1}^{n} c_{ij}\rho(x_i)\rho(x^j).$$

After relabeling indices in the first sum, these two sums cancel. Hence

$$[\rho(a), C_V] = 0.$$

Equivalently,

$$\rho(a)C_V = C_V\rho(a) \qquad \text{for all } a \in L.$$

□

Chapter 7

Lecture 7: Weyl's theorem

7.1 Conditions Defining Semisimplicity

Proposition 7.1.1. *Let F be a field, let A be an F-algebra (not necessarily with identity), and let V be a left A-module. Then the following are equivalent:*

1. *Every submodule $W \subseteq V$ is a direct summand of V.*

2. *There exists a family $\{V_i\}_{i\in I}$ of irreducible submodules of V such that*

$$V = \bigoplus_{i\in I} V_i.$$

Proof. We prove (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (1).

(1) $\Rightarrow$ (2). Assume that every submodule of V is a direct summand.

We first show that every nonzero submodule of V contains an irreducible submodule. Let $0 \neq W \subseteq V$ be a submodule, and choose $0 \neq w \in W$. Let P be the submodule of V generated by w, that is, the smallest submodule containing w. Since V is an F-vector space and every submodule is in particular an F-subspace, we have $Fw \subseteq P$, hence $w \in P$ and therefore $P \neq 0$.

Consider the set of proper submodules of P, partially ordered by inclusion. Since $\{0\}$ is a proper submodule of P, this set is nonempty. Moreover, the union of any chain of proper submodules is again a proper submodule of P. Hence, by Zorn's lemma, there exists a maximal proper submodule $Q \subsetneq P$.

By assumption, Q is a direct summand of P, so there exists a submodule $Q' \subseteq P$ such that

$$P = Q \oplus Q'.$$

We claim that Q' is irreducible. Indeed, let $0 \neq X \subseteq Q'$ be a submodule. Then

$$Q \oplus X$$

is a submodule of P properly containing Q. By the maximality of Q, we must have $Q \oplus X = P$. Since also $P = Q\oplus Q'$, it follows that $X = Q'$. Thus Q' has no nonzero proper submodule, so Q' is irreducible.

Therefore every nonzero submodule of V contains an irreducible submodule.

Now let

$$E := \sum\{\, S \subseteq V \mid S \text{ is an irreducible submodule of } V \,\}.$$

Then E is a submodule of V. We claim that $E = V$. Suppose not. Then E is a proper submodule of V, and by assumption there exists a submodule $U \subseteq V$ such that

$$V = E \oplus U.$$

Since $E \neq V$, we have $U \neq 0$. By the first part, U contains an irreducible submodule S. But then $S \subseteq E$ by the definition of E, and also $S \subseteq U$. Hence

$$S \subseteq E \cap U = 0,$$

a contradiction. Thus $E = V$, so V is the sum of irreducible submodules.

It remains to show that this sum is in fact direct. Write

$$V = \sum_{i \in I} V_i$$

with each V_i irreducible. Let $J \subseteq I$ be maximal such that

$$\sum_{j \in J} V_j$$

is a direct sum. Then

$$\sum_{j \in J} V_j = \bigoplus_{j \in J} V_j.$$

We claim that

$$V = \bigoplus_{j \in J} V_j.$$

Indeed, let $i \in I$. Since V_i is irreducible, the intersection

$$V_i \cap \left(\bigoplus_{j \in J} V_j \right)$$

is either 0 or V_i. If it were 0, then

$$\left(\bigoplus_{j \in J} V_j \right) \oplus V_i$$

would still be a direct sum, contradicting the maximality of J unless $i \in J$ already. Hence the intersection must be V_i, so

$$V_i \subseteq \bigoplus_{j \in J} V_j.$$

Since this holds for every $i \in I$, we get

$$V = \sum_{i \in I} V_i \subseteq \bigoplus_{j \in J} V_j.$$

The reverse inclusion is obvious, so

$$V = \bigoplus_{j \in J} V_j.$$

This proves (2).

(2) ⇒ (1). Assume that

$$V = \bigoplus_{i \in I} V_i$$

where each V_i is irreducible. Let $W \subseteq V$ be a submodule. Consider the collection

$$\mathcal{J} := \left\{ J \subseteq I \;\middle|\; W \cap \bigoplus_{j \in J} V_j = 0 \right\}.$$

By Zorn's lemma, there exists a maximal element $J \in \mathcal{J}$.

We claim that

$$V = W + \bigoplus_{j \in J} V_j.$$

Suppose not. Then there exists some $i \in I$ such that

$$V_i \nsubseteq W + \bigoplus_{j \in J} V_j.$$

Since V_i is irreducible, the intersection

$$V_i \cap \left(W + \bigoplus_{j \in J} V_j \right)$$

is either 0 or V_i. The second possibility would imply

$$V_i \subseteq W + \bigoplus_{j \in J} V_j,$$

contrary to our choice of i. Hence

$$V_i \cap \left(W + \bigoplus_{j \in J} V_j \right) = 0.$$

In particular,

$$W \cap \left(\bigoplus_{j \in J} V_j \oplus V_i \right) = 0,$$

which contradicts the maximality of J. Therefore

$$V = W + \bigoplus_{j \in J} V_j.$$

Finally, this sum is direct, because by the definition of J we have

$$W \cap \bigoplus_{j \in J} V_j = 0.$$

Thus

$$V = W \oplus \bigoplus_{j \in J} V_j,$$

so W is a direct summand of V. This proves (1). □

7.2 Weyl's theorem

I use my proof in a former note use short exact sequence.

Lemma 7.2.1. *Let L be a semisimple Lie algebra and $\phi : L \to \mathfrak{gl}(V)$ a representation. Then $\phi(L) \subseteq \mathfrak{sl}(V)$. In particular, L acts trivially on any one-dimensional L-module.*

Proof. Since L is semisimple, we have $L = [L, L]$. Hence

$$\phi(L) = \phi([L, L]) = [\phi(L), \phi(L)].$$

But $[\mathfrak{gl}(V), \mathfrak{gl}(V)] = \mathfrak{sl}(V)$, so $\phi(L) \subseteq \mathfrak{sl}(V)$.

Let W be a one-dimensional L-module, and let $\psi : L \to \mathfrak{gl}(W) \cong \mathbb{k}$. Then for all $x \in L$, we have $\mathrm{Tr}(\psi(x)) = 0$. Hence $\psi(x) = 0$, so L acts trivially on W. □

Theorem 7.2.2. *Let L be a semisimple Lie algebra and $\phi : L \to \mathfrak{gl}(V)$ a representation, where V is finite-dimensional. Then ϕ is completely reducible. In other words, if W is an L-submodule of V, then there exists a submodule W' such that*

$$V = W \oplus W'.$$

Remark 7.2.3. *If V is irreducible, then there is nothing to prove. Hence we may assume that V is reducible.*

Remark 7.2.4. *Being completely reducible is equivalent to the following: for any submodule $W \subseteq V$, there exists a submodule W' such that $V = W \oplus W'$. Furthermore, this is equivalent to saying that the short exact sequence*

$$0 \longrightarrow W \longrightarrow V \longrightarrow V/W \longrightarrow 0$$

splits.

Definition 7.2.5 (Definition 2.17). *The short exact sequence of a direct sum is called a* splitting short exact sequence*, i.e.*

$$0 \longrightarrow N \xrightarrow{f} N \oplus P \xrightarrow{g} P \longrightarrow 0$$

where $f : N \to N \oplus P$ is given by $n \mapsto (n, 0)$, and $g : N \oplus P \to P$ is given by $(n, p) \mapsto p$.

Definition 7.2.6 (Definition 2.18). *A short exact sequence*

$$0 \longrightarrow N \xrightarrow{u} M \xrightarrow{v} P \longrightarrow 0$$

is called a splitting short exact sequence *if there exists an R-module isomorphism*

$$\theta : M \to N \oplus P$$

such that the following diagram commutes:

$$\begin{tikzcd} 0 \arrow[r] & N \arrow[r, "u"] \arrow[d, equal] & M \arrow[r, "v"] \arrow[d, "\theta"] & P \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & N \arrow[r, "f"] & N \oplus P \arrow[r, "g"] & P \arrow[r] & 0 \end{tikzcd}$$

An intuition is that u and v behave like the standard embedding f and projection g, respectively.

Theorem 7.2.7 (Theorem 2.9). *Consider the following short exact sequence:*

$$0 \longrightarrow N \xrightarrow{u} M \xrightarrow{v} P \longrightarrow 0$$

Then the following are equivalent:

1. *There exists $u' : M \to N$ such that $u' \circ u = \mathrm{id}_N$.*
2. *There exists $v' : P \to M$ such that $v \circ v' = \mathrm{id}_P$.*
3. *There exists an isomorphism $\theta : M \to N \oplus P$ such that the diagram*

$$\begin{tikzcd} 0 \arrow[r] & N \arrow[r, "u"] \arrow[d, equal] & M \arrow[r, "v"] \arrow[d, "\theta"] & P \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & N \arrow[r, "f"] & N \oplus P \arrow[r, "g"] & P \arrow[r] & 0 \end{tikzcd}$$

   commutes.

In other words, the short exact sequence splits.

Proof. (1) $\Rightarrow$ (3): Define $\theta : M \to N \oplus P$ by

$$\theta(m) = (u'(m), v(m)).$$

Then the diagram commutes. We show θ is an isomorphism.

Injective: Suppose $\theta(m) = (0,0)$. Then $v(m) = 0$, so $m \in \operatorname{Im}(u)$, i.e. $m = u(n)$ for some $n \in N$. Hence

$$0 = u'(m) = u'(u(n)) = n,$$

so $m = u(n) = 0$.

Surjective: Let $(n,p) \in N \oplus P$. Since v is surjective, there exists $m \in M$ such that $v(m) = p$. Then

$$v(m + u(x)) = v(m) = p \quad \text{for all } x \in N.$$

Take $x = n - u'(m)$. Then

$$u'(m + u(x)) = u'(m) + x = n,$$

so

$$\theta(m + u(x)) = (n,p).$$

Thus θ is surjective.

(3) $\Rightarrow$ (1): Since θ is an isomorphism and the diagram commutes, write

$$\theta(m) = (u'(m), v(m)).$$

Then for $n \in N$,

$$\theta(u(n)) = (n,0),$$

so $u'(u(n)) = n$, hence $u' \circ u = \mathrm{id}_N$.

(3) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3) are similar. □

Back to the proof of Weyl Theorem:

Proof. What we want to show is $\forall W \subset V$ as submodule, the short exact sequence

$$0 \longrightarrow W \longrightarrow V \longrightarrow V/W \longrightarrow 0$$

splits. We begin with very special case, and step by step, we will prove the general case.

Step 1: At first we prove if W is irreducible with codimension 1 in V, then the sequence splits. i.e. W has a direct sum complement in V.

Let $\phi : L \to \mathfrak{gl}(V)$. Take the casimir element C_ϕ as before. Since C_ϕ commutes with $\phi(L)$, $\ker C_\phi$ is an L-submodule of V.

Since W is an L-submodule, we have $C_\phi(W) \subseteq W$.

Since W is of codimension 1, L acts trivially on V/W, we have C_ϕ acts trivially on V/W. Therefore $C_\phi|_{V/W} = 0$.

By Schur lemma, $C_\phi|_W = \lambda \cdot 1_W$. i.e. $\operatorname{Tr}(C_\phi|_W) = \lambda \cdot \dim W$. Recall $\operatorname{Tr} C_\phi = \dim L$ and $C_\phi|_{V/W} = 0$, we have $\lambda = \dim L / \dim W > 0$. Thus C_ϕ acts on W as $\operatorname{diag}\{\lambda, \ldots, \lambda\}$ where $\lambda > 0$. Thus $C_\phi(W) = W$, $C_\phi|_{V/W} = 0$. Take $\frac{1}{\lambda} \ker C_\phi$ then $\frac{1}{\lambda} \ker C_\phi$ is an L-submodule.

Now consider $\forall v \in V$, $v = v - \frac{1}{\lambda}C_\phi(v) + \frac{1}{\lambda}C_\phi(v)$. Remark, since C_ϕ acts trivially on V/W we have $C_\phi(V) \subseteq W$. Since $C_\phi\left(v - \frac{1}{\lambda}C_\phi(v)\right) = C_\phi(v) - \frac{1}{\lambda}C_\phi(C_\phi(v)) = C_\phi(v) - \frac{1}{\lambda}\cdot\lambda C_\phi(v) = 0$, $v - \frac{1}{\lambda}C_\phi(v) \in \ker C_\phi$. Thus $V = \frac{1}{\lambda}\ker C_\phi + \operatorname{im} C_\phi = \frac{1}{\lambda}\ker C_\phi + W$.

Finally, if $\exists v_0$ such that $v_0 \in \ker C_\phi \cap \operatorname{im} C_\phi$. Then $C_\phi(v_0) = \lambda v_0 = 0$ shows $v_0 = 0$. So $V = \frac{1}{\lambda}\ker C_\phi \oplus W$.

Step 2: If W is not irreducible. Then we can find a non-trivial irreducible submodule W_1 such that $0 \neq W_1 \subset W$. Take

$$0 \longrightarrow W/W_1 \longrightarrow V/W_1 \longrightarrow \frac{V/W_1}{W/W_1} \longrightarrow 0$$

Since $\operatorname{codim}_V W = 1$, we have $\operatorname{codim}_{V/W_1} W/W_1 = 1$. And we can induction on $\dim W$.

As $0 \neq W_1 \subset W$, $\dim W/W_1 < \dim W$. We can assume

$$0 \longrightarrow W/W_1 \longrightarrow V/W_1 \longrightarrow \frac{V/W_1}{W/W_1} \longrightarrow 0$$

splits. i.e. $V/W_1 = W/W_1 \oplus \tilde{W}/W_1$ for some $\tilde{W}$ and $\dim \tilde{W}/W_1 = 1$.

Now consider

$$0 \longrightarrow W_1 \longrightarrow \tilde{W} \longrightarrow \tilde{W}/W_1 \longrightarrow 0$$

Since $\dim W_1 < \dim W$, by induction $\exists X$ such that $\tilde{W} = W_1 \oplus X$ where $\dim X = 1$. Write $X = \operatorname{span}\{x\}$. $\forall v \in V, v + W_1 = w + W_1 + \tilde{w} + W_1 = w + W_1 + w_1 + kx + W_1 = w + kx + W_1 = kx + W$ where $k \in F$. In other words, $V = W \oplus X$ this is a direct sum since $W, X \subset V$ and $W \cap X = 0$.

Step 3: Take V as an arbitrary L-module (reducible) and $W \subset V$ non-trivial submodule. Consider $\operatorname{Hom}_F(V, W)$. Remark, $\operatorname{Hom}_F(V, W)$ is the set of linear maps but not L-module homomorphisms. $\operatorname{Hom}_F(V, W)$ itself is an L-module.

Define $\mathcal{V} \subset \operatorname{Hom}_F(V, W)$, $\mathcal{V} = \{f \in \operatorname{Hom}_F(V, W) \mid f|_W = a \cdot 1_W, a \in F\}$. Also define $\mathcal{W} = \{g \in \operatorname{Hom}_F(V, W) \mid g|_W = 0\}$. Then $\mathcal{V}, \mathcal{W}$ are L-modules and $\mathcal{W} \subset \mathcal{V}$.

Actually, $\forall f \in \mathcal{V}, v \in V, x \in L$, $(x \cdot f)(v) = x \cdot f(v) - f(x \cdot v) = x \cdot (av) - a(x \cdot v) = 0$, i.e. $x \cdot (f(v)) = f(x \cdot v)$. So f is an L-module homomorphism. Besides, it is obvious that $\operatorname{codim}_{\mathcal{V}} \mathcal{W} = 1$.

So we have

$$0 \longrightarrow \mathcal{W} \longrightarrow \mathcal{V} \longrightarrow \mathcal{V}/\mathcal{W} \longrightarrow 0$$

splits. i.e. $\mathcal{V} = \mathcal{W} \oplus \tilde{\mathcal{V}}$ for some $\tilde{\mathcal{V}}$ and $\tilde{\mathcal{V}} = \operatorname{span}\{f\}$ for some $f \in \operatorname{Hom}_F(V, W)$ where $f|_W = a \cdot 1_W$ for some $0 \neq a \in F$.

Now take $\frac{1}{a}f$ consider the origin short exact sequence

$$0 \longrightarrow W \underset{\frac{1}{a}f}{\overset{i}{\rightleftarrows}} V \xrightarrow{\pi} V/W \longrightarrow 0$$

$\frac{1}{a}f \circ i = \mathrm{id}_W$ shows this short exact sequence splits.

$\square$

Chapter 8

Lecture 8: $\mathfrak{sl}_2$-modules

Remark 8.0.1 (Maschke's theorem)**.** *Let G be a finite group, and let K be a field such that*

$$\mathrm{char}(K) \nmid |G|.$$

Then every $K[G]$-module is completely reducible.

Idea of the proof. It suffices to show that every submodule W of a $K[G]$-module V is a direct summand.

Choose a K-linear projection

$$P\colon V \to W$$

(which need not commute with the action of G). Averaging over the group, define

$$P' = \frac{1}{|G|} \sum_{g \in G} g P g^{-1}.$$

Then P' commutes with the action of G. Moreover, P' still maps V into W, and for every $w \in W$ one has $P'(w) = w$. Hence

$$\mathrm{Im}(P') = W.$$

Therefore $\ker P'$ is a G-submodule and

$$V = W \oplus \ker P'.$$

So W is a direct summand. It follows that V is completely reducible. □

Remark 8.0.2 (Relation with Weyl's theorem)**.** *Weyl's original proof of complete reducibility uses the same averaging idea for representations of compact Lie groups, replacing the finite sum by an integral over the group (with respect to Haar measure).*

8.1 $\mathfrak{sl}_2$-modules

Recall the standard $\mathfrak{sl}_2$-triple

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

satisfying

$$[e, f] = h, \qquad [h, e] = 2e, \qquad [h, f] = -2f.$$

Theorem 8.1.1. *Let F be an algebraically closed field of characteristic 0, and let V be a finite-dimensional irreducible $\mathfrak{sl}_2(F)$-module. Then there exist*

$$m \in \mathbb{Z}_{\geq 0}, \qquad 0 \neq v \in V$$

such that

$$hv = mv, \qquad ev = 0,$$

and

$$\{v, fv, f^2v, \ldots, f^m v\}$$

is a basis of V. In particular,

$$\dim V = m + 1,$$

and the weights of h on V are

$$m, m-2, m-4, \ldots, -m.$$

Moreover, for $0 \leq i \leq m$,

$$h(f^i v) = (m - 2i) f^i v,$$

and for $1 \leq i \leq m$,

$$e(f^i v) = i(m - i + 1) f^{i-1} v.$$

Proof. Choose an eigenvector $0 \neq u \in V$ of h, say

$$hu = \alpha u.$$

Then

$$h(eu) = [h, e]u + e(hu) = 2eu + \alpha eu = (\alpha + 2)eu,$$

and similarly

$$h(fu) = [h, f]u + f(hu) = -2fu + \alpha fu = (\alpha - 2)fu.$$

Thus e raises weights by 2, and f lowers weights by 2.

Since V is finite-dimensional, the vectors

$$u, eu, e^2u, \ldots$$

cannot all be nonzero. Let $r \geq 0$ be minimal such that

$$e^{r+1}u = 0,$$

and set

$$v = e^r u.$$

Then $v \neq 0$, $ev = 0$, and v is again an eigenvector of h; write

$$hv = \lambda v.$$

Now consider the subspace

$$W = \operatorname{span}\{v, fv, f^2 v, \dots\}.$$

It is stable under f by definition, and it is stable under h because

$$h(f^i v) = (\lambda - 2i) f^i v.$$

We claim that W is also stable under e.

For $i \geq 1$,

$$ef^i v = [e,f] f^{i-1} v + fef^{i-1} v = hf^{i-1} v + fef^{i-1} v.$$

Iterating this identity, we obtain

$$ef^i v = \sum_{j=0}^{i-1} f^j h f^{i-1-j} v.$$

Since

$$hf^{i-1-j} v = (\lambda - 2(i-1-j)) f^{i-1-j} v,$$

it follows that

$$ef^i v = \sum_{j=0}^{i-1} (\lambda - 2(i-1-j)) f^{i-1} v = i(\lambda - i + 1) f^{i-1} v.$$

Hence $eW \subseteq W$, so W is a nonzero submodule of V. By irreducibility of V, we have

$$W = V.$$

Now let $m \geq 0$ satisfy

$$f^m v \neq 0, \qquad f^{m+1} v = 0.$$

Then

$$V = \operatorname{span}\{v, fv, \dots, f^m v\}.$$

These vectors are linearly independent, since they are eigenvectors of h with distinct eigenvalues

$$\lambda, \lambda - 2, \ldots, \lambda - 2m.$$

Thus $\{v, fv, \ldots, f^m v\}$ is a basis of V.

Finally, applying the formula for $ef^i v$ with $i = m + 1$, we get

$$0 = ef^{m+1}v = (m + 1)(\lambda - m)f^m v.$$

Since $f^m v \neq 0$, we must have

$$\lambda = m.$$

Therefore

$$hv = mv, \qquad ev = 0,$$

and the weights are exactly

$$m, m - 2, \ldots, -m.$$

□

Corollary 8.1.2. *For each $m \in \mathbb{Z}_{\geq 0}$, there exists, up to isomorphism, a unique irreducible $\mathfrak{sl}_2(F)$-module of dimension $m + 1$.*

Proof. By Theorem 8.1.1, any such module admits a basis

$$v, fv, \ldots, f^m v$$

with

$$hv = mv, \qquad ev = 0,$$

and

$$h(f^i v) = (m - 2i)f^i v, \qquad e(f^i v) = i(m - i + 1)f^{i-1}v, \qquad f(f^i v) = f^{i+1}v.$$

Thus the action is completely determined by m. □

Remark 8.1.3. *Once complete reducibility of finite-dimensional $\mathfrak{sl}_2(F)$-modules is known, one obtains the following consequences.*

Corollary 8.1.4. *Let V be a finite-dimensional $\mathfrak{sl}_2(F)$-module. Then every eigenvalue of h on V is an integer.*

Proof. Decompose V as a direct sum of irreducible submodules. By Theorem 8.1.1, each irreducible summand has highest weight $m \in \mathbb{Z}_{\geq 0}$, hence all of its weights are

$$m, m - 2, \ldots, -m,$$

which are integers. □

Proposition 8.1.5. *Let V be a finite-dimensional $\mathfrak{sl}_2(F)$-module, and let $0 \neq w \in V$ be an eigenvector of h with eigenvalue α. Suppose*

$$f^{q+1}w = 0, \qquad f^q w \neq 0,$$

and

$$e^{p+1}w = 0, \qquad e^p w \neq 0.$$

Then

$$\alpha = q - p.$$

Proof. Write V as a direct sum of irreducible submodules:

$$V = \bigoplus_j V_j.$$

Then

$$w = \sum_j w_j, \qquad w_j \in V_j,$$

where each nonzero w_j is again an eigenvector of h of eigenvalue α.

Fix a nonzero w_j. By Theorem 8.1.1, there exists a highest weight vector $v_j \in V_j$ and an integer $i_j \geq 0$ such that

$$w_j = f^{i_j} v_j.$$

If the highest weight of V_j is m_j, then

$$\alpha = m_j - 2i_j.$$

Set

$$p_j = i_j, \qquad q_j = m_j - i_j.$$

Then

$$\alpha = q_j - p_j.$$

Because the decomposition is direct, the integers p and q attached to w satisfy

$$p = \max_j p_j, \qquad q = \max_j q_j.$$

Since $q_j - p_j = \alpha$ for every nonzero w_j, we get

$$q = \max_j(\alpha + p_j) = \alpha + \max_j p_j = \alpha + p,$$

hence

$$\alpha = q - p.$$

$\square$

Proposition 8.1.6. *For every $m \in \mathbb{Z}_{\geq 0}$, there exists an irreducible $\mathfrak{sl}_2(F)$-module of dimension $m+1$.*

Proof. Consider the polynomial algebra $F[x,y]$. Define operators on $F[x,y]$ by

$$e = y\frac{\partial}{\partial x}, \qquad f = x\frac{\partial}{\partial y}, \qquad h = [e,f] = y\frac{\partial}{\partial y} - x\frac{\partial}{\partial x}.$$

A direct computation shows that

$$[e,f] = h, \qquad [h,e] = 2e, \qquad [h,f] = -2f,$$

so this defines an $\mathfrak{sl}_2(F)$-action on $F[x,y]$.

Fix $m \geq 0$, and let

$$V_m = \operatorname{span}_F\{x^i y^j \mid i+j = m\}.$$

Then V_m is stable under e, f, h, so V_m is an $\mathfrak{sl}_2(F)$-submodule. Clearly,

$$\dim V_m = m+1.$$

Take

$$v = y^m.$$

Then

$$ev = 0, \qquad hv = my^m = mv,$$

so v is a highest weight vector of weight m. Moreover,

$$f^i(v) = m(m-1)\cdots(m-i+1)x^i y^{m-i} \qquad (0 \leq i \leq m),$$

so the vectors

$$v, fv, \ldots, f^m v$$

span V_m. Hence, by Corollary 8.1.2, V_m is the irreducible module of highest weight m. $\square$

Chapter 9

Lecture 9: Root space decomposition

Throughout, let F be an algebraically closed field of characteristic 0.

9.1 Jordan decomposition in a semisimple linear Lie algebra

Theorem 9.1.1. *Let $L \subseteq \mathfrak{gl}(V)$ be a semisimple Lie algebra, where V is finite-dimensional. Then for every $a \in L$, the semisimple and nilpotent parts of a in $\mathfrak{gl}(V)$ both lie in L. Hence the abstract Jordan decomposition in L coincides with the usual Jordan decomposition in $\mathfrak{gl}(V)$.*

Proof. Take $a \in L$, and write its usual Jordan decomposition in $\mathfrak{gl}(V)$ as

$$a = a_s + a_n, \qquad [a_s, a_n] = 0,$$

where a_s is semisimple and a_n is nilpotent.

Then

$$\mathrm{ad}_{\mathfrak{gl}(V)}(a) = \mathrm{ad}_{\mathfrak{gl}(V)}(a_s) + \mathrm{ad}_{\mathfrak{gl}(V)}(a_n)$$

is the Jordan decomposition of $\mathrm{ad}_{\mathfrak{gl}(V)}(a)$ in $\mathfrak{gl}(\mathfrak{gl}(V))$. In particular, $\mathrm{ad}_{\mathfrak{gl}(V)}(a_s)$ is a polynomial in $\mathrm{ad}_{\mathfrak{gl}(V)}(a)$ with no constant term. Since $\mathrm{ad}_{\mathfrak{gl}(V)}(a)$ preserves L, it follows that

$$[a_s, L] \subseteq L.$$

Thus $\mathrm{ad}(a_s)|_L$ is a derivation of L.

Because L is semisimple, every derivation of L is inner. Hence there exists $a_s' \in L$ such that

$$[a_s, x] = [a_s', x] \qquad \text{for all } x \in L.$$

Therefore

$$[a_s - a_s', L] = 0.$$

Now decompose the L-module V as

$$V = \bigoplus_i V_i,$$

where each V_i is irreducible. Since $a_s - a_s'$ commutes with the action of L, Schur's lemma shows that $a_s - a_s'$ acts by a scalar on each V_i.

On the other hand, $L = [L, L]$, so for every $x \in L$ and every i,

$$\operatorname{tr}_{V_i}(x) = 0.$$

Applying this to $x = a - a_s' \in L$, we get

$$\operatorname{tr}_{V_i}(a - a_s') = 0.$$

Since a_n is nilpotent, $\operatorname{tr}_{V_i}(a_n) = 0$, hence

$$\operatorname{tr}_{V_i}(a_s - a_s') = 0.$$

But $a_s - a_s'$ is scalar on V_i, so that scalar must be 0. Thus $a_s - a_s'$ acts trivially on every V_i, whence

$$a_s = a_s' \in L.$$

Finally,

$$a_n = a - a_s \in L.$$

So both parts lie in L, as required. □

Corollary 9.1.2. *Let L be a semisimple Lie algebra and $\varphi\colon L \to \mathfrak{gl}(V)$ a finite-dimensional representation. If*

$$x = s + n$$

is the abstract Jordan decomposition of $x \in L$, then

$$\varphi(x) = \varphi(s) + \varphi(n)$$

is the usual Jordan decomposition of $\varphi(x)$ in $\mathfrak{gl}(V)$.

Proof. It is enough to show that

$$\operatorname{ad}_{\varphi(L)}(\varphi(x)) = \operatorname{ad}_{\varphi(L)}(\varphi(s)) + \operatorname{ad}_{\varphi(L)}(\varphi(n))$$

is the abstract Jordan decomposition of $\operatorname{ad}_{\varphi(L)}(\varphi(x))$.

Since $\operatorname{ad}_L(s)$ is semisimple, L is spanned by its eigenvectors. Applying φ, we see that $\varphi(L)$ is spanned by eigenvectors of $\operatorname{ad}_{\varphi(L)}(\varphi(s))$. Hence $\operatorname{ad}_{\varphi(L)}(\varphi(s))$ is semisimple.

Since $\operatorname{ad}_L(n)$ is nilpotent, its induced action on the quotient $\varphi(L) \cong L/\ker(\varphi)$ is nilpotent. Hence $\operatorname{ad}_{\varphi(L)}(\varphi(n))$ is nilpotent.

Also, because $[s, n] = 0$,

$$\left[\mathrm{ad}_{\varphi(L)}(\varphi(s)), \mathrm{ad}_{\varphi(L)}(\varphi(n))\right] = 0.$$

Thus the displayed decomposition is the abstract Jordan decomposition in the semisimple linear Lie algebra $\varphi(L)$. By the theorem above, it is also the usual Jordan decomposition in $\mathfrak{gl}(V)$. $\square$

9.2 Root space decomposition

Definition 9.2.1. *A Lie algebra L is called* toral *if $\mathrm{ad}(a)$ is semisimple for every $a \in L$.*

Lemma 9.2.2. *Any toral Lie algebra is abelian.*

Proof. Fix $0 \neq a \in L$. Since $\mathrm{ad}\, a$ is semisimple,

$$L = \bigoplus_{\lambda} L_\lambda(a), \qquad L_\lambda(a) = \{x \in L \mid [a, x] = \lambda x\}.$$

We claim that $L_\alpha(a) = 0$ for all $\alpha \neq 0$.

Suppose instead that $L_\alpha(a) \neq 0$ for some $\alpha \neq 0$, and choose $0 \neq b \in L_\alpha(a)$. If $x \in L_\beta(a)$, then the Jacobi identity gives

$$[a, [b, x]] = [[a, b], x] + [b, [a, x]] = \alpha[b, x] + \beta[b, x] = (\alpha + \beta)[b, x].$$

Hence

$$[b, L_\beta(a)] \subseteq L_{\alpha+\beta}(a).$$

Thus $\mathrm{ad}\, b$ carries each eigenspace of $\mathrm{ad}\, a$ into the next one shifted by α. Since $\mathrm{ad}\, a$ has only finitely many eigenvalues, it follows that $\mathrm{ad}\, b$ is nilpotent.

On the other hand, L is toral, so $\mathrm{ad}\, b$ is semisimple. Hence $\mathrm{ad}\, b = 0$. But

$$(\mathrm{ad}\, b)(a) = [b, a] = -\alpha b \neq 0,$$

a contradiction. Therefore $L_\alpha(a) = 0$ whenever $\alpha \neq 0$.

So $L = L_0(a)$, i.e.

$$[a, x] = 0 \qquad \text{for all } x \in L.$$

Since a was arbitrary, L is abelian. $\square$

Now let L be semisimple, and let H be a maximal toral subalgebra of L. Since H is toral, it is abelian by the lemma above. Hence the commuting semisimple endomorphisms $\mathrm{ad}(h)$ $(h \in H)$ are simultaneously diagonalizable, and we obtain a decomposition

$$L = L_0 \oplus \bigoplus_{\alpha \in \Delta} L_\alpha,$$

where

$$L_\alpha = \{x \in L \mid [h, x] = \alpha(h)x \text{ for all } h \in H\}, \qquad \alpha \in H^*,$$

and $\Delta \subseteq H^* \setminus \{0\}$ is the set of nonzero weights.

Lemma 9.2.3. *For $\alpha, \beta \in \Delta \cup \{0\}$ one has*

$$[L_\alpha, L_\beta] \subseteq L_{\alpha+\beta},$$

where by convention $L_{\alpha+\beta} = 0$ if $\alpha + \beta \notin \Delta \cup \{0\}$.

Proof. Take $x_\alpha \in L_\alpha$, $y_\beta \in L_\beta$, and $h \in H$. Then

$$[h, [x_\alpha, y_\beta]] = [[h, x_\alpha], y_\beta] + [x_\alpha, [h, y_\beta]] = \alpha(h)[x_\alpha, y_\beta] + \beta(h)[x_\alpha, y_\beta].$$

So

$$[h, [x_\alpha, y_\beta]] = (\alpha + \beta)(h)[x_\alpha, y_\beta].$$

Hence $[x_\alpha, y_\beta] \in L_{\alpha+\beta}$. □

Let $(\cdot \mid \cdot)$ denote the Killing form of L.

Lemma 9.2.4. *For any $\alpha, \beta \in \Delta \cup \{0\}$, if $\alpha + \beta \neq 0$, then*

$$(L_\alpha \mid L_\beta) = 0.$$

Proof. Choose $h \in H$ such that $(\alpha + \beta)(h) \neq 0$. Let $x_\alpha \in L_\alpha$ and $y_\beta \in L_\beta$. Then

$$\alpha(h)(x_\alpha \mid y_\beta) = ([h, x_\alpha] \mid y_\beta) = (h \mid [x_\alpha, y_\beta]),$$

while

$$\beta(h)(x_\alpha \mid y_\beta) = (x_\alpha \mid [h, y_\beta]) = -([h, x_\alpha] \mid y_\beta).$$

Hence

$$(\alpha(h) + \beta(h))(x_\alpha \mid y_\beta) = 0.$$

Since $(\alpha + \beta)(h) \neq 0$, we obtain

$$(x_\alpha \mid y_\beta) = 0.$$

□

Lemma 9.2.5.

$$L_0 = H.$$

Proof. First note that the restriction of the Killing form to L_0 is nondegenerate. Indeed, if $x \in L_0$ and $(x \mid L_0) = 0$, then by the previous lemma

$$(x \mid L_\alpha) = 0 \qquad (\alpha \neq 0),$$

so x is orthogonal to all of L. Since the Killing form on L is nondegenerate, $x = 0$.

Now let $a \in L_0$, and write its abstract Jordan decomposition in L as

$$a = a_s + a_n.$$

Because $\mathrm{ad}(a)$ commutes with every $\mathrm{ad}(h)$ $(h \in H)$, the same is true of its semisimple and nilpotent parts. Hence

$$a_s, a_n \in L_0.$$

Since a_s is semisimple and centralizes H, the subalgebra

$$H + F a_s$$

is toral. By maximality of H, we must have $a_s \in H$.

We claim that $a_n = 0$. To prove this, it is enough — by the nondegeneracy of $(\cdot \mid \cdot)$ on L_0 — to show that

$$(a_n \mid L_0) = 0.$$

First observe that L_0 is solvable. Indeed, if $u \in [L_0, L_0]$, then $u \in L_0$, so its semisimple part u_s lies in H. On the other hand, since $u = [x, y]$ with $x, y \in L_0$ and H centralizes L_0, one sees from the Killing-form computation that u_s acts trivially on L; hence $u_s = 0$. Therefore every element of $[L_0, L_0]$ is nilpotent, so Engel's theorem implies that $[L_0, L_0]$ is nilpotent. Thus L_0 is solvable.

By Lie's theorem, there exists a basis of L with respect to which all operators $\mathrm{ad}(x)$ $(x \in L_0)$ are upper triangular. Since a_n is nilpotent, $\mathrm{ad}(a_n)$ is strictly upper triangular. Therefore, for every $x \in L_0$,

$$(a_n \mid x) = \mathrm{tr}\big(\mathrm{ad}(a_n)\,\mathrm{ad}(x)\big) = 0.$$

Hence $(a_n \mid L_0) = 0$, and so $a_n = 0$.

Thus $a = a_s \in H$. Since $a \in L_0$ was arbitrary, we have $L_0 \subseteq H$. The reverse inclusion is clear, so $L_0 = H$. ☐

Corollary 9.2.6. *The restriction of the Killing form to H is nondegenerate.*

Proof. This follows immediately from the lemma $L_0 = H$ and the nondegeneracy of the Killing form on L_0. ☐

Corollary 9.2.7. *For every $\alpha \in \Delta$, the pairing*

$$L_\alpha \times L_{-\alpha} \to F, \qquad (x, y) \mapsto (x \mid y),$$

is nondegenerate. In particular, $L_{-\alpha} \neq 0$.

Proof. Let $0 \neq x \in L_\alpha$ and assume $(x \mid L_{-\alpha}) = 0$. By the orthogonality lemma, x is also orthogonal to $H = L_0$ and to every L_β with $\beta \neq -\alpha$. Hence x is orthogonal to all of L, contradicting the nondegeneracy of the Killing form. ☐

We may therefore identify H with H^* via the Killing form:

$$\nu\colon H \longrightarrow H^*, \qquad h \longmapsto (h \mid \cdot)|_H.$$

Thus

$$\nu(h)(y) = (h \mid y) \qquad \text{for all } y \in H.$$

Let

$$\nu^{-1}\colon H^* \longrightarrow H$$

be its inverse. For $\alpha, \beta \in H^*$ define

$$(\alpha \mid \beta) := \big(\nu^{-1}(\alpha) \mid \nu^{-1}(\beta)\big).$$

Lemma 9.2.8. *For $x_\alpha \in L_\alpha$ and $y_{-\alpha} \in L_{-\alpha}$,*

$$[x_\alpha, y_{-\alpha}] = (x_\alpha \mid y_{-\alpha})\, \nu^{-1}(\alpha).$$

Proof. Since $[L_\alpha, L_{-\alpha}] \subseteq L_0 = H$, both sides lie in H. Let $h \in H$. Then

$$([h, x_\alpha] \mid y_{-\alpha}) = (h \mid [x_\alpha, y_{-\alpha}]).$$

But

$$[h, x_\alpha] = \alpha(h) x_\alpha,$$

so

$$\alpha(h)(x_\alpha \mid y_{-\alpha}) = (h \mid [x_\alpha, y_{-\alpha}]).$$

By definition of $\nu^{-1}(\alpha)$,

$$\alpha(h) = (\nu^{-1}(\alpha) \mid h).$$

Hence

$$(h \mid [x_\alpha, y_{-\alpha}]) = \big(h \mid (x_\alpha \mid y_{-\alpha})\, \nu^{-1}(\alpha)\big).$$

Since the Killing form on H is nondegenerate, we conclude that

$$[x_\alpha, y_{-\alpha}] = (x_\alpha \mid y_{-\alpha})\, \nu^{-1}(\alpha).$$

□

Lemma 9.2.9. *For every $\alpha \in \Delta$,*

$$(\alpha \mid \alpha) \neq 0.$$

Proof. Choose $0 \neq x_\alpha \in L_\alpha$. By the preceding corollary, there exists $y_{-\alpha} \in L_{-\alpha}$ such that

$$(x_\alpha \mid y_{-\alpha}) \neq 0.$$

Hence, by the previous lemma,

$$[x_\alpha, y_{-\alpha}] = (x_\alpha \mid y_{-\alpha})\, \nu^{-1}(\alpha) \neq 0.$$

Suppose $(\alpha \mid \alpha) = 0$. Then

$$\alpha\big(\nu^{-1}(\alpha)\big) = \big(\nu^{-1}(\alpha) \mid \nu^{-1}(\alpha)\big) = (\alpha \mid \alpha) = 0.$$

Set

$$L' := Fx_\alpha \oplus Fy_{-\alpha} \oplus F\nu^{-1}(\alpha).$$

Then

$$[\nu^{-1}(\alpha), x_\alpha] = \alpha(\nu^{-1}(\alpha))x_\alpha = 0,$$

and similarly

$$[\nu^{-1}(\alpha), y_{-\alpha}] = 0.$$

Also

$$[x_\alpha, y_{-\alpha}] \in F\nu^{-1}(\alpha).$$

Thus L' is a solvable subalgebra.

Consider the adjoint representation of L' on L. By Lie's theorem, the operators $\mathrm{ad}(L')$ are simultaneously upper triangularizable. Since

$$\nu^{-1}(\alpha) \in [L', L'],$$

its adjoint action is strictly upper triangular, hence nilpotent. But $\nu^{-1}(\alpha) \in H$, so $\mathrm{ad}(\nu^{-1}(\alpha))$ is semisimple. Therefore it must be 0. This implies $\nu^{-1}(\alpha) = 0$, hence $\alpha = 0$, a contradiction.

Therefore $(\alpha \mid \alpha) \neq 0$. □

Proposition 9.2.10. *For every root $\alpha \in \Delta$, one may choose root vectors*

$$x_\alpha \in L_\alpha, \qquad y_{-\alpha} \in L_{-\alpha},$$

such that if

$$h_\alpha := \frac{2}{(\alpha \mid \alpha)}\, \nu^{-1}(\alpha),$$

then

$$[x_\alpha, y_{-\alpha}] = h_\alpha, \qquad [h_\alpha, x_\alpha] = 2x_\alpha, \qquad [h_\alpha, y_{-\alpha}] = -2y_{-\alpha}.$$

Proof. Choose $0 \neq x_\alpha \in L_\alpha$. By the nondegeneracy of the pairing $L_\alpha \times L_{-\alpha}$, choose $y_{-\alpha} \in L_{-\alpha}$ so that

$$(x_\alpha \mid y_{-\alpha}) = \frac{2}{(\alpha \mid \alpha)}.$$

Then the preceding lemma gives

$$[x_\alpha, y_{-\alpha}] = \frac{2}{(\alpha \mid \alpha)}\, \nu^{-1}(\alpha) = h_\alpha.$$

Now

$$[h_\alpha, x_\alpha] = \alpha(h_\alpha)x_\alpha = \frac{2}{(\alpha \mid \alpha)}\alpha(\nu^{-1}(\alpha))x_\alpha = \frac{2}{(\alpha \mid \alpha)}(\alpha \mid \alpha)x_\alpha = 2x_\alpha.$$

Similarly,

$$[h_\alpha, y_{-\alpha}] = -\alpha(h_\alpha)y_{-\alpha} = -2y_{-\alpha}.$$

□

Chapter 10

Lecture 10

Proposition 10.0.1. *Let L be a finite-dimensional semisimple Lie algebra over an algebraically closed field F of characteristic 0. Let*

$$L = H \oplus \bigoplus_{\alpha \in \Delta} L_\alpha$$

be the root space decomposition relative to a Cartan subalgebra H. For each $\alpha \in \Delta$, choose root vectors

$$0 \neq x_\alpha \in L_\alpha, \qquad 0 \neq y_\alpha \in L_{-\alpha},$$

and set

$$h_\alpha = [x_\alpha, y_\alpha], \qquad \alpha(h_\alpha) = 2.$$

Then

$$\mathfrak{s}_\alpha := Fx_\alpha \oplus Fh_\alpha \oplus Fy_\alpha \cong \mathfrak{sl}_2(F).$$

Moreover:

1. $$F\alpha \cap \Delta = \{-\alpha, \alpha\}, \qquad \dim L_\alpha = 1.$$

2. *If $\alpha, \beta \in \Delta$ and $\beta \neq \pm\alpha$, then*

$$\frac{2(\alpha,\beta)}{(\alpha,\alpha)} \in \mathbb{Z}.$$

Furthermore, there exist $p, q \in \mathbb{Z}_{\geq 0}$ such that the roots of the form $\beta + i\alpha$ are precisely

$$\beta - q\alpha,\ \beta - (q-1)\alpha,\ \ldots,\ \beta - \alpha,\ \beta,\ \beta + \alpha,\ \ldots,\ \beta + p\alpha,$$

and

$$q - p = \frac{2(\alpha,\beta)}{(\alpha,\alpha)}.$$

In particular, if $\alpha + \beta \in \Delta$, then

$$[L_\alpha, L_\beta] = L_{\alpha+\beta}.$$

Proof. We regard L as an $\mathfrak{s}_\alpha$-module via the adjoint representation.

First consider

$$M_\alpha := \bigoplus_{c \in F} L_{c\alpha},$$

where $L_0 = H$, and where $L_{c\alpha} = 0$ if $c\alpha \notin \Delta \cup \{0\}$. This is an $\mathfrak{s}_\alpha$-submodule of L. If $x \in L_{c\alpha}$, then

$$[h_\alpha, x] = c\alpha(h_\alpha)x = 2c\,x.$$

Hence $L_{c\alpha}$ is the h_α-weight space of weight $2c$. By the finite-dimensional representation theory of $\mathfrak{sl}_2$, all weights are integers. Therefore, whenever $L_{c\alpha} \neq 0$, we have

$$2c \in \mathbb{Z}.$$

Now consider the submodule

$$M_{\text{even}} := \bigoplus_{i \in \mathbb{Z}} L_{i\alpha}.$$

All its h_α-weights are even. Since every irreducible $\mathfrak{sl}_2$-module with even weights contains weight 0, each irreducible submodule S of M_{even} satisfies

$$S \cap H \neq 0.$$

Choose $0 \neq h \in S \cap H$.

If $h \in \operatorname{Ker} \alpha$, then

$$[x_\alpha, h] = [y_\alpha, h] = [h_\alpha, h] = 0,$$

so Fh is a trivial $\mathfrak{s}_\alpha$-submodule. Since S is irreducible, it follows that

$$S = Fh.$$

If $h \notin \operatorname{Ker} \alpha$, then

$$[x_\alpha, h] = -\alpha(h)x_\alpha \neq 0, \qquad [y_\alpha, h] = \alpha(h)y_\alpha \neq 0.$$

Thus

$$x_\alpha, y_\alpha \in S.$$

Consequently

$$h_\alpha = [x_\alpha, y_\alpha] \in S,$$

and hence

$$\mathfrak{s}_\alpha \subseteq S.$$

Since $\mathfrak{s}_\alpha$ is the irreducible adjoint module for itself, we get

$$S = \mathfrak{s}_\alpha.$$

Therefore

$$M_{\text{even}} = H + \mathfrak{s}_\alpha.$$

It follows in particular that

$$2\alpha \notin \Delta \qquad \text{and} \qquad \dim L_\alpha = 1.$$

Applying this to the root $\frac{1}{2}\alpha$, if it existed, would imply $\alpha \notin \Delta$, a contradiction. Hence

$$\frac{1}{2}\alpha \notin \Delta.$$

Next consider the submodule

$$M_{\text{odd}} := \bigoplus_{i \in \mathbb{Z}} L_{(i+\frac{1}{2})\alpha}.$$

All its h_α-weights are odd. If this module were nonzero, then some irreducible summand would have odd weights, hence would contain weight 1. This would force

$$L_{\frac{1}{2}\alpha} \neq 0,$$

contradicting the previous paragraph. Therefore

$$M_{\text{odd}} = 0.$$

Combining this with $2c \in \mathbb{Z}$, we obtain

$$F\alpha \cap \Delta = \{-\alpha, \alpha\}, \qquad \dim L_\alpha = 1.$$

Now let $\alpha, \beta \in \Delta$ with $\beta \neq \pm\alpha$. Consider

$$M_{\alpha,\beta} := \bigoplus_{i \in \mathbb{Z}} L_{\beta + i\alpha}.$$

This is again an $\mathfrak{s}_\alpha$-submodule of L. If $x \in L_{\beta+i\alpha}$, then

$$[h_\alpha, x] = (\beta + i\alpha)(h_\alpha)x.$$

Using

$$\alpha(h_\alpha) = 2 \qquad \text{and} \qquad \beta(h_\alpha) = \frac{2(\alpha,\beta)}{(\alpha,\alpha)},$$

we get

$$(\beta + i\alpha)(h_\alpha) = \frac{2(\alpha,\beta)}{(\alpha,\alpha)} + 2i.$$

Since the weights of a finite-dimensional $\mathfrak{sl}_2$-module are integers, it follows that

$$\frac{2(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}.$$

Moreover, all these weights have the same parity. By the first part, each nonzero root space $L_{\beta+i\alpha}$ is one-dimensional. Hence $M_{\alpha,\beta}$ is irreducible as an $\mathfrak{s}_\alpha$-module. Therefore the weights occurring in $M_{\alpha,\beta}$ form one complete $\mathfrak{sl}_2$-string. Thus there exist $p, q \in \mathbb{Z}_{\geq 0}$ such that the roots of the form $\beta + i\alpha$ are exactly

$$\beta - q\alpha,\ \beta - (q-1)\alpha,\ \dots,\ \beta - \alpha,\ \beta,\ \beta + \alpha,\ \dots,\ \beta + p\alpha.$$

The highest weight is

$$\beta(h_\alpha) + 2p,$$

and the lowest weight is

$$\beta(h_\alpha) - 2q.$$

Since the weights of an irreducible $\mathfrak{sl}_2$-module are symmetric about 0, we have

$$\beta(h_\alpha) - 2q = -\big(\beta(h_\alpha) + 2p\big).$$

Hence

$$q - p = \beta(h_\alpha) = \frac{2(\alpha, \beta)}{(\alpha, \alpha)}.$$

Finally, if $\alpha + \beta \in \Delta$, then $L_{\alpha+\beta} \neq 0$. In the irreducible $\mathfrak{s}_\alpha$-module $M_{\alpha,\beta}$, the raising operator x_α maps

$$L_\beta \longrightarrow L_{\beta+\alpha}.$$

Since $\beta + \alpha$ is a weight, this map is nonzero. As $L_{\beta+\alpha}$ is one-dimensional, we obtain

$$[L_\alpha, L_\beta] = L_{\alpha+\beta}.$$

This proves the proposition. □

Let Δ be a root system in a Euclidean space E. For $\alpha \in \Delta$, define the reflection

$$s_\alpha : E \longrightarrow E$$

by

$$s_\alpha(x) = x - \frac{2(x, \alpha)}{(\alpha, \alpha)}\alpha.$$

Then

$$s_\alpha(\alpha) = \alpha - \frac{2(\alpha, \alpha)}{(\alpha, \alpha)}\alpha = -\alpha.$$

Moreover, if

$$H_\alpha := \{\, x \in E \mid (x, \alpha) = 0 \,\},$$

then

$$s_\alpha(x) = x \qquad \text{for all } x \in H_\alpha.$$

Thus s_α is the reflection with respect to the hyperplane H_α, which has codimension 1 in E.

Lemma 10.0.2. *For every $\alpha \in \Delta$, one has*

$$s_\alpha(\Delta) = \Delta.$$

Proof. Since $s_\alpha^2 = \mathrm{id}_E$, the map s_α is a bijection. Hence it suffices to prove that

$$s_\alpha(\Delta) \subseteq \Delta.$$

Let $\beta \in \Delta$. By the root-string property, there exist $p, q \in \mathbb{Z}_{\geq 0}$ such that the roots of the form $\beta + i\alpha$ are precisely

$$\beta - q\alpha,\ \beta - (q-1)\alpha,\ \ldots,\ \beta,\ \ldots,\ \beta + p\alpha,$$

and

$$q - p = \frac{2(\beta, \alpha)}{(\alpha, \alpha)}.$$

Therefore

$$s_\alpha(\beta) = \beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)}\alpha = \beta - (q-p)\alpha = \beta + (p-q)\alpha.$$

Since

$$-q \leq p - q \leq p,$$

the root-string property implies that

$$\beta + (p-q)\alpha \in \Delta.$$

Hence $s_\alpha(\beta) \in \Delta$. Since $\beta \in \Delta$ was arbitrary, we have

$$s_\alpha(\Delta) \subseteq \Delta.$$

As s_α is bijective, it follows that

$$s_\alpha(\Delta) = \Delta.$$

□

Chapter 11

Lecture 11

Let V be the Euclidean space containing the root system Δ, and let $\alpha \in \Delta$. Since

$$(\alpha, \alpha) \neq 0,$$

we have

$$V = \ker(\alpha) \oplus F\alpha,$$

where

$$\ker(\alpha) := \{x \in V \mid (x, \alpha) = 0\}.$$

We define the reflection $s_\alpha : V \to V$ by

$$s_\alpha(x) = x - \frac{2(x, \alpha)}{(\alpha, \alpha)}\alpha.$$

From the previous discussion, we already know that

$$s_\alpha(\Delta) = \Delta.$$

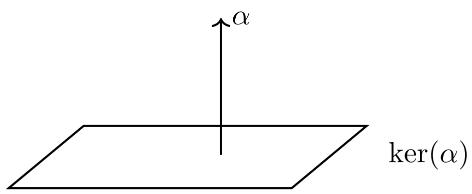


Figure 11.1: The hyperplane $\ker(\alpha)$ and the normal vector α.

Lemma 11.0.1. *One has*

$$\operatorname{span}(\Delta) = H^*.$$

Proof. Suppose $\operatorname{span}(\Delta) \neq H^*$. Then there exists

$$0 \neq h \in H$$

such that

$$\gamma(h) = 0 \qquad \text{for all } \gamma \in \operatorname{span}(\Delta).$$

In particular,

$$\alpha(h) = 0 \qquad \text{for all } \alpha \in \Delta.$$

Hence h commutes with all root vectors x_α. Since $h \in H$, it also commutes with H. Therefore h commutes with all of L, so

$$h \in Z(L).$$

But L is semisimple, hence $Z(L) = 0$. This is a contradiction. Therefore

$$\operatorname{span}(\Delta) = H^*.$$

□

Lemma 11.0.2. *For all $\alpha, \beta \in \Delta$, one has*

$$(\alpha, \beta) \in \mathbb{Q}.$$

Proof. Recall that

$$\frac{2(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}.$$

Thus it suffices to prove that

$$(\alpha, \alpha) \in \mathbb{Q} \qquad \text{for all } \alpha \in \Delta.$$

Let $\nu : H \to H^*$ be the isomorphism induced by the Killing form B, and write

$$t_\alpha := \nu^{-1}(\alpha) \in H.$$

Then

$$h_\alpha = \frac{2}{(\alpha, \alpha)}\, t_\alpha.$$

Now

$$(\alpha, \alpha) = B(t_\alpha, t_\alpha) = \operatorname{Tr}\big((\operatorname{ad} t_\alpha)^2\big) = \sum_{\beta \in \Delta} \beta(t_\alpha)^2.$$

Since

$$\beta(t_\alpha) = \frac{(\alpha, \alpha)}{2}\, \beta(h_\alpha),$$

we obtain

$$(\alpha, \alpha) = \frac{(\alpha, \alpha)^2}{4} \sum_{\beta \in \Delta} \beta(h_\alpha)^2.$$

Therefore

$$\frac{4}{(\alpha, \alpha)} = \sum_{\beta \in \Delta} \beta(h_\alpha)^2 \in \mathbb{Q},$$

because each $\beta(h_\alpha) \in \mathbb{Z}$. Hence

$$(\alpha, \alpha) \in \mathbb{Q}.$$

It now follows that

$$(\alpha, \beta) = \frac{(\alpha, \alpha)}{2}\, \beta(h_\alpha) \in \mathbb{Q}.$$

$\square$

Now consider

$$E := \sum_{\alpha \in \Delta} \mathbb{Q}\alpha \subseteq H^*.$$

Choose $\alpha_1, \dots, \alpha_n \in \Delta$ such that

$$\{\alpha_1, \dots, \alpha_n\}$$

is an F-basis of H^*. For any $\beta \in \Delta$, write

$$\beta = \sum_{i=1}^{n} k_i \alpha_i, \qquad k_i \in F.$$

Lemma 11.0.3. *Each coefficient k_i lies in $\mathbb{Q}$.*

Proof. For $j = 1, \dots, n$, we have

$$(\beta, \alpha_j) = \sum_{i=1}^{n} k_i (\alpha_i, \alpha_j).$$

This is a system of n linear equations in the unknowns $k_1, \dots, k_n$, whose coefficients lie in $\mathbb{Q}$, since by the previous lemma

$$(\beta, \alpha_j) \in \mathbb{Q}, \qquad (\alpha_i, \alpha_j) \in \mathbb{Q}.$$

Because the form $(\cdot, \cdot)$ is nondegenerate, the Gram matrix

$$\big((\alpha_i, \alpha_j)\big)_{1 \le i,j \le n}$$

is invertible. Hence the system has a unique solution, and this solution lies in $\mathbb{Q}$. Therefore

$$k_i \in \mathbb{Q} \qquad \text{for all } i.$$

$\square$

Corollary 11.0.4. *One has*

$$\dim_{\mathbb{Q}} \left(\sum_{\alpha \in \Delta} \mathbb{Q}\alpha \right) = \dim_F H^*.$$

Proof. Let

$$n = \dim_F H^*.$$

Since

$$\operatorname{span}_F(\Delta) = H^*,$$

we may choose roots

$$\alpha_1, \ldots, \alpha_n \in \Delta$$

which form an F-basis of H^*.

First, $\alpha_1, \ldots, \alpha_n$ are $\mathbb{Q}$-linearly independent. Indeed, if

$$q_1\alpha_1 + \cdots + q_n\alpha_n = 0, \qquad q_i \in \mathbb{Q},$$

then this is also an F-linear relation. Since $\alpha_1, \ldots, \alpha_n$ are F-linearly independent, we get

$$q_1 = \cdots = q_n = 0.$$

Hence

$$\dim_{\mathbb{Q}}\left(\sum_{\alpha\in\Delta} \mathbb{Q}\alpha\right) \geq n.$$

Conversely, by the preceding lemma, every $\beta \in \Delta$ can be written as

$$\beta = \sum_{i=1}^{n} k_i\alpha_i$$

with

$$k_i \in \mathbb{Q}.$$

Therefore

$$\sum_{\alpha\in\Delta} \mathbb{Q}\alpha = \sum_{i=1}^{n} \mathbb{Q}\alpha_i.$$

Thus

$$\dim_{\mathbb{Q}}\left(\sum_{\alpha\in\Delta} \mathbb{Q}\alpha\right) \leq n.$$

Combining the two inequalities gives

$$\dim_{\mathbb{Q}}\left(\sum_{\alpha\in\Delta} \mathbb{Q}\alpha\right) = n = \dim_F H^*.$$

□

Lemma 11.0.5. *The restriction of* $(\cdot, \cdot)$ *to*

$$\sum_{\alpha\in\Delta} \mathbb{Q}\alpha$$

is positive definite.

Proof. Let

$$0 \neq \lambda \in \sum_{\alpha \in \Delta} \mathbb{Q}\alpha,$$

and let $t_\lambda := \nu^{-1}(\lambda) \in H$. Then

$$(\lambda, \lambda) = B(t_\lambda, t_\lambda) = \operatorname{Tr}\big((\operatorname{ad} t_\lambda)^2\big) = \sum_{\alpha \in \Delta} \alpha(t_\lambda)^2 \geq 0.$$

If $(\lambda, \lambda) = 0$, then

$$\alpha(t_\lambda) = 0 \qquad \text{for all } \alpha \in \Delta.$$

By the lemma $\operatorname{span}(\Delta) = H^*$, this implies $t_\lambda = 0$, hence $\lambda = 0$, a contradiction. Therefore

$$(\lambda, \lambda) > 0 \qquad \text{for all } 0 \neq \lambda \in \sum_{\alpha \in \Delta} \mathbb{Q}\alpha.$$

Thus the form is positive definite on

$$\sum_{\alpha \in \Delta} \mathbb{Q}\alpha.$$

□

Let

$$E_{\mathbb{Q}} := \sum_{\alpha \in \Delta} \mathbb{Q}\alpha \subseteq H^*.$$

By the preceding corollary,

$$\dim_{\mathbb{Q}} E_{\mathbb{Q}} = \dim_F H^*.$$

We now extend $E_{\mathbb{Q}}$ to a real vector space by setting

$$E := E_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R}.$$

Thus E is a finite-dimensional real vector space.

The symmetric bilinear form $(\cdot, \cdot)$ on $E_{\mathbb{Q}}$ extends naturally to an $\mathbb{R}$-bilinear form on E by

$$\left(\sum_i v_i \otimes r_i, \sum_j w_j \otimes r'_j \right) := \sum_{i,j} (v_i, w_j) r_i r'_j,$$

where

$$v_i, w_j \in E_{\mathbb{Q}}, \qquad r_i, r'_j \in \mathbb{R}.$$

This form is positive definite. Indeed, if

$$\alpha_1, \ldots, \alpha_n$$

is a $\mathbb{Q}$-basis of $E_{\mathbb{Q}}$, then it is also an $\mathbb{R}$-basis of E. With respect to this basis, the Gram matrix of the extended form is still

$$\big((\alpha_i, \alpha_j)\big)_{1 \leq i,j \leq n}.$$

Since the original form is positive definite on $E_{\mathbb{Q}}$, this matrix is positive definite. Hence the extended form is positive definite on E. Therefore E is a Euclidean space.

11.0.1 Abstract Root Systems

Definition 11.0.6 (Root system). *Let $E = \mathbb{R}^n$ be a Euclidean space with inner product $(\cdot, \cdot)$. A subset $\Delta \subseteq E$ is called a* root system *if the following conditions hold:*

1. $$\Delta \subseteq E \setminus \{0\}, \qquad |\Delta| < \infty.$$

2. $$\mathrm{span}_{\mathbb{R}}(\Delta) = E.$$

3. *For every $\alpha \in \Delta$,*
$$s_\alpha(\Delta) = \Delta,$$
where
$$s_\alpha(x) = x - \frac{2(x, \alpha)}{(\alpha, \alpha)}\alpha$$
is the reflection with respect to the hyperplane orthogonal to α.

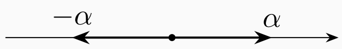
4. *For all $\alpha, \beta \in \Delta$,*
$$\frac{2(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}.$$

5. *For every $\alpha \in \Delta$,*
$$\mathbb{R}\alpha \cap \Delta = \{\alpha, -\alpha\}.$$

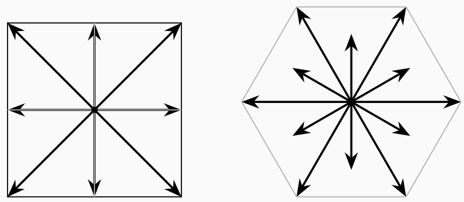
For the moment, we forget the Lie algebra and work only with abstract root systems.

Definition 11.0.7 (Irreducible root system). *Let $\Delta \subseteq E$ be a root system in a Euclidean space E. We say that Δ is* irreducible *if it cannot be written as a disjoint union*
$$\Delta = \Delta_1 \cup \Delta_2,$$
where Δ_1, Δ_2 are nonempty subsets satisfying
$$(\Delta_1, \Delta_2) = 0.$$
Equivalently, Δ is irreducible if E cannot be decomposed as an orthogonal direct sum
$$E = \mathrm{span}_{\mathbb{R}}(\Delta_1) \oplus \mathrm{span}_{\mathbb{R}}(\Delta_2)$$
with roots split between the two summands.

Example 11.0.8. *In dimension 1, there is only one root system:*
$$\Delta = \{\alpha, -\alpha\}.$$



Example 11.0.9. *Typical irreducible root systems in dimension* 2 *include the following pictures.*



We now study possible angles between two roots. Let

$$\alpha, \beta \in \Delta$$

satisfy

$$(\alpha, \beta) \neq 0, \qquad \beta \neq \pm\alpha.$$

Let θ be the angle between α and β. Then

$$(\alpha, \beta) = \|\alpha\| \, \|\beta\| \cos\theta.$$

By the axioms of a root system,

$$\frac{2(\beta, \alpha)}{(\alpha, \alpha)} \in \mathbb{Z}, \qquad \frac{2(\alpha, \beta)}{(\beta, \beta)} \in \mathbb{Z}.$$

Multiplying these two integers gives

$$\frac{2(\beta, \alpha)}{(\alpha, \alpha)} \cdot \frac{2(\alpha, \beta)}{(\beta, \beta)} = \frac{4(\alpha, \beta)^2}{(\alpha, \alpha)(\beta, \beta)} = 4\cos^2\theta.$$

Since $\beta \neq \pm\alpha$, this product cannot be 4. Since $(\alpha, \beta) \neq 0$, it is nonzero. Hence

$$4\cos^2\theta \in \{1, 2, 3\}.$$

Therefore

$$\cos^2\theta \in \left\{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right\}.$$

Thus the possible non-right, nonzero angles are

$$\theta \in \left\{\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{5\pi}{6}\right\}.$$

In particular, at least one of the two integers

$$\frac{2(\beta, \alpha)}{(\alpha, \alpha)}, \qquad \frac{2(\alpha, \beta)}{(\beta, \beta)}$$

has absolute value 1. Replacing β by $-\beta$ if necessary, we may assume

$$(\alpha, \beta) > 0.$$

Without loss of generality, suppose

$$\frac{2(\beta, \alpha)}{(\alpha, \alpha)} = 1.$$

Then

$$1 = \frac{2\|\beta\|\|\alpha\| \cos\theta}{\|\alpha\|^2} = \frac{2\|\beta\| \cos\theta}{\|\alpha\|}.$$

Hence

$$\frac{\|\alpha\|}{\|\beta\|} = 2\cos\theta.$$

Consequently,

$$\cos\theta = \frac{1}{2} \quad \Longrightarrow \quad \|\alpha\| = \|\beta\|,$$

$$\cos\theta = \frac{\sqrt{2}}{2} \quad \Longrightarrow \quad \|\alpha\| = \sqrt{2}\,\|\beta\|,$$

and

$$\cos\theta = \frac{\sqrt{3}}{2} \quad \Longrightarrow \quad \|\alpha\| = \sqrt{3}\,\|\beta\|.$$

Thus the possible ratios of root lengths are

$$1, \qquad \sqrt{2}, \qquad \sqrt{3}.$$

Let $E = \mathbb{R}^n$ be a Euclidean space with inner product $(\cdot, \cdot)$, and let $\Delta \subset E$ be a root system.

Definition 11.0.10 (Base)**.** *A subset*

$$B = \{\beta_1, \dots, \beta_n\} \subset \Delta$$

is called a base *of Δ if the following two conditions hold:*

1. *B is a basis of E;*
2. *for every $\alpha \in \Delta$, one has*

$$\alpha = k_1\beta_1 + \cdots + k_n\beta_n$$

where either all $k_i \in \mathbb{Z}_{\geq 0}$, or all $k_i \in \mathbb{Z}_{\leq 0}$.

In this case one obtains a decomposition

$$\Delta = \Delta^+ \dot{\cup} \Delta^-,$$

where

$$\Delta^+ = \Delta \cap \sum_{i=1}^{n} \mathbb{Z}_{\geq 0}\beta_i, \qquad \Delta^- = -\Delta^+.$$

Lemma 11.0.11. *Let B be a base of Δ. If $\beta_1, \beta_2 \in B$ and $\beta_1 \neq \beta_2$, then*

$$(\beta_1, \beta_2) \leq 0.$$

Equivalently, the angle between two distinct elements of B is obtuse or right.

Proof. Suppose, to the contrary, that

$$(\beta_1, \beta_2) > 0.$$

Then the Cartan integer

$$\frac{2(\beta_1, \beta_2)}{(\beta_2, \beta_2)}$$

is a positive integer. Since Δ is stable under reflections, we have

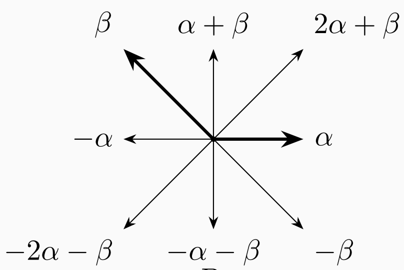
$$s_{\beta_2}(\beta_1) = \beta_1 - \frac{2(\beta_1, \beta_2)}{(\beta_2, \beta_2)}\beta_2 \in \Delta.$$

But relative to the basis B, this root has coefficient $1 > 0$ on β_1 and a negative coefficient on β_2. This contradicts the definition of a base, since every root must be either a nonnegative or a nonpositive integral linear combination of the elements of B. Therefore

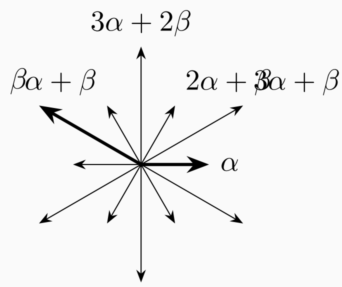
$$(\beta_1, \beta_2) \leq 0.$$

$\square$

Example 11.0.12. *The following diagrams illustrate possible choices of bases in rank 2.*



B_2



G_2

Lemma 11.0.13. *Let F be an infinite field and let V be a finite-dimensional vector space over F. Then V is not the union of finitely many proper hyperplanes.*

Proof. Suppose, to the contrary, that

$$V = V_1 \cup \cdots \cup V_m$$

where each V_i is a proper hyperplane. We may assume that no V_i is contained in the union of the others.

Choose

$$y \in V_1 \setminus \bigcup_{i=2}^{m} V_i.$$

Since $V_1 \neq V$, choose $x \notin V_1$. Consider the affine line

$$y + Fx = \{y + ax \mid a \in F\}.$$

This line meets V_1 only at y, because if $y + ax \in V_1$ and $a \neq 0$, then $x \in V_1$, a contradiction. Moreover, for each $i \geq 2$, the line $y + Fx$ meets V_i in at most one point, since $y \notin V_i$.

Thus the finite union $V_1 \cup \cdots \cup V_m$ meets the line $y + Fx$ in only finitely many points. Since F is infinite, there exists $a \in F$ such that

$$y + ax \notin V_1 \cup \cdots \cup V_m,$$

a contradiction. □

Lemma 11.0.14. *Let $\alpha, \beta \in \Delta$ with $\alpha \neq \beta$. If*

$$\alpha - \beta \notin \Delta,$$

then

$$(\alpha, \beta) \leq 0.$$

Proof. By the root-string property, the β-string through α has the form

$$\alpha - r\beta, \ldots, \alpha, \ldots, \alpha + q\beta,$$

where $r, q \in \mathbb{Z}_{\geq 0}$, and

$$r - q = \frac{2(\alpha, \beta)}{(\beta, \beta)}.$$

Since $\alpha - \beta \notin \Delta$, one has $r = 0$. Hence

$$\frac{2(\alpha, \beta)}{(\beta, \beta)} = -q \leq 0.$$

Since $(\beta, \beta) > 0$, it follows that

$$(\alpha, \beta) \leq 0.$$

□

Lemma 11.0.15. *Let $S = \{s_1, \dots, s_m\}$ be a finite set of nonzero vectors in a Euclidean space. Suppose that*

$$(s_i, s_j) \leq 0 \qquad \textit{for all } i \neq j,$$

and suppose that there exists $f \neq 0$ such that

$$(f, s_i) > 0 \qquad \textit{for all } i.$$

Then S is linearly independent.

Proof. Suppose that

$$\sum_{i=1}^{m} k_i s_i = 0$$

for some $k_i \in \mathbb{R}$, not all zero. If all nonzero k_i have the same sign, then pairing with f gives an immediate contradiction.

Thus, after separating positive and negative coefficients, we may write

$$\sum_{i \in I} k_i s_i = \sum_{j \in J} h_j s_j,$$

where $I \cap J = \varnothing$ and all $k_i, h_j > 0$. Set

$$v = \sum_{i \in I} k_i s_i = \sum_{j \in J} h_j s_j.$$

Then

$$0 \leq (v, v) = \left(\sum_{i \in I} k_i s_i, \sum_{j \in J} h_j s_j \right) = \sum_{i \in I,\ j \in J} k_i h_j (s_i, s_j) \leq 0.$$

Hence $(v, v) = 0$, so $v = 0$. But then

$$0 = (f, v) = \sum_{i \in I} k_i (f, s_i) > 0,$$

a contradiction. Therefore S is linearly independent. ☐

Theorem 11.0.16 (Existence of a base)**.** *Every root system $\Delta \subset E$ admits a base.*

Proof. First note that

$$\bigcup_{\alpha \in \Delta} \alpha^\perp \neq E.$$

Indeed, Δ is finite, and each $\alpha^\perp$ is a proper hyperplane. Hence the previous lemma applies.

Choose

$$\gamma \in E \setminus \bigcup_{\alpha \in \Delta} \alpha^\perp.$$

Such a vector γ is called *regular*. Thus

$$(\gamma, \alpha) \neq 0 \qquad \text{for all } \alpha \in \Delta.$$

Define

$$\Delta^+ = \{\alpha \in \Delta \mid (\gamma, \alpha) > 0\}, \qquad \Delta^- = \{\alpha \in \Delta \mid (\gamma, \alpha) < 0\}.$$

Then

$$\Delta = \Delta^+ \dot{\cup} \Delta^-, \qquad \Delta^- = -\Delta^+.$$

Now define

$$B = \left\{\beta \in \Delta^+ \;\middle|\; \beta \text{ is not a sum of two elements of } \Delta^+\right\}.$$

We claim that B is a base of Δ.

First, $B \neq \varnothing$. Indeed, since Δ^+ is finite, choose $\beta \in \Delta^+$ such that

$$(\gamma, \beta) = \min\{(\gamma, \alpha) \mid \alpha \in \Delta^+\}.$$

If $\beta = \alpha_1 + \alpha_2$ with $\alpha_1, \alpha_2 \in \Delta^+$, then

$$(\gamma, \alpha_i) > 0 \qquad (i = 1, 2),$$

and hence

$$(\gamma, \alpha_i) < (\gamma, \beta),$$

contradicting the minimality of β. Thus $\beta \in B$.

Next, every element of Δ^+ is a nonnegative integral linear combination of elements of B. Indeed, if $\alpha \in \Delta^+$ and $\alpha \notin B$, then

$$\alpha = \alpha_1 + \alpha_2$$

for some $\alpha_1, \alpha_2 \in \Delta^+$. Repeating this process, and using the finiteness of Δ^+, we eventually express α as a sum of elements of B. Since $\Delta^- = -\Delta^+$, every negative root is a nonpositive integral linear combination of elements of B.

It remains to prove that B is linearly independent. Let $\beta_1, \beta_2 \in B$ with $\beta_1 \neq \beta_2$. We first show that

$$\beta_1 - \beta_2 \notin \Delta.$$

If $\beta_1 - \beta_2 \in \Delta^+$, then

$$\beta_1 = \beta_2 + (\beta_1 - \beta_2),$$

contradicting the definition of B. If $\beta_1 - \beta_2 \in \Delta^-$, then

$$\beta_2 = \beta_1 + (\beta_2 - \beta_1),$$

again contradicting the definition of B. Hence

$$\beta_1 - \beta_2 \notin \Delta.$$

By the previous lemma,

$$(\beta_1, \beta_2) \leq 0.$$

Moreover, since $B \subset \Delta^+$, one has

$$(\gamma, \beta) > 0 \qquad \text{for all } \beta \in B.$$

Therefore, by the linear independence lemma, B is linearly independent.

Finally, B spans Δ, and Δ spans E. Hence B is a basis of E. We have already shown that every root is either a nonnegative or a nonpositive integral linear combination of elements of B. Thus B is a base of Δ. $\square$

Chapter 12

Lecture 12

Let Δ be a root system in a Euclidean space E, with inner product $(\cdot\,,\cdot)$. Recall that a vector $\gamma \in E$ is called *regular* if

$$(\gamma, \alpha) \neq 0 \qquad \text{for all } \alpha \in \Delta.$$

Equivalently, γ does not lie on any reflecting hyperplane.

From the previous discussion, bases of Δ correspond to regular decompositions

$$\Delta = \Delta^+ \sqcup \Delta^-.$$

Moreover, for any base B of Δ, there exists a vector $\gamma \in E$ such that

$$(\beta, \gamma) > 0 \qquad \text{for all } \beta \in B.$$

Definition 12.0.1 (Weyl chamber)**.** *For each $\alpha \in \Delta$, define the hyperplane*

$$P_\alpha = \{\, \gamma \in E \mid (\gamma, \alpha) = 0 \,\}.$$

The hyperplanes P_α, $\alpha \in \Delta$, partition E into finitely many regions. The connected components of

$$E \setminus \bigcup_{\alpha\in\Delta} P_\alpha$$

are called the open Weyl chambers *of E.*

Each regular vector $\gamma \in E$ belongs to precisely one Weyl chamber. Thus the choice of a regular vector determines a decomposition

$$\Delta = \Delta^+(\gamma) \sqcup \Delta^-(\gamma),$$

where

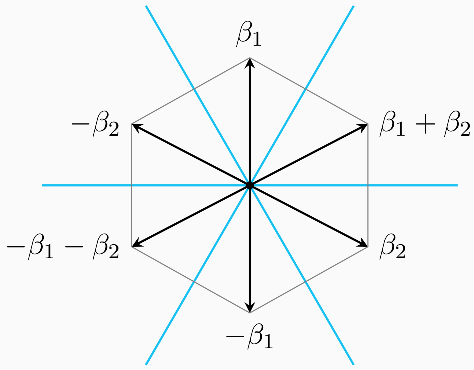
$$\Delta^+(\gamma) = \{\alpha \in \Delta \mid (\alpha, \gamma) > 0\}, \qquad \Delta^-(\gamma) = \{\alpha \in \Delta \mid (\alpha, \gamma) < 0\}.$$

Consequently, one obtains a natural correspondence

$$\{\text{bases of } \Delta\} \longleftrightarrow \{\text{Weyl chambers of } E\}.$$

Example 12.0.2 (The root system of type A_2)**.** *The following picture illustrates the Weyl chambers for the root system*

$$\Delta = \{\pm\beta_1,\ \pm\beta_2,\ \pm(\beta_1 + \beta_2)\}.$$



Definition 12.0.3 (Weyl group)**.** *For $\alpha \in \Delta$, let s_α denote the reflection*

$$s_\alpha(\lambda) = \lambda - \frac{2(\lambda, \alpha)}{(\alpha, \alpha)}\alpha.$$

The subgroup of $\mathrm{GL}(E)$ *generated by all reflections s_α, $\alpha \in \Delta$, is called the* Weyl group *of Δ, and is denoted by*

$$W.$$

Since every $w \in W$ preserves the root system,

$$w(\Delta) = \Delta,$$

we obtain an injective homomorphism

$$W \hookrightarrow \mathrm{Sym}(\Delta).$$

Hence

$$|W| < \infty.$$

For any base B of Δ, denote by

$$W_B = \langle\, s_\beta \mid \beta \in B \,\rangle$$

the subgroup generated by the simple reflections corresponding to B.

Lemma 12.0.4. *Let B be a base of Δ, and let $\alpha \in B$. Then s_α permutes the set*

$$\Delta^+ \setminus \{\alpha\}.$$

Proof. Let

$$\gamma \in \Delta^+ \setminus \{\alpha\}.$$

Write

$$\gamma = \sum_{\beta \in B} k_\beta \beta, \qquad k_\beta \in \mathbb{Z}_{\geq 0}.$$

Since $\gamma \neq \alpha$, there exists some $\beta \in B$, $\beta \neq \alpha$, such that

$$k_\beta > 0.$$

Now

$$s_\alpha(\gamma) = \gamma - \langle \gamma, \alpha^\vee \rangle \alpha,$$

where

$$\langle \gamma, \alpha^\vee \rangle = \frac{2(\gamma, \alpha)}{(\alpha, \alpha)} \in \mathbb{Z}.$$

Thus $s_\alpha(\gamma)$ has the same coefficients as γ on all simple roots $\beta \neq \alpha$. In particular, the coefficient of the above β remains positive.

Since $s_\alpha(\gamma) \in \Delta$, it must be either positive or negative with respect to the base B. But it cannot be negative, because a negative root is a nonpositive integral combination of the simple roots in B, whereas $s_\alpha(\gamma)$ has a positive coefficient on some $\beta \neq \alpha$. Therefore

$$s_\alpha(\gamma) \in \Delta^+.$$

Moreover, $s_\alpha(\gamma) \neq \alpha$, since otherwise

$$\gamma = s_\alpha(\alpha) = -\alpha,$$

contradicting $\gamma \in \Delta^+$. Hence

$$s_\alpha(\gamma) \in \Delta^+ \setminus \{\alpha\}.$$

Since $s_\alpha^2 = 1$, the map s_α is an involution. Therefore it permutes $\Delta^+ \setminus \{\alpha\}$. □

Let Δ be a root system in a Euclidean space E, and let B be a base of Δ. Denote by $\Delta^+ = \Delta^+(B)$ the corresponding set of positive roots, and put

$$\rho = \frac{1}{2} \sum_{\alpha \in \Delta^+} \alpha.$$

Recall that, for $\beta \in B$,

$$s_\beta(\rho) = \rho - \beta.$$

Indeed, s_β sends β to $-\beta$ and permutes $\Delta^+ \setminus \{\beta\}$.

Lemma 12.0.5. *Given any two bases B and B' of Δ, there exists $w \in W_B$ such that*

$$w(B') = B,$$

where

$$W_B = \langle s_\beta \mid \beta \in B \rangle.$$

Proof. Choose a regular vector $\gamma' \in E$ corresponding to the base B', so that

$$(\beta', \gamma') > 0 \qquad \text{for all } \beta' \in B'.$$

We want to find $w \in W_B$ such that

$$(\beta, w(\gamma')) > 0 \qquad \text{for all } \beta \in B.$$

Since W_B is finite, we may choose $w \in W_B$ such that

$$(\rho, w(\gamma'))$$

is maximal among all values

$$(\rho, u(\gamma')), \qquad u \in W_B.$$

Let $\beta \in B$. Since $s_\beta w \in W_B$, the maximality of w gives

$$(\rho, s_\beta w(\gamma')) \le (\rho, w(\gamma')).$$

Because s_β is an orthogonal reflection, we have

$$(\rho, s_\beta w(\gamma')) = (s_\beta \rho, w(\gamma')).$$

Using $s_\beta(\rho) = \rho - \beta$, we obtain

$$(\rho - \beta, w(\gamma')) \le (\rho, w(\gamma')).$$

Hence

$$(\beta, w(\gamma')) \ge 0.$$

In fact, the inequality is strict. If

$$(\beta, w(\gamma')) = 0,$$

then

$$(w^{-1}\beta, \gamma') = 0.$$

But $w^{-1}\beta \in \Delta$, while γ' is regular, a contradiction. Therefore

$$(\beta, w(\gamma')) > 0 \qquad \text{for all } \beta \in B.$$

Thus $w(\gamma')$ lies in the Weyl chamber corresponding to B.

On the other hand, the base corresponding to $w(\gamma')$ is $w(B')$. Therefore

$$w(B') = B.$$

$\square$

Lemma 12.0.6. *Every root $\alpha \in \Delta$ is contained in some base of Δ.*

Proof. Let $\alpha \in \Delta$. Consider the hyperplane

$$\alpha^{\perp} = \{x \in E \mid (x, \alpha) = 0\}.$$

For every $\beta \in \Delta \setminus \{\pm\alpha\}$, the subspace

$$\alpha^{\perp} \cap \beta^{\perp}$$

is a proper subspace of $\alpha^{\perp}$. Since Δ is finite and $\alpha^{\perp}$ cannot be a finite union of proper subspaces, we may choose

$$\rho \in \alpha^{\perp}$$

such that

$$\rho \notin \beta^{\perp} \qquad \text{for all } \beta \in \Delta \setminus \{\pm\alpha\}.$$

Thus

$$(\rho, \beta) \neq 0 \qquad \text{for all } \beta \in \Delta \setminus \{\pm\alpha\}.$$

Now choose a regular vector ρ' sufficiently close to ρ such that

$$(\rho', \alpha) > 0$$

and

$$|(\rho', \alpha)| < |(\rho', \beta)| \qquad \text{for all } \beta \in \Delta \setminus \{\pm\alpha\}.$$

Let $B_{\rho'}$ be the base corresponding to the regular vector ρ'. Then α is positive with respect to $B_{\rho'}$.

We claim that $\alpha \in B_{\rho'}$. Suppose not. Then α is a positive root which is not simple. Hence there exists a simple root $\beta \in B_{\rho'}$ such that

$$\alpha - \beta \in \Delta^{+}.$$

Therefore

$$(\rho', \alpha) = (\rho', \beta) + (\rho', \alpha - \beta),$$

where both terms on the right are positive. Since

$$\beta \neq \pm\alpha,$$

our choice of ρ' gives

$$(\rho', \beta) > (\rho', \alpha),$$

which is impossible. Hence $\alpha \in B_{\rho'}$.

Therefore every root is contained in some base. $\square$

Corollary 12.0.7. *For any base B of Δ,*

$$\Delta = W_B B.$$

Equivalently, every root of Δ is W_B-conjugate to a simple root in B.

Proof. Let $\alpha \in \Delta$. By the preceding lemma, there exists a base B_1 such that

$$\alpha \in B_1.$$

By the first lemma, there exists $w \in W_B$ such that

$$w(B_1) = B.$$

Hence

$$w(\alpha) \in B.$$

Therefore

$$\alpha \in W_B B.$$

Thus

$$\Delta = W_B B.$$

$\square$

Corollary 12.0.8. *For any base B of Δ,*

$$W = W_B.$$

Proof. By definition,

$$W_B \subseteq W.$$

It remains to prove the reverse inclusion.

Let $\alpha \in \Delta$. By the preceding corollary, there exist $w \in W_B$ and $\beta \in B$ such that

$$w(\alpha) = \beta.$$

Using the conjugation formula for reflections,

$$s_{w(\alpha)} = w s_\alpha w^{-1},$$

we get

$$s_\beta = w s_\alpha w^{-1}.$$

Hence

$$s_\alpha = w^{-1} s_\beta w.$$

Since $w \in W_B$ and $s_\beta \in W_B$, it follows that

$$s_\alpha \in W_B.$$

Thus every reflection s_α, $\alpha \in \Delta$, lies in W_B. Since W is generated by all such reflections, we obtain

$$W \subseteq W_B.$$

Therefore

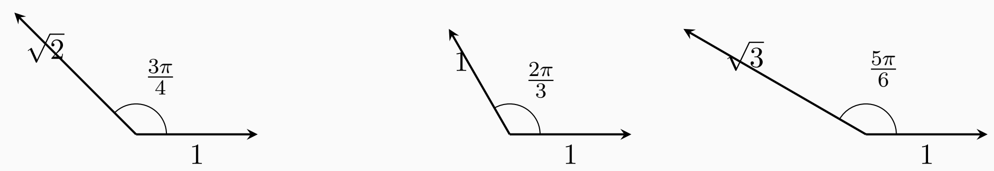
$$W = W_B.$$

$\square$

Example 12.0.9 (Irreducible root systems of rank 2)**.** *Up to isomorphism, the irreducible root systems of rank 2 are of types*

$$A_2, \qquad B_2, \qquad G_2.$$

They may be represented by two simple roots whose angles and relative lengths are as follows.



Corollary 12.0.10. *Let $\alpha, \beta \in \Delta$ be roots with $\beta \neq \pm\alpha$. If $k \in \mathbb{R}$ and*

$$\beta + k\alpha \in \Delta,$$

then

$$k \in \mathbb{Z}.$$

Proof. By the preceding results, α is contained in some base B_1 of Δ. Write

$$B_1 = \{\alpha_1, \alpha_2, \ldots, \alpha_\ell\}, \qquad \alpha_1 = \alpha.$$

Since $\beta \in \Delta$, we may write

$$\beta = \sum_{i=1}^{\ell} k_i \alpha_i, \qquad k_i \in \mathbb{Z}.$$

Then

$$\beta + k\alpha = (k_1 + k)\alpha_1 + \sum_{i=2}^{\ell} k_i \alpha_i.$$

But $\beta + k\alpha \in \Delta$. Hence its coordinates with respect to the base B_1 are integers. Therefore

$$k_1 + k \in \mathbb{Z}.$$

Since $k_1 \in \mathbb{Z}$, it follows that

$$k \in \mathbb{Z}.$$

$\square$

Chapter 13

Lecture 13

Remark 13.0.1. *Last time we proved that if B and B' are two bases of a root system Δ, then there exists $w \in W_B$ such that*

$$w(B') = B.$$

We now prove that such an element is unique.

Proposition 13.0.2. *Let $w \in W_B$. Then*

$$w(B) = B \qquad \Longleftrightarrow \qquad w = 1.$$

Proof. The implication $w = 1 \Rightarrow w(B) = B$ is clear. Conversely, suppose that $w(B) = B$. We prove that $w = 1$.

Assume, to the contrary, that $w \neq 1$. Write

$$w = S_{\beta_1} S_{\beta_2} \cdots S_{\beta_m}, \qquad \beta_i \in B,$$

as a shortest expression. Thus $m \geq 1$.

We claim that

$$w(\beta_m) \in \Delta^-.$$

If $m = 1$, this is immediate, since

$$S_{\beta_1}(\beta_1) = -\beta_1 \in \Delta^-.$$

Suppose now that $m > 1$, and suppose for contradiction that

$$w(\beta_m) \in \Delta^+.$$

Since

$$S_{\beta_m}(\beta_m) = -\beta_m \in \Delta^-,$$

we have

$$S_{\beta_1} S_{\beta_2} \cdots S_{\beta_{m-1}}(\beta_m) \in \Delta^-.$$

Consider the sequence

$$\beta_m, \quad S_{\beta_{m-1}}(\beta_m), \quad S_{\beta_{m-2}}S_{\beta_{m-1}}(\beta_m), \quad \ldots, \quad S_{\beta_1}S_{\beta_2}\cdots S_{\beta_{m-1}}(\beta_m).$$

Its first term lies in Δ^+, while its last term lies in Δ^-. Hence there exists k such that

$$S_{\beta_{k+1}}\cdots S_{\beta_{m-1}}(\beta_m) \in \Delta^+$$

but

$$S_{\beta_k}S_{\beta_{k+1}}\cdots S_{\beta_{m-1}}(\beta_m) \in \Delta^-.$$

By the standard property of simple reflections, if $\alpha \in \Delta^+$, $\beta \in B$, and $S_\beta(\alpha) \in \Delta^-$, then $\alpha = \beta$. Therefore

$$S_{\beta_{k+1}}\cdots S_{\beta_{m-1}}(\beta_m) = \beta_k.$$

Set

$$u = S_{\beta_{k+1}}\cdots S_{\beta_{m-1}}.$$

Then

$$\beta_k = u(\beta_m),$$

and hence

$$S_{\beta_k} = uS_{\beta_m}u^{-1}.$$

It follows that

$$\begin{aligned} w &= S_{\beta_1}\cdots S_{\beta_{k-1}}S_{\beta_k}S_{\beta_{k+1}}\cdots S_{\beta_{m-1}}S_{\beta_m} \\ &= S_{\beta_1}\cdots S_{\beta_{k-1}}\big(uS_{\beta_m}u^{-1}\big)uS_{\beta_m} \\ &= S_{\beta_1}\cdots S_{\beta_{k-1}}u \\ &= S_{\beta_1}\cdots S_{\beta_{k-1}}S_{\beta_{k+1}}\cdots S_{\beta_{m-1}}. \end{aligned}$$

This is a shorter expression for w, contradicting the choice of

$$w = S_{\beta_1}\cdots S_{\beta_m}.$$

Thus

$$w(\beta_m) \in \Delta^-.$$

On the other hand, since $w(B) = B$ and $\beta_m \in B$, we have

$$w(\beta_m) \in B \subseteq \Delta^+,$$

a contradiction. Therefore $w = 1$. □

Corollary 13.0.3. *If B and B' are two bases of Δ, then there is at most one element $w \in W_B$ such that*

$$w(B') = B.$$

Proof. Suppose that $w_1, w_2 \in W_B$ satisfy

$$w_1(B') = B \qquad \text{and} \qquad w_2(B') = B.$$

Then

$$w_2 w_1^{-1}(B) = B.$$

By the proposition,

$$w_2 w_1^{-1} = 1.$$

Hence $w_1 = w_2$. $\square$

Definition 13.0.4 (Normal system)**.** *Let E be a Euclidean space. A set of vectors*

$$v_1, \ldots, v_k \in E$$

is called a normal system *if the following conditions hold:*

1. *$v_1, \ldots, v_k$ are linearly independent;*
2. $$\|v_i\| = 1 \qquad \textit{for all } i;$$
3. *for $i \neq j$,*
$$(v_i \mid v_j) \in \left\{0, -\frac{1}{2}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{3}}{2}\right\}.$$

Example 13.0.5. *Let*

$$B = \{\beta_1, \ldots, \beta_n\}$$

be a base of a root system Δ. Then

$$\left\{\frac{\beta_1}{\|\beta_1\|}, \ldots, \frac{\beta_n}{\|\beta_n\|}\right\}$$

is a normal system.

Definition 13.0.6 (Coxeter graph associated with a normal system)**.** *Let*

$$\{v_1, \ldots, v_k\}$$

be a normal system. Its Coxeter graph *is defined as follows:*

- *the vertices are $v_1, \ldots, v_k$;*
- *for $i \neq j$, put*
$$4(v_i \mid v_j)^2$$
edges between v_i and v_j.

Thus two vertices are connected precisely when

$$(v_i \mid v_j) \neq 0.$$

Proposition 13.0.7 (Properties of normal systems and Coxeter graphs)**.** *Let*

$$\{v_1, \dots, v_k\}$$

be a normal system. Then the following hold.

1. *Every subset of a normal system is again a normal system.*

2. *The number of connected pairs $v_i - v_j$ is at most $k-1$.*

3. *The Coxeter graph contains no cycles.*

4. *Fix v_i. Then the total number of edges adjacent to v_i is strictly less than 4. Equivalently, it is at most 3.*

5. *Suppose $v_1, \dots, v_k$ form a simple chain in the Coxeter graph:*

$$v_1 - v_2 - \cdots - v_{k-1} - v_k,$$

namely

$$(v_i \mid v_j) = \begin{cases} -\dfrac{1}{2}, & j = i+1, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$v = v_1 + \cdots + v_k.$$

Replacing $v_1, \dots, v_k$ by v gives a new normal system.

Proof. 1. This is immediate from the definition of a normal system.

2. Let

$$v = v_1 + \cdots + v_k.$$

Since $v_1, \dots, v_k$ are linearly independent, we have $v \neq 0$. Hence

$$(v \mid v) > 0.$$

On the other hand,

$$\begin{aligned} (v \mid v) &= \sum_{i=1}^{k} (v_i \mid v_i) + 2 \sum_{i<j} (v_i \mid v_j) \\ &\leq k - \#\{\text{connected pairs}\}, \end{aligned}$$

because if v_i and v_j are connected, then

$$(v_i \mid v_j) \leq -\frac{1}{2}.$$

Therefore

$$\#\{\text{connected pairs}\} < k.$$

Since this number is an integer, it is at most $k-1$.

3. If the Coxeter graph contained a cycle with r vertices, then the corresponding r vectors would form a normal subsystem whose Coxeter graph has at least r connected pairs. This contradicts (2), which says that a normal system of r vectors has at most $r-1$ connected pairs.

4. Let

$$v_{j_1}, \ldots, v_{j_l}$$

be the vertices adjacent to v_i, and let k_s denote the number of edges between v_i and v_{j_s}. Thus

$$k_s = 4(v_i \mid v_{j_s})^2.$$

By (3), the vertices $v_{j_1}, \ldots, v_{j_l}$ must be pairwise orthogonal. Otherwise we would obtain a cycle.

Consider

$$u = v_i - \sum_{s=1}^{l} (v_i \mid v_{j_s}) v_{j_s}.$$

Since the vectors in a normal system are linearly independent, we have $u \neq 0$. Hence

$$(u \mid u) > 0.$$

Since $v_{j_1}, \ldots, v_{j_l}$ are pairwise orthogonal and have norm 1, we get

$$\begin{aligned}(u \mid u) &= (v_i \mid v_i) - \sum_{s=1}^{l} (v_i \mid v_{j_s})^2 \\ &= 1 - \frac{1}{4} \sum_{s=1}^{l} k_s.\end{aligned}$$

Therefore

$$\sum_{s=1}^{l} k_s < 4.$$

Hence the total number of edges adjacent to v_i is strictly less than 4.

5. Let

$$S = \{v_1, \ldots, v_k\} \cup T$$

be the original normal system, where T denotes the remaining vectors. Set

$$v = v_1 + \cdots + v_k.$$

We show that

$$S' = \{v\} \cup T$$

is again a normal system.

First, S' is linearly independent. Indeed, if

$$av + \sum_{w \in T} a_w w = 0,$$

then

$$a(v_1 + \cdots + v_k) + \sum_{w \in T} a_w w = 0.$$

Since the original system S is linearly independent, we get

$$a = 0 \qquad \text{and} \qquad a_w = 0 \quad \text{for all } w \in T.$$

Next,

$$\begin{aligned}
(v \mid v) &= (v_1 + \cdots + v_k \mid v_1 + \cdots + v_k) \\
&= \sum_{i=1}^{k} (v_i \mid v_i) + 2 \sum_{i=1}^{k-1} (v_i \mid v_{i+1}) \\
&= k + 2(k-1)\left(-\frac{1}{2}\right) \\
&= 1.
\end{aligned}$$

Hence $\|v\| = 1$.

Finally, let $w \in T$. By (3), the vertex w can be connected to at most one of $v_1, \ldots, v_k$; otherwise the Coxeter graph would contain a cycle. Hence

$$(w \mid v) = (w \mid v_1) + \cdots + (w \mid v_k)$$

is either 0, or equal to one of the inner products $(w \mid v_i)$. Therefore

$$(w \mid v) \in \left\{0, -\frac{1}{2}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{3}}{2}\right\}.$$

All other inner products are unchanged. Thus S' is a normal system.

□

Chapter 14

Lecture 14

14.1 Classification of Coxeter Graphs

Let Γ be the Coxeter graph associated with a set of simple roots. We normalize the corresponding vectors v_i so that

$$(v_i, v_i) = 1.$$

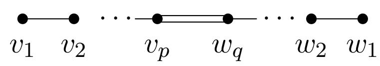
Recall that, from the previous discussion, for every vertex v_i, the number of edges adjacent to v_i, counted with multiplicity, is at most 4.

We shall classify the possible connected Coxeter graphs. The following reductions are used.

1. If a triple edge occurs, then the graph must be the whole graph:

   This gives the graph of type G_2.

2. If a double edge occurs, then the graph is a chain with a double edge somewhere in the chain. Thus it has the form

3. If all edges are single, then the graph is either a chain, or a tree with one branching point of degree 3.

14.1.1 The case of a double edge

Assume that the graph contains a double edge. Let the two chains attached to the double edge have lengths p and q, respectively. Define

$$v = v_1 + 2v_2 + \cdots + pv_p, \qquad w = w_1 + 2w_2 + \cdots + qw_q.$$

Since v and w are linearly independent, by the strict Cauchy–Schwarz inequality,

$$(v,w)^2 < (v,v)(w,w).$$

We compute:

$$\begin{aligned}(v,v) &= 1^2 + 2^2 + \cdots + p^2 - 2 \cdot \frac{1}{2}\big(1\cdot 2 + 2\cdot 3 + \cdots + (p-1)p\big) \\ &= p^2 - \frac{p(p-1)}{2} = \frac{p(p+1)}{2}.\end{aligned}$$

Similarly,

$$(w,w) = \frac{q(q+1)}{2}.$$

Since v_p and w_q are joined by a double edge,

$$(v_p, w_q) = -\frac{\sqrt{2}}{2}.$$

Hence

$$(v,w) = -pq\frac{\sqrt{2}}{2},$$

and therefore

$$(v,w)^2 = \frac{p^2q^2}{2}.$$

Thus

$$\frac{p^2q^2}{2} < \frac{p(p+1)}{2} \cdot \frac{q(q+1)}{2}.$$

Cancelling $pq > 0$, we obtain

$$2pq < (p+1)(q+1).$$

Equivalently,

$$pq - p - q + 1 < 2,$$

that is,

$$(p-1)(q-1) < 2.$$

Hence either $p = 1$, or $q = 1$, or

$$p = q = 2.$$

The first two possibilities give the Coxeter graph of type B_n or C_n, while the last possibility gives the Coxeter graph of type F_4.

14.1.2 The simply-laced branching case

Now assume that all edges are single and that the graph has a branching point. Let the branching vertex be γ, and let the three arms have lengths p, q, r. Write

$$v = v_1 + 2v_2 + \cdots + pv_p,$$

$$w = w_1 + 2w_2 + \cdots + qw_q,$$

$$u = u_1 + 2u_2 + \cdots + ru_r.$$

The vectors v, w, u are pairwise orthogonal. Extend

$$\frac{v}{\sqrt{(v,v)}}, \qquad \frac{w}{\sqrt{(w,w)}}, \qquad \frac{u}{\sqrt{(u,u)}}$$

to an orthonormal basis. Since $(\gamma, \gamma) = 1$, we have

$$\frac{(\gamma,v)^2}{(v,v)} + \frac{(\gamma,w)^2}{(w,w)} + \frac{(\gamma,u)^2}{(u,u)} < 1.$$

Here the inequality is strict because γ, v, w, u are linearly independent.

As before,

$$(v,v) = \frac{p(p+1)}{2}, \qquad (w,w) = \frac{q(q+1)}{2}, \qquad (u,u) = \frac{r(r+1)}{2}.$$

Since γ is joined to v_p, w_q, u_r by single edges,

$$(\gamma,v) = -\frac{p}{2}, \qquad (\gamma,w) = -\frac{q}{2}, \qquad (\gamma,u) = -\frac{r}{2}.$$

Therefore

$$\frac{p}{2(p+1)} + \frac{q}{2(q+1)} + \frac{r}{2(r+1)} < 1.$$

Equivalently,

$$\frac{1}{p+1} + \frac{1}{q+1} + \frac{1}{r+1} > 1.$$

Assume, without loss of generality, that

$$p \le q \le r.$$

Then

$$\frac{3}{p+1} > 1,$$

so $p = 1$.

Now

$$\frac{1}{2} + \frac{1}{q+1} + \frac{1}{r+1} > 1.$$

Hence

$$\frac{1}{q+1} + \frac{1}{r+1} > \frac{1}{2}.$$

Since $q \le r$, this forces $q = 1$ or $q = 2$.

- If $q = 1$, then r may be arbitrary. This gives the family D_n.
- If $q = 2$, then

  $$\frac{1}{2} + \frac{1}{3} + \frac{1}{r+1} > 1,$$

  so

  $$\frac{1}{r+1} > \frac{1}{6}.$$

Hence

$$r + 1 < 6,$$

and therefore

$$r = 2, 3, 4.$$

These give the exceptional types

$$E_6, \qquad E_7, \qquad E_8.$$

14.1.3 Complete classification

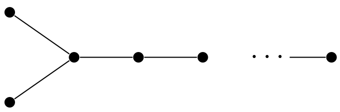
Combining the preceding cases, the connected Coxeter graphs are precisely the following.

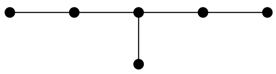
G_2

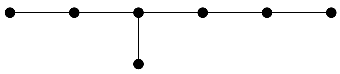
A_n

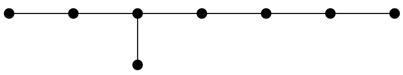
B_n or C_n

F_4

D_n 

E_6 

E_7 

E_8 

Definition 14.1.1 (Dynkin diagram)**.** *Let Φ be a root system. Whenever a double or triple edge occurs in the Coxeter graph of Φ, we add an arrow pointing to the shorter of the two roots. The resulting oriented graph is called the* Dynkin diagram *of Φ.*

Thus the Coxeter graphs of B_n and C_n have the same underlying unoriented graph, but their Dynkin diagrams are distinguished by the direction of the arrow:

B_n :

C_n :

14.2 Realization of Root Systems

Let

$$E = \mathbb{R}^n$$

with the usual inner product, and let

$$w_1, \ldots, w_n$$

be the usual orthonormal basis of $\mathbb{R}^n$. The $\mathbb{Z}$-span of this basis is a lattice, denoted by

$$I = \sum_{i=1}^{n} \mathbb{Z} w_i.$$

The root systems below will be defined as sets of vectors in a suitable lattice having prescribed length or lengths.

Remark 14.2.1. *To check that such a set Δ is a root system, one usually verifies the following points. First, Δ is finite, spans the ambient Euclidean space, and does not contain 0. Also,*

$$\mathbb{R}\alpha \cap \Delta = \{\alpha, -\alpha\} \qquad \textit{for all } \alpha \in \Delta.$$

Moreover, for $\alpha, \beta \in \Delta$,

$$\langle \beta, \alpha^\vee \rangle = \frac{2(\beta, \alpha)}{(\alpha, \alpha)} \in \mathbb{Z}.$$

Finally, one checks that the reflection

$$s_\alpha(\beta) = \beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)} \alpha$$

maps Δ into itself. Since s_α preserves lengths, it is often enough to check that $s_\alpha(\beta)$ lies in the relevant lattice.

14.2.1 The root system A_n

Let E be the n-dimensional subspace of $\mathbb{R}^{n+1}$ orthogonal to

$$w_1 + \cdots + w_{n+1}.$$

Set

$$I' = I \cap E = \left\{ \sum_{i=1}^{n+1} k_i w_i \;\middle|\; k_i \in \mathbb{Z},\ \sum_{i=1}^{n+1} k_i = 0 \right\}.$$

Define

$$A_n = \left\{ \alpha \in I' \;\middle|\; (\alpha, \alpha) = 2 \right\}.$$

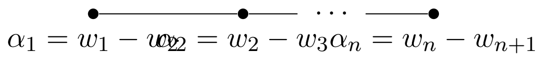
Then

$$A_n = \{\, w_i - w_j \mid 1 \leq i \neq j \leq n+1 \,\}.$$

A choice of simple roots is

$$\alpha_1 = w_1 - w_2, \quad \alpha_2 = w_2 - w_3, \quad \ldots, \quad \alpha_n = w_n - w_{n+1}.$$

Thus the Dynkin diagram is



The corresponding positive and negative roots are

$$\Delta^+ = \{\, w_i - w_j \mid i < j \,\}, \qquad \Delta^- = \{\, w_i - w_j \mid i > j \,\}.$$

For $\alpha_i = w_i - w_{i+1}$, the reflection s_{α_i} interchanges the subscripts i and $i+1$, and leaves all other subscripts fixed. Hence s_{α_i} corresponds to the transposition $(i\ i+1)$ in S_{n+1}. Since the Weyl group is generated by the simple reflections, we obtain

$$W(A_n) \cong S_{n+1}.$$

14.2.2 The root system D_n

Let $E = \mathbb{R}^n$. Define

$$D_n = \{\alpha \in I \mid (\alpha, \alpha) = 2\}.$$

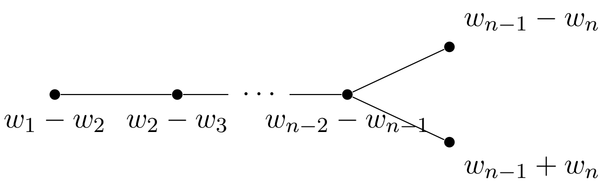
Then

$$D_n = \{\, \pm w_i \pm w_j \mid 1 \le i < j \le n \,\}.$$

A choice of simple roots is

$$\alpha_1 = w_1 - w_2, \quad \alpha_2 = w_2 - w_3, \quad \ldots, \quad \alpha_{n-1} = w_{n-1} - w_n, \quad \alpha_n = w_{n-1} + w_n.$$

The Dynkin diagram is



14.2.3 The root system B_n

Let $E = \mathbb{R}^n$. Define

$$B_n = \{\alpha \in I \mid (\alpha, \alpha) = 1 \text{ or } 2\}.$$

Then

$$B_n = \{\, \pm w_i,\ \pm w_i \pm w_j \mid 1 \le i < j \le n \,\}.$$

A choice of simple roots is

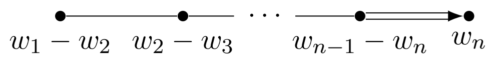
$$\alpha_1 = w_1 - w_2, \quad \alpha_2 = w_2 - w_3, \quad \ldots, \quad \alpha_{n-1} = w_{n-1} - w_n, \quad \alpha_n = w_n.$$

Notice that every short root is expressible in this basis. For example,

$$w_i = (w_i - w_{i+1}) + (w_{i+1} - w_{i+2}) + \cdots + (w_{n-1} - w_n) + w_n.$$

Similarly, the long roots $w_i - w_j$ and $w_i + w_j$ are expressible in this basis.

The Dynkin diagram is



The arrow points toward the shorter root w_n.

14.2.4 The root system C_n

The root system C_n is dual to B_n. Thus

$$C_n = \{ \pm 2w_i,\ \pm w_i \pm w_j \mid 1 \le i < j \le n \}.$$

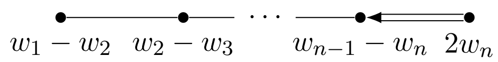
Equivalently, it is obtained from B_n by replacing each root α by its coroot

$$\alpha^\vee = \frac{2\alpha}{(\alpha, \alpha)}.$$

A choice of simple roots is

$$\alpha_1 = w_1 - w_2, \quad \alpha_2 = w_2 - w_3, \quad \ldots, \quad \alpha_{n-1} = w_{n-1} - w_n, \quad \alpha_n = 2w_n.$$

The Dynkin diagram is



Here the arrow points toward the shorter root $w_{n-1} - w_n$.

14.2.5 The root system F_4

Let $E = \mathbb{R}^4$, and set

$$I' = I + \frac{\mathbb{Z}}{2}(w_1 + w_2 + w_3 + w_4).$$

Define

$$F_4 = \left\{ \alpha \in I' \;\middle|\; (\alpha, \alpha) = 1 \text{ or } 2 \right\}.$$

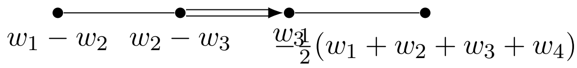
Then

$$F_4 = \left\{ \pm w_i,\ \pm w_i \pm w_j,\ \frac{1}{2}(\pm w_1 \pm w_2 \pm w_3 \pm w_4) \;\middle|\; 1 \le i < j \le 4 \right\}.$$

A choice of simple roots is

$$\alpha_1 = w_1 - w_2, \qquad \alpha_2 = w_2 - w_3, \qquad \alpha_3 = w_3, \qquad \alpha_4 = -\frac{1}{2}(w_1 + w_2 + w_3 + w_4).$$

The Dynkin diagram is



14.2.6 The root system G_2

There is also an exceptional realization of G_2 in terms of the octonions:

$$G_2 = \mathrm{Der}(\mathbb{O}).$$

Over $\mathbb{C}$, one obtains the complex Lie algebra of type G_2 by complexifying:

$$\mathrm{Der}(\mathbb{O} \otimes_{\mathbb{R}} \mathbb{C}).$$

For the abstract root system, take E to be the 2-dimensional subspace of $\mathbb{R}^3$ orthogonal to

$$w_1 + w_2 + w_3.$$

Let

$$I' = I \cap E.$$

Define

$$G_2 = \left\{ \alpha \in I' \;\middle|\; (\alpha, \alpha) = 2 \text{ or } 6 \right\}.$$

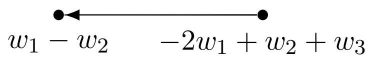
Then

$$G_2 = \{\, \pm(w_i - w_j) \mid 1 \leq i < j \leq 3 \,\} \cup \{\, \pm(2w_i - w_j - w_k) \mid \{i, j, k\} = \{1, 2, 3\} \,\}.$$

A choice of simple roots is

$$\alpha_1 = w_1 - w_2, \qquad \alpha_2 = -2w_1 + w_2 + w_3.$$

The Dynkin diagram is



The arrow points toward the shorter root $w_1 - w_2$.

Chapter 15

Homework

15.1 Homework week 1

Exercise 15.1.1. *Prove the simplicity of Lie algebras $\mathfrak{sl}_n$, $\mathfrak{o}_n$, $\mathfrak{sp}_{2n}$ and $W(n)$.*

I can only prove when base field is of characteristic 0 and $\neq 2$. (Or $\operatorname{Char} F \nmid n$, which is essentially the same. If $\operatorname{Char} F \mid n$ or $\operatorname{Char} F = 2$, then either I will contained in the center of L, or the alternativity become symmetry, things could be more complicated).

The simplicity of $\mathfrak{sl}_n$

Proof. Let $I \neq 0$ be an ideal of $\mathfrak{sl}_n(F)$. We prove that $I = \mathfrak{sl}_n(F)$.

Write

$$\mathfrak{sl}_n(F) = \mathfrak{h} \oplus \bigoplus_{i \neq j} F e_{ij},$$

where $\mathfrak{h}$ is the subalgebra of diagonal trace-zero matrices. Choose $0 \neq x \in I$ and write

$$x = y + \sum_{t=1}^{s} c_t e_{i_t j_t}, \qquad y \in \mathfrak{h},$$

where the pairs (i_t, j_t) are distinct and $c_t \neq 0$ for all t.

We first show that I contains some off-diagonal matrix unit e_{kl}.

If $s = 0$, then $x = y \in \mathfrak{h}$ is non-zero. Writing

$$y = \operatorname{diag}(d_1, \ldots, d_n),$$

we can choose $k \neq l$ such that $d_k \neq d_l$. Then

$$[y, e_{kl}] = (d_k - d_l) e_{kl} \in I,$$

hence $e_{kl} \in I$.

Assume now that $s \geq 1$. For

$$h = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \in \mathfrak{h},$$

define

$$\alpha_{ij}(h) = \lambda_i - \lambda_j.$$

Since $\operatorname{char} F = 0$, the field F is infinite, so we may choose $h \in \mathfrak{h}$ such that

$$\mu_t := \alpha_{i_t j_t}(h) \neq 0 \qquad \text{and} \qquad \mu_1, \dots, \mu_s \text{ are pairwise distinct.}$$

Because $\mathfrak{h}$ is abelian, for every $r \geq 1$,

$$(\operatorname{ad} h)^r(x) = \sum_{t=1}^{s} c_t \mu_t^r e_{i_t j_t}.$$

Put

$$z_r := (\operatorname{ad} h)^r(x) \in I, \qquad r = 1, \dots, s.$$

Then

$$\begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_s \end{pmatrix} = \begin{pmatrix} \mu_1 & \mu_2 & \cdots & \mu_s \\ \mu_1^2 & \mu_2^2 & \cdots & \mu_s^2 \\ \vdots & \vdots & \ddots & \vdots \\ \mu_1^s & \mu_2^s & \cdots & \mu_s^s \end{pmatrix} \begin{pmatrix} c_1 e_{i_1 j_1} \\ c_2 e_{i_2 j_2} \\ \vdots \\ c_s e_{i_s j_s} \end{pmatrix}.$$

The coefficient matrix is invertible: it is the product of $\operatorname{diag}(\mu_1, \dots, \mu_s)$ and a Vandermonde matrix in $\mu_1, \dots, \mu_s$. Therefore each $c_t e_{i_t j_t}$ is a linear combination of $z_1, \dots, z_s$, so $e_{i_t j_t} \in I$ for every t. In particular, I contains some e_{kl} with $k \neq l$.

Fix such an element $e_{kl} \in I$. Since I is an ideal,

$$h_{kl} := [e_{kl}, e_{lk}] = e_{kk} - e_{ll} \in I.$$

For every $j \neq k$,

$$[h_{kl}, e_{kj}] = \begin{cases} 2e_{kl}, & j = l, \\ e_{kj}, & j \neq l, \end{cases}$$

so $e_{kj} \in I$ for all $j \neq k$. Similarly, for every $i \neq k$,

$$[e_{ik}, h_{kl}] = \begin{cases} 2e_{lk}, & i = l, \\ e_{ik}, & i \neq l, \end{cases}$$

hence $e_{ik} \in I$ for all $i \neq k$.

Now let $i \neq j$. If $i = k$ or $j = k$, then we already know $e_{ij} \in I$. If $i, j \neq k$, then

$$e_{ij} = [e_{ik}, e_{kj}] \in I.$$

Thus every off-diagonal matrix unit e_{ij} belongs to I.

Finally, for $i \neq j$,

$$[e_{ij}, e_{ji}] = e_{ii} - e_{jj} \in I.$$

These elements span $\mathfrak{h}$, hence $\mathfrak{h} \subseteq I$. Therefore

$$\mathfrak{sl}_n(F) = \mathfrak{h} \oplus \bigoplus_{i \neq j} Fe_{ij} \subseteq I,$$

so $I = \mathfrak{sl}_n(F)$. Hence $\mathfrak{sl}_n(F)$ is simple. $\square$

The simplicity of $\mathfrak{o}_n$ 9‘=’

Proof. Let $\mathfrak{g} = \mathfrak{o}_n(F)$. For $i \neq j$, set

$$X_{ij} = E_{ij} - E_{ji},$$

so $X_{ji} = -X_{ij}$ and $\{X_{ij} \mid 1 \leq i < j \leq n\}$ is a basis of $\mathfrak{g}$. The bracket is

$$[X_{ij}, X_{kl}] = \delta_{jk} X_{il} - \delta_{ik} X_{jl} - \delta_{jl} X_{ik} + \delta_{il} X_{jk}.$$

If $n = 1$ or $n = 2$, then $\mathfrak{g}$ is abelian, hence not simple. For $n = 4$, let

$$I_+ = \operatorname{span}\{X_{12} + X_{34},\ X_{13} - X_{24},\ X_{14} + X_{23}\},$$

$$I_- = \operatorname{span}\{X_{12} - X_{34},\ X_{13} + X_{24},\ X_{14} - X_{23}\}.$$

A direct computation shows that I_+ and I_- are ideals, $[I_+, I_-] = 0$, and

$$\mathfrak{o}_4 = I_+ \oplus I_-.$$

Therefore $\mathfrak{o}_4$ is not simple.

Now assume $n = 3$ or $n \geq 5$, and let $I \neq 0$ be an ideal of $\mathfrak{g}$. It is enough to show that I contains some basis element X_{pq}. Indeed, if $X_{pq} \in I$ and $r \notin \{p, q\}$, then

$$[X_{pq}, X_{qr}] = X_{pr}, \qquad [X_{rp}, X_{pq}] = X_{rq},$$

so repeated bracketing gives every basis element of $\mathfrak{g}$.

For $i \neq j$, define

$$P_{ij}(x) = -[X_{ij}, [X_{ij}, x]].$$

Since I is an ideal, $P_{ij}(I) \subseteq I$. Moreover,

$$P_{ij}(X_{kl}) = -[X_{ij}, [X_{ij}, X_{kl}]]$$

and this can be computed directly from the bracket formula.

If $|\{i, j\} \cap \{k, l\}| = 0$, then all Kronecker deltas vanish, so

$$[X_{ij}, X_{kl}] = 0, \qquad \text{hence} \qquad P_{ij}(X_{kl}) = 0.$$

If $|\{i, j\} \cap \{k, l\}| = 2$, then $\{k, l\} = \{i, j\}$, so $X_{kl} = \pm X_{ij}$. Hence

$$[X_{ij}, X_{kl}] = 0, \qquad \text{and again} \qquad P_{ij}(X_{kl}) = 0.$$

Now assume $|\{i,j\}\cap\{k,l\}|=1$. There are four possibilities:

$$[X_{ij},X_{jl}]=X_{il},\qquad P_{ij}(X_{jl})=-[X_{ij},X_{il}]=X_{jl},$$

$$[X_{ij},X_{il}]=-X_{jl},\qquad P_{ij}(X_{il})=-[X_{ij},-X_{jl}]=X_{il},$$

$$[X_{ij},X_{ki}]=X_{jk},\qquad P_{ij}(X_{ki})=-[X_{ij},X_{jk}]=-X_{ik}=X_{ki},$$

$$[X_{ij},X_{kj}]=-X_{ki},\qquad P_{ij}(X_{kj})=-[X_{ij},-X_{ki}]=X_{kj}.$$

Therefore

$$P_{ij}(X_{kl})=\begin{cases}X_{kl}, & |\{i,j\}\cap\{k,l\}|=1,\\ 0, & |\{i,j\}\cap\{k,l\}|=0 \text{ or } 2.\end{cases}$$

If $n=3$, write

$$x=c_{12}X_{12}+c_{13}X_{13}+c_{23}X_{23}\in I,\qquad x\neq 0.$$

Then

$$x-P_{12}(x)=c_{12}X_{12},\qquad x-P_{13}(x)=c_{13}X_{13},\qquad x-P_{23}(x)=c_{23}X_{23}.$$

Since not all c_{ij} vanish, one of these is a non-zero scalar multiple of a basis element. Hence I contains a basis element, so $I=\mathfrak{o}_3$.

Assume now $n\geq 5$. Choose

$$0\neq x=\sum_{1\leq a<b\leq n}c_{ab}X_{ab}\in I,$$

and relabel indices so that $c_{12}\neq 0$. Set

$$x_1=P_{13}(x),\qquad x_2=P_{24}(x_1),\qquad x_3=P_{15}(x_2),\qquad x_4=P_{25}(x_3).$$

By the description of P_{ij},

$$x_2\in\operatorname{span}\{X_{12},X_{14},X_{23},X_{34}\},$$

$$x_3\in\operatorname{span}\{X_{12},X_{14}\},$$

and therefore

$$x_4=c_{12}X_{12}\in I.$$

Thus $X_{12}\in I$, hence $I=\mathfrak{o}_n$.

Therefore $\mathfrak{o}_n$ is simple exactly when $n=3$ or $n\geq 5$. □

The simplicity of $\mathfrak{sp}_{2n}$

Proof. If $n=1$, then $\mathfrak{sp}_2(F)\cong\mathfrak{sl}_2(F)$, which is simple. So assume $n\geq 2$, and let

$$\mathfrak{g}=\mathfrak{sp}_{2n}(F)=\left\{\begin{pmatrix}A & B\\ C & -A^T\end{pmatrix}\;\middle|\; B^T=B,\ C^T=C\right\}.$$

For $1 \le i, j \le n$, define

$$A_{ij} = E_{ij} - E_{n+j,n+i}, \qquad B_{ij} = E_{i,n+j} + E_{j,n+i}, \qquad C_{ij} = E_{n+i,j} + E_{n+j,i}.$$

Then $B_{ij} = B_{ji}$ and $C_{ij} = C_{ji}$, and a basis of $\mathfrak{g}$ is

$$\{A_{ij} \mid i \neq j\} \cup \{H_i := A_{ii} \mid 1 \le i \le n\} \cup \{B_{ij}, C_{ij} \mid 1 \le i \le j \le n\}.$$

Let $\mathfrak{h} = \operatorname{span}\{H_1, \dots, H_n\}$. A direct computation gives

$$[H_r, A_{ij}] = (\delta_{ri} - \delta_{rj})A_{ij},$$

$$[H_r, B_{ij}] = (\delta_{ri} + \delta_{rj})B_{ij}, \qquad [H_r, C_{ij}] = -(\delta_{ri} + \delta_{rj})C_{ij}.$$

Thus every basis element outside $\mathfrak{h}$ is a common eigenvector for $\operatorname{ad}\mathfrak{h}$.

Let $I \neq 0$ be an ideal of $\mathfrak{g}$. Choose $0 \neq x \in I$. If $x \notin \mathfrak{h}$, then exactly as in the proof for $\mathfrak{sl}_n$, a Vandermonde argument with a suitable $h \in \mathfrak{h}$ isolates a non-zero basis vector in I. If $x \in \mathfrak{h}$, write

$$x = \lambda_1 H_1 + \cdots + \lambda_n H_n,$$

and choose i with $\lambda_i \neq 0$. Then

$$[x, B_{ii}] = 2\lambda_i B_{ii} \in I.$$

Hence in all cases I contains one of the basis elements A_{ij} $(i \neq j)$, B_{ij}, or C_{ij}.

We now show that this already implies $I = \mathfrak{g}$.

First suppose that $B_{ii} \in I$ for some i. Then for every $j \neq i$,

$$A_{ij} = \frac{1}{2}[B_{ii}, C_{ij}] \in I, \qquad B_{ij} = \frac{1}{2}[A_{ij}, B_{jj}] \in I, \qquad C_{ij} = -\frac{1}{2}[A_{ij}, C_{ii}] \in I.$$

Also,

$$H_i = \frac{1}{4}[B_{ii}, C_{ii}] \in I.$$

For $p, q \neq i$ with $p \neq q$, we then have

$$A_{pq} = [B_{pi}, C_{iq}] \in I, \qquad B_{pq} = [A_{pi}, B_{iq}] \in I, \qquad C_{pq} = -[A_{ip}, C_{iq}] \in I.$$

For $p \neq i$, the same computations give

$$B_{pp} = [A_{pi}, B_{ip}] \in I, \qquad C_{pp} = -[A_{ip}, C_{ip}] \in I.$$

Finally, for $p \neq i$,

$$H_i + H_p = [B_{ip}, C_{ip}] \in I,$$

so $H_p \in I$. Therefore every basis element of $\mathfrak{g}$ lies in I, and hence $I = \mathfrak{g}$.

If $B_{ij} \in I$ with $i \neq j$, then

$$H_i + H_j = [B_{ij}, C_{ij}] \in I,$$

and hence

$$B_{ii} = \frac{1}{2}[H_i + H_j, B_{ii}] \in I.$$

So we again reduce to the first case.

If $A_{ij} \in I$ with $i \neq j$, then

$$B_{ij} = \frac{1}{2}[A_{ij}, B_{jj}] \in I,$$

and so

$$H_i + H_j = [B_{ij}, C_{ij}] \in I.$$

Consequently,

$$B_{ii} = \frac{1}{2}[H_i + H_j, B_{ii}] \in I,$$

and we reduce to the previous case.

If $C_{ii} \in I$, then

$$H_i = -\frac{1}{4}[C_{ii}, B_{ii}] \in I,$$

so

$$B_{ii} = \frac{1}{2}[H_i, B_{ii}] \in I.$$

Again we reduce to the first case.

If $C_{ij} \in I$ with $i \neq j$, then

$$H_i + H_j = [B_{ij}, C_{ij}] \in I,$$

hence

$$C_{ii} = -\frac{1}{2}[H_i + H_j, C_{ii}] \in I.$$

Also,

$$H_i = -\frac{1}{4}[C_{ii}, B_{ii}] \in I,$$

so

$$B_{ii} = \frac{1}{2}[H_i, B_{ii}] \in I.$$

Again we reduce to the first case.

Therefore every non-zero ideal of $\mathfrak{g}$ is equal to $\mathfrak{g}$, so $\mathfrak{sp}_{2n}(F)$ is simple. $\square$

The simplicity of $W(n)$

Proof. Let

$$W(n) = \mathrm{Der}\big(F[x_1, \ldots, x_n]\big) = \left\{ \sum_{i=1}^{n} f_i \partial_i \;\middle|\; f_i \in F[x_1, \ldots, x_n] \right\}, \qquad \partial_i = \frac{\partial}{\partial x_i},$$

where $\operatorname{char} F = 0$. This Lie algebra is non-abelian, since

$$[\partial_i, x_i] = 1 \neq 0.$$

Let $I \neq 0$ be an ideal of $W(n)$. We prove that $I = W(n)$.

Choose

$$0 \neq D = \sum_{i=1}^{n} f_i \partial_i \in I.$$

By replacing D with repeated brackets $[\partial_k, D]$, we may successively differentiate the coefficients. Since $\operatorname{char} F = 0$, differentiating a non-constant polynomial lowers its degree without annihilating its highest term. Hence, after finitely many such brackets, we obtain a non-zero constant vector field

$$C = \sum_{i=1}^{n} c_i \partial_i \in I.$$

Choose k with $c_k \neq 0$. Then for every j,

$$[x_k \partial_j, C] = \sum_{i=1}^{n} c_i [x_k \partial_j, \partial_i] = -c_k \partial_j \in I.$$

Therefore $\partial_j \in I$ for all $j = 1, \ldots, n$.

Now let $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$ and fix j. Since $\operatorname{char} F = 0$, the scalar $\alpha_1 + 1$ is invertible in F. Set

$$E = -\frac{1}{\alpha_1 + 1} x_1^{\alpha_1 + 1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} \partial_j \in W(n).$$

Because $\partial_1 \in I$ and I is an ideal,

$$[E, \partial_1] = -\partial_1 \left(-\frac{1}{\alpha_1 + 1} x_1^{\alpha_1 + 1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} \right) \partial_j = x_1^{\alpha_1} \cdots x_n^{\alpha_n} \partial_j \in I.$$

Thus every monomial derivation $x^\alpha \partial_j$ lies in I. These elements form a basis of $W(n)$, so $I = W(n)$.

Therefore $W(n)$ is simple. □

15.2 Homework week 2

Exercise 15.2.1. *Prove that if A is a finite-dimensional associative algebra and for every element $a \in A$ one has $a^n = 0$ for some n, then the algebra A is nilpotent. Do not use Wedderburn's theorems. Use Engel's theorem.*

Proof. Let A be a finite-dimensional associative algebra over a field, such that every element $a \in A$ is nilpotent. We want to show that A is nilpotent as an algebra, meaning there exists some integer $N > 0$ such that $A^N = 0$.

Consider the left regular representation of A on itself. For each $a \in A$, define the left multiplication map $L_a : A \to A$ by $L_a(x) = ax$ for all $x \in A$. Since A is an associative algebra, the mapping $a \mapsto L_a$ is an algebra homomorphism from A to the algebra of endomorphisms $\operatorname{End}(A)$.

We can view $\operatorname{End}(A)$ as a Lie algebra $\mathfrak{gl}(A)$ with the standard commutator bracket $[T, S] = T \circ S - S \circ T$. The set of all left multiplication maps $\mathfrak{g} = \{L_a \mid a \in A\}$ forms a Lie subalgebra of $\mathfrak{gl}(A)$, because

$$[L_a, L_b] = L_a \circ L_b - L_b \circ L_a = L_{ab} - L_{ba} = L_{[a,b]},$$

and $a, b \in A \implies [a, b] \in A$, so $L_{[a,b]} \in \mathfrak{g}$.

By hypothesis, for every $a \in A$, there exists an integer $n > 0$ such that $a^n = 0$. Consequently, the endomorphism L_a is strictly nilpotent, since $L_a^n(x) = a^n x = 0$ for all $x \in A$. Therefore, $\mathfrak{g}$ is a Lie subalgebra of $\mathfrak{gl}(A)$ consisting entirely of nilpotent endomorphisms.

According to Engel's Theorem, if a Lie subalgebra of $\mathfrak{gl}(V)$ (where V is a non-zero finite-dimensional vector space) consists of nilpotent elements, then there exists a flag of subspaces

$$0 = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_k = A$$

such that $\mathfrak{g}(V_i) \subseteq V_{i-1}$ for all $1 \leq i \leq k$.

Translating this back to the associative algebra A, we have $L_a(V_i) \subseteq V_{i-1}$ for all $a \in A$, which is precisely equivalent to the statement $A \cdot V_i \subseteq V_{i-1}$.

Using this property repeatedly, we obtain:

$$A \cdot V_k \subseteq V_{k-1}$$

$$A^2 \cdot V_k = A \cdot (A \cdot V_k) \subseteq A \cdot V_{k-1} \subseteq V_{k-2}$$

$$\cdots$$

$$A^k \cdot V_k \subseteq V_0 = 0.$$

Since $V_k = A$, we conclude that $A^{k+1} = A^k \cdot A \subseteq 0$. Thus, A is a nilpotent associative algebra. □

Exercise 15.2.2. *Let* $\operatorname{char} F = 0$*; let* L *be a finite-dimensional solvable Lie algebra; let* $e_1, \ldots, e_m$ *be a basis of the algebra* L*. Let* M *be a finite-dimensional* L*-module. Suppose that every element* e_i*, for* $1 \leq i \leq m$*, acts on* M *nilpotently. Prove that the algebra* L *acts on* M *nilpotently.*

Proof. Let $\bar{F}$ be the algebraic closure of F. We consider the scalar extensions of the Lie algebra and the module: $\bar{L} = L \otimes_F \bar{F}$ and $\bar{M} = M \otimes_F \bar{F}$.

Since L is a finite-dimensional solvable Lie algebra over F, its scalar extension $\bar{L}$ remains a solvable Lie algebra over $\bar{F}$. Furthermore, the elements $e_1, \ldots, e_m$ naturally form a basis for $\bar{L}$ as a vector space over $\bar{F}$.

Because $\operatorname{char} \bar{F} = 0$ and $\bar{F}$ is algebraically closed, we can apply Lie's Theorem to the finite-dimensional $\bar{L}$-module $\bar{M}$. Lie's Theorem guarantees the existence of a basis for $\bar{M}$ such that the representation $\rho : \bar{L} \to \mathrm{gl}(\bar{M})$ maps every element of $\bar{L}$ to an upper triangular matrix.

Consequently, for each $x \in \bar{L}$, the diagonal entries of the matrix $\rho(x)$ are given by a set of linear functionals $\lambda_1(x), \ldots, \lambda_n(x)$, where $n = \dim_{\bar{F}} \bar{M}$ and $\lambda_k \in \bar{L}^*$.

We are given that each basis element e_i $(1 \leq i \leq m)$ acts nilpotently on M. This property is preserved under scalar extension, so each e_i also acts nilpotently on $\bar{M}$. An upper triangular matrix that is nilpotent must have all its diagonal entries equal to zero. Thus, for each $i \in \{1, \ldots, m\}$ and each $k \in \{1, \ldots, n\}$, we have:

$$\lambda_k(e_i) = 0.$$

Since the linear functionals λ_k vanish on the basis $e_1, \ldots, e_m$, and linear transformations are determined by their action on a basis, they must vanish identically on the entire Lie algebra $\bar{L}$. That is, $\lambda_k(x) = 0$ for all $x \in \bar{L}$.

This implies that for any $x \in \bar{L}$, the matrix $\rho(x)$ is strictly upper triangular. Consequently, every $x \in \bar{L}$ acts as a nilpotent endomorphism on $\bar{M}$.

Restricting this result back to the original base field F and the original module M, we conclude that every element $x \in L$ acts nilpotently on M. By Engel's Theorem, a Lie algebra of nilpotent endomorphisms acts nilpotently, which completes the proof. □

Exercise 15.2.3. *For every prime p, find an example showing that Lie's theorem fails.*

Proof. Let F be an algebraically closed field of characteristic $p > 0$. We will construct a finite-dimensional solvable Lie algebra $\mathfrak{g}$ and a finite-dimensional representation V over F such that $\mathfrak{g}$ has no common eigenvector in V, thus showing that Lie's theorem fails.

Let V be a p-dimensional vector space over F with a chosen basis $v_0, v_1, \ldots, v_{p-1}$. We define two linear transformations $x, y \in \mathfrak{gl}(V)$ by defining their actions on the basis vectors:

$$x(v_i) = v_{i-1} \quad \text{for } 0 \leq i \leq p-1,$$
$$y(v_i) = iv_i \quad \text{for } 0 \leq i \leq p-1,$$

where the subscripts for the action of x are taken modulo p (so $x(v_0) = v_{p-1}$).

Let $\mathfrak{g}$ be the Lie subalgebra of $\mathfrak{gl}(V)$ generated by x and y. First, we compute the commutator $[x, y]$ acting on an arbitrary basis vector v_i:

$$[x, y](v_i) = x(y(v_i)) - y(x(v_i)) = x(iv_i) - y(v_{i-1}) = iv_{i-1} - (i-1)v_{i-1} = v_{i-1} = x(v_i).$$

Since this holds for all basis vectors, we have $[x, y] = x$. This shows that the vector space spanned by x and y is closed under the Lie bracket, so $\mathfrak{g} = \mathrm{span}_F\{x, y\}$ is a 2-dimensional Lie algebra. Furthermore, its derived algebra is $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}] = \mathrm{span}_F\{x\}$. Since $\mathfrak{g}'$ is 1-dimensional, it is abelian, which implies that $\mathfrak{g}$ is a solvable Lie algebra.

Next, we show that $\mathfrak{g}$ has no common eigenvector in V. Suppose, for the sake of contradiction, that $v = \sum_{i=0}^{p-1} c_i v_i$ is a common non-zero eigenvector for both x and y. Since v is an eigenvector of y, and the eigenvalues of y are exactly $0, 1, \ldots, p-1$ (which are all distinct in F because the characteristic is p), v must belong to one of the 1-dimensional eigenspaces of y. Thus, v must be a non-zero scalar multiple of v_k for some $0 \leq k \leq p-1$.

However, v must also be an eigenvector for x. We check the action of x on v_k:

$$x(v_k) = v_{k-1} \pmod{p}.$$

Because p is a prime (and thus $p \geq 2$), v_{k-1} is linearly independent from v_k. Therefore, v_k is not an eigenvector for x, which is a contradiction.

Consequently, the solvable Lie algebra $\mathfrak{g}$ has no common eigenvector in the representation V, demonstrating that Lie's theorem does not hold in characteristic p. □

15.3 Homework week 3

Exercise 15.3.1. *Let $x, y \in \operatorname{End} V$ commute. Prove that*

$$(x+y)_s = x_s + y_s, \qquad (x+y)_n = x_n + y_n.$$

Show by example that this can fail if x, y fail to commute. [Show first that x, y semisimple (resp. nilpotent) implies $x+y$ semisimple (resp. nilpotent).]

Lemma 15.3.2. *Let $a, b \in \operatorname{End}(V)$ commute. If both a and b are semisimple, then $a+b$ is semisimple.*

Proof. Since a is semisimple,

$$V = \bigoplus_{\lambda} V_\lambda(a).$$

Because $ab = ba$, each eigenspace $V_\lambda(a)$ is stable under b. Since b is semisimple, its restriction to each $V_\lambda(a)$ is again semisimple; hence each $V_\lambda(a)$ decomposes into eigenspaces of b. Thus there is a basis of V consisting of common eigenvectors of a and b. Relative to this basis, both a and b are diagonal, hence so is $a+b$. Therefore $a+b$ is semisimple. □

Lemma 15.3.3. *Let $a, b \in \operatorname{End}(V)$ commute. If both a and b are nilpotent, then $a+b$ is nilpotent.*

Proof. Choose $r, s \geq 1$ such that

$$a^r = 0, \qquad b^s = 0.$$

Since $ab = ba$, for any N we have

$$(a+b)^N = \sum_{k=0}^{N} \binom{N}{k} a^k b^{N-k}.$$

Take $N = r+s-1$. Then for each k, either $k \geq r$ or $N-k \geq s$, so every term in the above sum is zero. Hence

$$(a+b)^{r+s-1} = 0,$$

and therefore $a+b$ is nilpotent. □

Proof. Write

$$x = x_s + x_n, \qquad y = y_s + y_n$$

for the Jordan decompositions of x and y. Since x and y commute, and since each semisimple part and nilpotent part is given by a polynomial in the operator, it follows that

$$x_s,\ x_n,\ y_s,\ y_n$$

all commute with one another.

Now x_s and y_s are commuting semisimple operators, hence $x_s + y_s$ is semisimple. Also x_n and y_n are commuting nilpotent operators, hence $x_n + y_n$ is nilpotent. Therefore

$$x + y = (x_s + y_s) + (x_n + y_n)$$

is a decomposition of $x + y$ into commuting semisimple and nilpotent parts. By uniqueness of Jordan decomposition,

$$(x + y)_s = x_s + y_s, \qquad (x + y)_n = x_n + y_n.$$

To see that this may fail without the hypothesis $xy = yx$, take

$$x = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad y = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then x and y are semisimple, but

$$xy \neq yx,$$

and

$$x + y = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

is not semisimple. Thus

$$(x + y)_s \neq x_s + y_s.$$

Since $x_n = y_n = 0$ while $(x + y)_n \neq 0$, we also have

$$(x + y)_n \neq x_n + y_n.$$

□

Exercise 15.3.4. *Let $L = \mathfrak{sl}(2, F)$. Compute the basis of L dual to the standard basis, relative to the Killing form.*

Proof. Let

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the standard basis of $L = \mathfrak{sl}(2, F)$. We compute the Killing form

$$\kappa(x, y) = \mathrm{Tr}(\mathrm{ad}(x)\,\mathrm{ad}(y))$$

relative to the ordered basis (h, e, f).

Using

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h,$$

we get

$$\operatorname{ad}(h) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \qquad \operatorname{ad}(e) = \begin{pmatrix} 0 & 0 & 1 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \operatorname{ad}(f) = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

Hence

$$\kappa(h,h) = \operatorname{Tr}(\operatorname{ad}(h)^2) = 8,$$

$$\kappa(e,f) = \operatorname{Tr}(\operatorname{ad}(e)\operatorname{ad}(f)) = 4, \qquad \kappa(f,e) = 4,$$

and

$$\kappa(e,e) = \kappa(f,f) = \kappa(h,e) = \kappa(h,f) = 0.$$

Thus the Gram matrix of κ relative to (h,e,f) is

$$\begin{pmatrix} 8 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}.$$

It follows at once that the dual basis $(h^\vee, e^\vee, f^\vee)$ is given by

$$h^\vee = \frac{1}{8}h, \qquad e^\vee = \frac{1}{4}f, \qquad f^\vee = \frac{1}{4}e.$$

Therefore the basis of L dual to the standard basis with respect to the Killing form is

$$\boxed{\frac{1}{8}h,\ \frac{1}{4}f,\ \frac{1}{4}e.}$$

□

Exercise 15.3.5. *Relative to the standard basis of $\mathfrak{sl}(3,F)$, compute the determinant of κ. Which primes divide it?*

Proof. Let

$$e_{ij} \qquad (i \neq j)$$

be the usual matrix units, and let

$$h_1 = e_{11} - e_{22}, \qquad h_2 = e_{22} - e_{33}.$$

Take the standard basis of $\mathfrak{sl}(3,F)$ in the order

$$e_{12},\ e_{13},\ e_{21},\ e_{23},\ e_{31},\ e_{32},\ h_1,\ h_2.$$

We compute the Gram matrix of the Killing form

$$\kappa(x,y) = \operatorname{Tr}(\operatorname{ad}(x)\operatorname{ad}(y))$$

relative to this basis.

First consider the six root vectors. For $i \neq j$, one has

$$[e_{ij}, e_{ji}] = e_{ii} - e_{jj},$$

and $\mathrm{ad}(e_{ij})\,\mathrm{ad}(e_{ji})$ preserves each of the root spaces. A direct check shows

$$\kappa(e_{12}, e_{21}) = \kappa(e_{13}, e_{31}) = \kappa(e_{23}, e_{32}) = 6,$$

while all other pairings among the e_{ij} vanish except for the symmetric ones. Hence the root-vector part of the Gram matrix is block diagonal with three blocks

$$\begin{pmatrix} 0 & 6 \\ 6 & 0 \end{pmatrix},$$

so its determinant is

$$(-36)^3.$$

Next compute the Cartan part. Since

$$[h, e_{ij}] = (\varepsilon_i - \varepsilon_j)(h)e_{ij},$$

the operator $\mathrm{ad}(h)$ acts diagonally on the six root spaces and trivially on the Cartan subalgebra. Thus

$$\kappa(h, h') = \sum_{\alpha} \alpha(h)\alpha(h'),$$

the sum extending over the six roots. For

$$h_1 = e_{11} - e_{22}, \qquad h_2 = e_{22} - e_{33},$$

this gives

$$\kappa(h_1, h_1) = 12, \qquad \kappa(h_2, h_2) = 12, \qquad \kappa(h_1, h_2) = -6.$$

Therefore the Cartan block is

$$\begin{pmatrix} 12 & -6 \\ -6 & 12 \end{pmatrix},$$

whose determinant is

$$108.$$

Finally, each h_r is orthogonal to each e_{ij}, since $\mathrm{ad}(h_r)$ preserves root spaces while $\mathrm{ad}(e_{ij})$ shifts weights, hence

$$\mathrm{Tr}(\mathrm{ad}(h_r)\,\mathrm{ad}(e_{ij})) = 0.$$

So the full Gram matrix is block diagonal, and

$$\det(\kappa) = (-36)^3 \cdot 108 = -2^8\, 3^9.$$

Hence the primes dividing $\det(\kappa)$ are exactly

$$2 \quad \text{and} \quad 3.$$

$\square$

Exercise 15.3.6. *Using the standard basis for $L = \mathfrak{sl}(2, F)$, write down the Casimir element of the adjoint representation of L. Do the same thing for the usual (3-dimensional) representation of $\mathfrak{sl}(3, F)$, first computing dual bases relative to the trace form.*

Proof. Let

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

be the standard basis of $L = \mathfrak{sl}(2, F)$.

For the adjoint representation, use the Killing form

$$\kappa(x, y) = \mathrm{Tr}(\mathrm{ad}(x)\,\mathrm{ad}(y)).$$

A routine calculation gives

$$\kappa(e, f) = 4, \qquad \kappa(h, h) = 8,$$

and all other pairings among e, f, h are 0. Hence the basis dual to (e, h, f) relative to κ is

$$\frac{1}{4}f, \qquad \frac{1}{8}h, \qquad \frac{1}{4}e.$$

Therefore the Casimir element is

$$\Omega_{\mathrm{ad}} = \frac{1}{4}\,ef + \frac{1}{8}\,h^2 + \frac{1}{4}\,fe = \frac{1}{8}\,h^2 + \frac{1}{4}(ef + fe).$$

If one lets this act in the adjoint representation, then it acts as the scalar 1 on L.

Now consider the usual 3-dimensional representation of $\mathfrak{sl}(3, F)$. Take the standard basis

$$e_{ij} \quad (i \neq j), \qquad h_1 = e_{11} - e_{22}, \qquad h_2 = e_{22} - e_{33},$$

and use the trace form

$$(x, y) = \mathrm{Tr}(xy).$$

Since

$$\mathrm{Tr}(e_{ij}e_{kl}) = \delta_{jk}\delta_{il},$$

the elements dual to e_{ij} are e_{ji}.

For h_1, h_2 we have

$$\mathrm{Tr}(h_1^2) = 2, \qquad \mathrm{Tr}(h_2^2) = 2, \qquad \mathrm{Tr}(h_1h_2) = -1.$$

Thus the Gram matrix is

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

whose inverse is

$$\frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Hence the dual basis elements are

$$h_1^* = \frac{2}{3}h_1 + \frac{1}{3}h_2, \qquad h_2^* = \frac{1}{3}h_1 + \frac{2}{3}h_2.$$

Therefore the Casimir element for the natural representation is

$$\Omega_{\mathrm{nat}} = \sum_{i \neq j} e_{ij}e_{ji} + h_1h_1^* + h_2h_2^*,$$

that is,

$$\Omega_{\mathrm{nat}} = \sum_{i \neq j} e_{ij}e_{ji} + \frac{2}{3}h_1^2 + \frac{1}{3}h_1h_2 + \frac{1}{3}h_2h_1 + \frac{2}{3}h_2^2.$$

Since h_1 and h_2 commute, this may be written as

$$\Omega_{\mathrm{nat}} = \sum_{i \neq j} e_{ij}e_{ji} + \frac{2}{3}h_1^2 + \frac{2}{3}h_2^2 + \frac{2}{3}h_1h_2.$$

In the usual 3-dimensional representation one computes

$$\sum_{i \neq j} e_{ij}e_{ji} = 2I_3,$$

while

$$\frac{2}{3}h_1^2 + \frac{1}{3}h_1h_2 + \frac{1}{3}h_2h_1 + \frac{2}{3}h_2^2 = \frac{2}{3}I_3.$$

Hence

$$\Omega_{\mathrm{nat}} = \frac{8}{3}I_3.$$

So the Casimir operator for the natural module is the scalar $\frac{8}{3}$. $\square$

Exercise 15.3.7. *Let L be a simple Lie algebra. Let $\beta(x,y)$ and $\gamma(x,y)$ be two symmetric associative bilinear forms on L. If β and γ are nondegenerate, prove that β and γ are proportional. (Use Schur's Lemma.)*

Proof. Since β is nondegenerate, for each $x \in L$ there exists a unique linear map

$$T : L \to L$$

such that

$$\gamma(x,y) = \beta(Tx,y) \qquad \text{for all } x, y \in L.$$

We show that T commutes with all $\operatorname{ad} a$.

For $a, x, y \in L$, using associativity of γ and then of β, we get

$$\beta\big(T([a,x]),y\big) = \gamma([a,x],y) = \gamma(a,[x,y]) = \gamma(x,[y,a]) = \beta\big(Tx,[y,a]\big) = \beta([a,Tx],y).$$

Hence

$$\beta\big(T([a,x]) - [a,Tx], y\big) = 0 \qquad \text{for all } y \in L.$$

Since β is nondegenerate,

$$T([a,x]) = [a,Tx] \qquad \text{for all } a, x \in L.$$

Thus T is an L-module endomorphism of the adjoint module L.

Now L is simple, so its adjoint module is irreducible: indeed, any submodule is an ideal of L. By Schur's Lemma, $T = c\,\mathrm{id}_L$ for some $c \in F$. Therefore

$$\gamma(x,y) = \beta(Tx,y) = \beta(cx,y) = c\,\beta(x,y) \qquad \text{for all } x, y \in L.$$

So β and γ are proportional. □

Exercise 15.3.8. *It will be seen later on that $\mathfrak{sl}(n,F)$ is actually simple. Assuming this, and using 15.3.7, prove that the Killing form κ on $\mathfrak{sl}(n,F)$ is related to the ordinary trace form by*

$$\kappa(x,y) = 2n\ \mathrm{Tr}(xy).$$

Proof. Let

$$\beta(x,y) = \mathrm{Tr}(xy) \qquad (x, y \in \mathfrak{sl}(n,F)).$$

Since

$$\mathrm{Tr}([a,x]y) = \mathrm{Tr}(axy - xay) = \mathrm{Tr}(axy - ayx) = \mathrm{Tr}(a[x,y]),$$

the form β is associative. It is also symmetric. Moreover, β is nondegenerate on $\mathfrak{sl}(n,F)$: if $x \in \mathfrak{sl}(n,F)$ satisfies $\mathrm{Tr}(xy) = 0$ for all $y \in \mathfrak{sl}(n,F)$, then $x = 0$.

Assuming $\mathfrak{sl}(n,F)$ is simple, Exercise 15.3.7 shows that κ and β are proportional. Thus there exists $c \in F$ such that

$$\kappa(x,y) = c\ \mathrm{Tr}(xy) \qquad \text{for all } x, y \in \mathfrak{sl}(n,F).$$

It remains to determine c.

Take

$$h = e_{11} - e_{22}.$$

Then

$$\mathrm{Tr}(h^2) = 2.$$

Also, $\mathrm{ad}\,h$ acts on the standard matrix units by

$$[h, e_{ij}] = (\delta_{1i} - \delta_{2i} - \delta_{1j} + \delta_{2j})e_{ij}.$$

Hence the eigenvalues of $\operatorname{ad} h$ are:

$$2 \text{ on } e_{12}, \qquad -2 \text{ on } e_{21},$$

$$1 \text{ on } e_{1j}, e_{j2}\ (j \geq 3), \qquad -1 \text{ on } e_{2j}, e_{j1}\ (j \geq 3),$$

and 0 on all remaining basis elements of $\mathfrak{sl}(n, F)$. Therefore

$$\kappa(h,h) = \operatorname{Tr}\big((\operatorname{ad} h)^2\big) = 2^2 + (-2)^2 + 4(n-2) \cdot 1^2 = 4n.$$

Comparing with

$$\kappa(h,h) = c\ \operatorname{Tr}(h^2) = 2c,$$

we get $2c = 4n$, so $c = 2n$. Thus

$$\kappa(x,y) = 2n\ \operatorname{Tr}(xy) \qquad \text{for all } x, y \in \mathfrak{sl}(n, F).$$

□